MATRIX FUNCTIONS

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Preface

Computing functions of matrices - one of the main problems in numerical linear algebra - occurs in a wide variety of applications in science and engineering. The authoritative textbook in this field is undoubtedly

> N. J. Higham, *Functions of Matrices*, SIAM, 2008.

The following quote, which outlines the importance of the topic also comes from there.

> "*From their origin in pure mathematics, matrix functions have broadened into a subject of study in applied mathematics, with widespread applications in science and engineering. Research on matrix functions involves matrix theory, numerical analysis, approximation theory, and the development of algorithms and software, so it employs a wide range of theory and methods and promotes an appreciation of all these important topics.*"

This lecture notes complement the 10 ECTS advanced course *Matrix Functions* at Technische Universität Berlin, which I first taught in summer 2024. I would like to emphasize that this script is *not* a textbook, but it is intended as a supplementary resource for following the lectures and should not replace attending them. Students who plan to attend this course need a solid background in (numerical) linear algebra. In addition, basic knowledge of complex analysis is helpful.

The very first version of this lecture notes were typed by Anna Barkova, Lennart Helmstädter, and Andreas Hygrell in 2024. They have not only made their work available to me and their fellow students, but also pointed out several inaccuracies and inconsistencies, which led to an improvement of this text. I would like to thank them for this. I also wish to thank Errol Bucy, Emrik Erikson, and Adrian Stoof for reading previous versions of this script and for providing comments and corrections. However, it is very likely that the present version also contains inaccuracies or errors and leaves room for improvements of the presentation. Therefore, please send any errors you find or comments you have to me by email at

`zur@math.tu-berlin.de`.

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1 Introduction

The aim of this course is to formally define matrix functions and study their properties. Moreover, we will discuss the numerical computation of matrix functions $f(A)$ as well as the computation of $f(A)b$. The course roughly follows the book *Functions of Matrices* by Nicholas J. Higham [12].

1.1 The Intuitive Definition of a Matrix Function

Matrix functions answer the question of whether there exists a canonical extension of a given (scalar) function

$$f : \Omega \to \mathbb{C} \tag{1.1}$$

with $\Omega \subseteq$ to (a subset of) square matrices, i.e., to

$$f : U \to \mathbb{C}^{n\times n} \tag{1.2}$$

with $U \subseteq \mathbb{C}^{n\times n}$. For some types of functions, we already know this canonical extension.

(1) If f is a polynomial, i.e., $f(z) = \sum_{k=0}^{m} \alpha_k z^k$, then $f(A)$ is defined via additions, scalar multiplications and powers of A

$$f : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}, \qquad f(A) \coloneqq \sum_{k=0}^{m} \alpha_k A^k, \tag{1.3}$$

where $A^0 \coloneqq I_n$ and $A^k \coloneqq A^{k-1}A$.

Example 1.1. Let $f(z) = z^2 - 1$ and $A = \begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix}$. Then

$$f(A) = \begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix}^2 - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & 3 \end{bmatrix}.$$

(2) If f is a rational function, i.e., $f(z) = \frac{p(z)}{q(z)}$ with (coprime) polynomials p and q, we additionally require inverses of square matrices and define

$$f(A) \coloneqq p(A)q(A)^{-1} = q(A)^{-1}p(A), \tag{1.4}$$

if $q(A)$ is nonsingular, i.e., if no eigenvalue of A is a zero of q, or equivalently, a pole of f. Note that the equality in (1.4) holds by the following lemma.

Lemma 1.2. *Let $A \in \mathbb{C}^{n\times n}$ and let $B \in \mathbb{C}^{n\times n}$ be nonsingular such that $AB = BA$ holds. Then $AB^{-1} = B^{-1}A$.*

Proof. We have $AB^{-1} = B^{-1}BAB^{-1} = B^{-1}ABB^{-1} = B^{-1}A$. □

Alternative proof. We have $B^{-1} = p(B)$ for some polynomial p of degree at most $n-1$ by the Cayley-Hamilton theorem. Then

$$AB^{-1} = Ap(B) = p(B)A = B^{-1}A,$$

where we used again that A and B commute. □

Since $p(A)$ and $q(A)$ commute, we conclude equation (1.4) from the above lemma. Later, we will see that, if $AB = BA$, then $f(A)B = Bf(A)$ holds for any function f (for which $f(A)$ is well-defined).

Example 1.3. Let $f(z) = \dfrac{z-2}{z}$ and $A = \begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix}$. Then

$$f(A) = \left(\begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} -3 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 0 & 0 \end{bmatrix}.$$

(3) If f has a power series representation, i.e., with radius of convergence $r > 0$

$$f(z) = \sum_{k=0}^{\infty} \alpha_k z^k, \qquad |z| < r, \tag{1.5}$$

and the *spectral radius*

$$\rho(A) \coloneqq \max\{|\lambda| \mid \lambda \in \Lambda(A)\} \tag{1.6}$$

of $A \in \mathbb{C}^{n\times n}$ fulfills $\rho(A) < r$, then the matrix series

$$f(A) \coloneqq \sum_{k=0}^{\infty} \alpha_k A^k \tag{1.7}$$

converges; see Section 2 below.

Example 1.4. Let $f(z) = \exp(z) = \displaystyle\sum_{k=0}^{\infty} \frac{z^k}{k!}$ (here $r = \infty$) and $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Then $A^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $A^k = 0$ for $k \geq 3$. Therefore

$$f(A) = \sum_{k=0}^{\infty} \frac{1}{k!} A^k = \frac{1}{0!} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \frac{1}{1!} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} + \frac{1}{2!} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & \frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

In all three examples, A was an upper triangular matrix and we have received an upper triangular matrix $f(A)$ as well. Later, we will see that a matrix function of an upper (lower) triangular matrix is indeed always upper (lower) triangular again.

Note that we do *not* refer to other types of functions involving matrices as matrix functions, e.g.,

- element-wise operations: $f : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$, $[a_{ij}] \mapsto [f(a_{ij})]$.

  Example: $f : \mathbb{C}^{2\times 2} \to \mathbb{C}^{2\times 2}$, $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} a^2 & b^2 \\ c^2 & d^2 \end{bmatrix}$.

- maps $f : \mathbb{C}^{n\times n} \to \mathbb{C}^{m\times m}$ with $n \neq m$.

  Example: The *trace* $\operatorname{tr} : \mathbb{C}^{n\times n} \to \mathbb{C}$, $[a_{ij}] \mapsto \sum_{k=1}^{n} a_{kk}$.

  Example: The *resolvent* $R_A : \mathbb{C} \setminus \Lambda(A) \to \mathbb{C}^{n\times n}$, $z \mapsto (zI_n - A)^{-1}$.

1.2 Applications of Matrix Functions

We consider four applications of matrix functions.

1.2.1 Linear Ordinary Differential Equations

In an initial value problem (IVP) we search for a function $y : [0,1] \to \mathbb{C}^n$ which satisfies

$$\begin{cases} y'(t) = Ay(t) + g(t, y(t)), & t \in [0,1], \\ y(0) = y_0. \end{cases} \tag{1.8}$$

for given $A \in \mathbb{C}^{n\times n}$, $g : [0,1] \times \mathbb{C}^n \to \mathbb{C}^n$ and $y_0 \in \mathbb{C}^n$.

Theorem 1.5. *The unique solution of* (1.8) *is given by* Duhamel's formula

$$y(t) = \exp(tA)y_0 + \exp(tA) \int_0^t \exp(-sA)g(s, y_0)\,\mathrm{d}s. \tag{1.9}$$

In particular, for $g = 0$ (homogeneous case) we have

$$y(t) = \exp(tA)y_0. \tag{1.10}$$

1.2.2 Network analysis

We consider an undirected graph $G = (V, E)$ with *vertices* $V = \{v_1, \dots, v_n\}$ and *edges* $E \subseteq V \times V$. The *adjacency matrix* $A = [a_{ij}] \in \mathbb{R}^{n\times n}$ of G is defined by

$$a_{ij} = \begin{cases} 1, & (v_i, v_j) \in E, \\ 0, & \text{otherwise.} \end{cases} \tag{1.11}$$

Note, that A is symmetric since G is undirected. Moreover, $(A^k)_{ij}$ is the number of distinct *walks* from v_i to v_j of exact length k (neither vertices nor edges have to be distinct).

In network analysis, we are interested in certain properties of the graph, e.g.,

- the *centrality* of a vertex v_i, i.e., how important is v_i for the whole network. To measure this, we can count the number of walks from the vertex to itself (weighted by the number of edges used):

$$\left(I_n + A + \tfrac{1}{2!}A^2 + \tfrac{1}{3!}A^3 + \tfrac{1}{4!}A^4 + \cdots\right)_{ii} = \exp(A)_{ii}.$$

- the *communicability* between v_i and v_j. Similar to centrality, we ask how well v_i and v_j can communicate with each other. This can be measure by the number of walks from v_i to v_j (weighted by the number of edges used), e.g.,

$$\left(I_n + A + \tfrac{1}{2!}A^2 + \tfrac{1}{3!}A^3 + \tfrac{1}{4!}A^4 + \cdots\right)_{ij} = \exp(A)_{ij}.$$

- the closeness of the network to being *bipartite*.

  Recall that a graph is *bipartite* if V can be divided into two disjoint sets V_1, V_2 such that any edge connects a vertex in V_1 and in V_2. An undirected graph G is bipartite, if and only if it does not contain a *closed walk* (start = end) of odd length.

  Consequently, G is close to being bipartite, if *most* of the closed walks have an even number of edges, i.e., if

$$0 \leq \frac{\sum_{i=1}^{n}\left(I_n + \frac{1}{2!}A^2 + \frac{1}{4!}A^4 + \ldots\right)_{ii}}{\sum_{i=1}^{n}\left(I_n + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \ldots\right)_{ii}} = \frac{\operatorname{tr}(\cosh(A))}{\operatorname{tr}(\exp(A))} \leq 1 \tag{1.12}$$

  is close to 1.

For an introduction on network analysis with matrix functions, we refer to [11].

Example 1.6. We consider the undirected graph $G(V, E)$ with adjacency matrix

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix};$$

see Figure 1. The first powers of the adjacency matrix are

$$A^2 = \begin{bmatrix} 3 & 2 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 \\ 1 & 1 & 2 & 3 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}, \qquad A^3 = \begin{bmatrix} 4 & 5 & 5 & 6 & 1 \\ 5 & 4 & 5 & 6 & 1 \\ 5 & 5 & 2 & 2 & 2 \\ 6 & 6 & 2 & 2 & 3 \\ 1 & 1 & 2 & 3 & 0 \end{bmatrix}, \qquad A^4 = \begin{bmatrix} 16 & 15 & 9 & 10 & 6 \\ 15 & 16 & 9 & 10 & 6 \\ 9 & 9 & 10 & 12 & 2 \\ 10 & 10 & 12 & 15 & 2 \\ 6 & 6 & 2 & 2 & 3 \end{bmatrix}$$

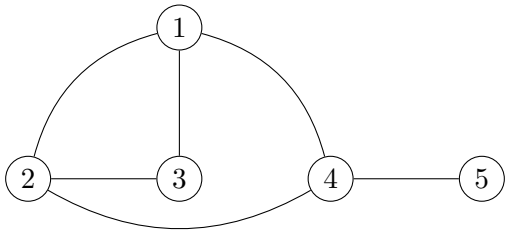


Figure 1: Graph from Example 1.6.

and the matrix exponential of the adjacency matrix is[1]

$$\exp(A) = I_n + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \ldots = \begin{bmatrix} 4.3333 & 3.9654 & 3.1207 & 3.4031 & 1.0752 \\ 3.9654 & 4.3333 & 3.1207 & 3.4031 & 1.0752 \\ 3.1207 & 3.1207 & 3.0274 & 2.1505 & 0.5648 \\ 3.4031 & 3.4031 & 2.1505 & 3.8203 & 1.7631 \\ 1.0752 & 1.0752 & 0.5648 & 1.7631 & 1.6698 \end{bmatrix}.$$

We see that the most central vertices are v_1 and v_2. The communication is best between v_1 and v_2 and worst between v_1 and v_4, and between v_2 and v_4.

The hyperbolic cosine of A is

$$\cosh(A) = I_n + \frac{1}{2!}A^2 + \frac{1}{4!}A^4 + \ldots = \begin{bmatrix} 3.3284 & 1.7854 & 0.9843 & 1.0420 & 0.8079 \\ 1.7854 & 3.3284 & 0.9843 & 1.0420 & 0.8079 \\ 0.9843 & 0.9843 & 2.5138 & 1.6157 & 0.1154 \\ 1.0420 & 1.0420 & 1.6157 & 3.2639 & 0.1187 \\ 0.8079 & 0.8079 & 0.1154 & 0.1187 & 1.6482 \end{bmatrix}.$$

and the above mentioned measure for the bipartiteness is

$$\frac{\operatorname{tr}(\cosh(A))}{\operatorname{tr}(\exp(A))} \approx 0.8195.$$

Note that other weightings may modeled by other matrix functions, e.g., if the adjacency matrix $A \in \mathbb{R}^{n\times n}$ fulfills $\rho(A) < 1$, then we have

$$f(A) = (I_n - A)^{-1} = I_n + A + A^2 + A^3 + \ldots. \tag{1.13}$$

for $f(z) = \frac{1}{1-z}$. In this case, the graph G must of course be directed, as otherwise $\rho(A) \geq 1$ holds potentially. As above, the entries of $f(A)$ give us informations about centrality and communicability, but for a uniform weighting, i.e., all walks count the same despite of there length. This was originally done in [17]. Recall the geometric series, i.e.,

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q} \quad \text{for} \quad |q| < 1, \tag{1.14}$$

which we used in (1.13).

[1]The computation in Example 1.6 were performed using the `expm` routine in `MATLAB`. Results are rounded to four decimal places.

1.2.3 Multivariate Sampling in Statistics

We continue with an application of the matrix square root $A^{\frac{1}{2}}$. Suppose, we want to generate a random vector $y \sim \mathcal{N}(\mu, C)$ where $\mathcal{N}(\mu, C)$ is the multivariate normal distribution with mean $\mu \in \mathbb{R}^n$ and a symmetric positive definite (SPD) covariance matrix $C \in \mathbb{R}^{n\times n}$. For this, we usually generate $x \in \mathcal{N}(0, I_n)$ and set $y = \mu + Lx$ where $C = LL^T$ is a Cholesky decomposition of C. Indeed,

$$\mathbb{E}[y] = \mathbb{E}[\mu + Lx] = \mathbb{E}[\mu] + \mathbb{E}[Lx] = \mathbb{E}[\mu] + L\mathbb{E}[x] = \mu$$

and

$$\begin{aligned}\mathbb{V}[y] &= \mathbb{E}\big[(y - \mathbb{E}[y])(y - \mathbb{E}[y])^T\big] = \mathbb{E}\big[(\mu + Lx - \mu)(\mu + Lx - \mu)^T\big] \\ &= \mathbb{E}\big[Lxx^TL^T\big] = L\mathbb{E}\big[xx^T\big]L^T = L\mathbb{E}\big[(x-0)(x-0)^T\big]L^T \\ &= L\mathbb{E}\big[(x - \mathbb{E}[x])(x - \mathbb{E}[x])^T\big]L^T = L\mathbb{V}[x]L^T = L \cdot I_n \cdot L^T = LL^T = C.\end{aligned}$$

Here, we have not used that L is lower triangular, but only $LL^T = C$. If it is not suitable to compute a Cholesky decomposition of C, e.g., if n is too large, we can generate y by setting $y = \mu + C^{\frac{1}{2}}x$ where $C^{\frac{1}{2}}$ is the (principal) square root of C. Since $C^{\frac{1}{2}}$ is symmetric, we also have $C^{\frac{1}{2}}(C^{\frac{1}{2}})^T = C$ and thus the above calculations hold for $y = \mu + C^{\frac{1}{2}}x$ as well. Note, that this approach does not require the full matrix $C^{\frac{1}{2}}$ but only the ability to evalaute $C^{\frac{1}{2}}x$, which allow to use Krylov subspace approaches (evaluation of $f(A)b$). We refer to [5, Sect. 5.1] for more information.

Lemma 1.7. *Let $A \in \mathbb{C}^{n\times n}$ be Hermitian positive definite (HPD). Then there exists a* unique *HPD matrix $A^{\frac{1}{2}} \in \mathbb{R}^{n\times n}$ with $A^{\frac{1}{2}}A^{\frac{1}{2}} = A$.*

Proof. Let $A = U\,\mathrm{diag}(d_1, \dots, d_n)U^H$ be a unitary diagonalization of the Hermitian matrix A, i.e., $U \in \mathbb{C}^{n\times n}$ with $U^HU = I_n$. Moreover, we have $d_j > 0$ since A is HPD. Define

$$A^{\frac{1}{2}} := U\,\mathrm{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})U^H.$$

Then $A^{\frac{1}{2}}$ is Hermitian and also positive definite since $\sqrt{d_i} > 0$. Moreover,

$$\begin{aligned}(A^{\frac{1}{2}})^2 &= U\,\mathrm{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})U^HU\,\mathrm{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})U^H \\ &= U\,\mathrm{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})\,\mathrm{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})U^H = U\,\mathrm{diag}(d_1, \dots, d_n)U^H = A.\end{aligned}$$

The proof of the uniqueness of $A^{\frac{1}{2}}$ is somewhat more involved and requires the simultaneous diagonalization of commuting normal matrices; see [16, Thm. 7.2.6] for details. □

1.2.4 Markov Transition Matrices

Markov transition matrices model the transition between a set of discrete states, e.g.,

- state 1: full employment,
- state 2: part-time employment,
- state 3: unemployed,

by a transition matrix $T = [t_{ij}] \in \mathbb{R}^{n \times n}$ where $t_{ij} \in [0, 1]$ is the probability of a transition from state i to state j. Since we are working with probabilities, we need to have $\sum_{j=1}^{n} t_{ij} = 1$ for $i = 1, \ldots, n$. Such a matrix is called *row-stochastic*.

In the example above, the transition matrix for changes within one year might be

$$T_A = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.25 & 0.6 & 0.15 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}.$$

Given this annual transition matrix, we may want to know the monthly transition matrix T_M which contains the transition probabilities for changes within one month. This matrix needs to fulfill

$$T_A = T_M^{12},$$

i.e., we are looking for a p-th root $T_A^{\frac{1}{12}}$ of A. In order to model transition probabilities, the matrix $T_A^{\frac{1}{12}}$ has to be row-stochastic as well. Conditions under which the p-th roots of stochastic matrices are (almost) stochastic themselves has been studied in [13] or the more recent article [10].

2 Definitions of $f(A)$

In this section, we consider the definition of matrix functions via the
(1) Jordan decomposition,
(2) Hermite interpolation polynomial,
(3) Taylor series,
(4) Cauchy integral formula.
The latter two definitions require the analyticity of f.

2.1 Jordan Form Definition

Recall that for any $A \in \mathbb{C}^{n\times n}$ there exists a *Jordan decomposition* of the form

$$A = XJX^{-1}, \tag{2.1}$$

where $X \in \mathbb{C}^{n\times n}$ is nonsingular and

$$J = \operatorname{diag}\left(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k)\right) \in \mathbb{C}^{n\times n} \tag{2.2}$$

is a *Jordan form* of A with *Jordan blocks*

$$J_{d_i}(\lambda_i) = \begin{bmatrix} \lambda_i & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix} \in \mathbb{C}^{d_i\times d_i}, \qquad i = 1, \ldots, k. \tag{2.3}$$

Some properties of the Jordan decomposition and related definitions are:

- $d_1 + \cdots + d_k = n$
- J is unique up to permutations of the Jordan blocks.
- Let $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_m$ be the distinct eigenvalues of A and let $\tilde{d}_i$ be the maximum size of the Jordan blocks corresponding to the eigenvalue $\tilde{\lambda}_i$. Then $\tilde{d}_i$ is called the *index* of $\tilde{\lambda}_i$.
- Recall that the *minimal polynomial* M_A of A is the unique monic polynomial of minimal degree that annihilates A, i.e., $M_A(A) = 0$. It is given by

$$M_A(z) = \prod_{i=1}^{m} (z - \tilde{\lambda}_i)^{\tilde{d}_i}. \tag{2.4}$$

- Any other polynomial which annihilates A is divided by M_A. In particular, M_A divides the *characteristic polynomial* P_A of A.

- If $d_1 = \cdots = d_k = 1$, then A is *diagonalizable* and the Jordan decomposition reduces to an *eigendecomposition*, i.e.,

$$A = XDX^{-1} \quad \text{with} \quad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n). \tag{2.5}$$

  In particular, the j-th column of X is an eigenvector of A corresponding to the eigenvalue λ_j.

For further details on the Jordan decomposition, we refer to the textbook [20, Ch. 16].

Remark 2.1. In numerical computations, we have to avoid the Jordan form since it does not depend continuously on A, i.e., small perturbations in the entries of A, e.g., caused by rounding errors, can completely change its Jordan form. For example, the Jordan form of

$$A_\varepsilon = \begin{bmatrix} 1+\varepsilon & 1 \\ 0 & 1 \end{bmatrix} \in \mathbb{C}^{2\times 2}$$

is given by

$$J_\varepsilon = \begin{cases} \begin{bmatrix} 1+\varepsilon & 0 \\ 0 & 1 \end{bmatrix}, & \varepsilon \neq 0, \\ \begin{bmatrix} 1+\varepsilon & 1 \\ 0 & 1 \end{bmatrix}, & \varepsilon = 0, \end{cases}$$

but we have $\lim_{\varepsilon\to 0} A_\varepsilon = A_0$, and

$$\lim_{\varepsilon\to 0} J_\varepsilon = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = J_0.$$

Instead, we should use a *Schur decomposition* $A = URU^H$, i.e., with $R \in \mathbb{C}^{n\times n}$ upper triangular and $U \in \mathbb{C}^{n\times n}$ unitary, whenever it is possible.

Definition 2.2. Let $A \in \mathbb{C}^{n\times n}$ have the Jordan form

$$\operatorname{diag}\big(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k)\big).$$

Then a function $f : \Omega \to \mathbb{C}$ with $\Omega \subseteq \mathbb{C}$ is *defined on the spectrum* of A, if the values

$$f^{(j)}(\lambda_i), \qquad j = 0, \ldots, d_i - 1, \qquad i = 1, \ldots, k$$

exist. Here, $f^{(j)}(\lambda_i)$ denotes the j-th derivative of f at λ_i.

For any f that is defined on the spectrum of A, we now can define $f(A)$ via the Jordan form of A.

Definition 2.3. Let $f : \Omega \to \mathbb{C}$ be defined on the spectrum of $A \in \mathbb{C}^{n\times n}$ and let

$$A = XJX^{-1} = X \operatorname{diag}\big(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k)\big)X^{-1} \tag{2.6}$$

be a Jordan decomposition of A. Then the (primary) *matrix function* $f(A)$ is defined as

$$f(A) \coloneqq X f(J) X^{-1} \quad \text{where} \quad f(J) \coloneqq \operatorname{diag}\big(f(J_{d_1}(\lambda_1)), \dots, f(J_{d_k}(\lambda_k))\big), \tag{2.7}$$

where

$$f(J_{d_i}(\lambda_i)) \coloneqq \begin{bmatrix} f(\lambda_i) & f'(\lambda_i) & \frac{f''(\lambda_i)}{2!} & \dots & \frac{f^{(d_i-1)}(\lambda_i)}{(d_i-1)!} \\ & f(\lambda_i) & f'(\lambda_i) & \ddots & \vdots \\ & & \ddots & \ddots & \frac{f''(\lambda_i)}{2!} \\ & & & \ddots & f'(\lambda_i) \\ & & & & f(\lambda_i) \end{bmatrix} \in \mathbb{C}^{d_i \times d_i}, \quad i = 1, \dots, k. \tag{2.8}$$

Remark 2.4. (1) In order to motivate the first definition in (2.7), note that for any polynomial p, we have $p(XAX^{-1}) = Xp(A)X^{-1}$, since

$$(XAX^{-1})^k = XA\underbrace{X^{-1}X}_{=I_n}AX^{-1} \cdot \ldots \cdot XAX^{-1} = XA^kX^{-1}.$$

(2) To justify (2.8), consider $J_{d_i}(\lambda_i) = \lambda_i I_{d_i} + N$ where

$$N = \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix} \in \mathbb{C}^{d_i \times d_i}.$$

Then

$$N^2 = \begin{bmatrix} 0 & 0 & 1 & & \\ & 0 & 0 & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ & & & & 0 \end{bmatrix}, \quad \dots \quad N^{d_i-1} = \begin{bmatrix} 0 & \cdots & 1 \\ & \ddots & \vdots \\ & & 0 \end{bmatrix}, \quad \text{and} \quad N^k = 0$$

for $k \geq d_i$. Next let f be a function with a Taylor series around λ_i, i.e.,

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda_i)}{k!}(z - \lambda_i)^k, \qquad |z - \lambda_i| < r.$$

Inserting $J_{d_i}(\lambda_i)$ in the series yields

$$\begin{aligned} f(J_{d_i}(\lambda_i)) &= \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda_i)}{k!}(\lambda_i I_{d_i} + N - \lambda_i I_{d_i})^k \\ &= \sum_{k=0}^{\infty} \frac{f^{(k)}(\lambda_i)}{k!} N^k = \sum_{k=0}^{d_i-1} \frac{f^{(k)}(\lambda_i)}{k!} N^k \\ &= f(\lambda_i) I_{d_i} + \begin{bmatrix} 0 & f'(\lambda_i) & & \\ & \ddots & \ddots & \\ & & \ddots & f'(\lambda_i) \\ & & & 0 \end{bmatrix} + \ldots + \begin{bmatrix} 0 & \cdots & \frac{f^{(d_i-1)}(\lambda_i)}{(d_i-1)!} \\ & \ddots & \vdots \\ & & 0 \end{bmatrix}, \end{aligned}$$

which is the same as in (2.8).

(3) If A is diagonalizable, i.e.,

$$A = X \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} X^{-1}, \quad \text{then} \quad f(A) = X \begin{bmatrix} f(\lambda_1) & & \\ & \ddots & \\ & & f(\lambda_n) \end{bmatrix} X^{-1}.$$

In particular, $f(A)$ is also diagonalizable with eigenvalues $f(\lambda_1), \ldots, f(\lambda_n)$. Moreover, the eigenvectors of A and $f(A)$ coincide.

Note, that this not an equivalence. Even though A is not diagonalizable, $f(A)$ can still be diagonal, e.g., for

$$f(z) = \cos(z) \qquad \text{and} \qquad A = \begin{bmatrix} \pi & 1 \\ 0 & \pi \end{bmatrix},$$

the function f is defined on the spectrum of A and

$$f(A) = \begin{bmatrix} f(\pi) & f'(\pi) \\ 0 & f(\pi) \end{bmatrix} = \begin{bmatrix} \cos(\pi) & -\sin(\pi) \\ 0 & \cos(\pi) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

is diagonal.

Next, we will discuss that $f(A)$ in Definition 2.3 is well-defined, i.e., it is independent of the particular Jordan decomposition that is used.

Theorem 2.5. *Let $A \in \mathbb{C}^{n\times n}$ with Jordan decompositions*

$$A = X J_1 X^{-1} \quad \textit{and} \quad A = Y J_2 Y^{-1}. \tag{2.9}$$

Moreover, let f be defined on the spectrum of A and define

$$f_1(A) \coloneqq X f(J_1) X^{-1} \quad \textit{and} \quad f_2(A) \coloneqq Y f(J_2) Y^{-1}. \tag{2.10}$$

Then $f_1(A) = f_2(A)$.

Proof. By (2.9), we have $J_1 = ZJ_2Z^{-1}$ with $Z = X^{-1}Y$. In particular, J_1 is a Jordan form of ZJ_2Z^{-1}. Hence,

$$\begin{aligned} f_1(A) &= Xf(J_1)X^{-1} = Xf(ZJ_2Z^{-1})X^{-1} \\ &= XZf(J_2)Z^{-1}X^{-1} = Yf(J_2)Y^{-1} = f_2(A). \end{aligned} \tag{2.11}$$

Note, that strictly speaking we can not use $f(ZJ_2Z^{-1}) = Zf(J_2)Z^{-1}$ at this point. However, the well-definedness of Definition 2.3 will later follow from the equivalence of the Jordan form definition with the Hermite interpolation definition (see Theorem 2.17) because the latter is invariant under similarity transformations. □

Remark 2.6. The definition of $f(A)$ does not require a globally defined (smooth) function f, but only the discrete data points

$$f^{(j)}(\lambda_i), \qquad j = 0, \ldots, d_i - 1, \qquad i = 1, \ldots, k.$$

The assumptions on f depend on the eigenstructure of A, e.g., if A is diagonalizable, then f does not have to be differentiable.

2.2 Definition via Polynomial Interpolation

Consider the *Hermite interpolation problem.* Given
(i) $\lambda_1, \ldots, \lambda_k \in \mathbb{C}$ mutually distinct,
(ii) $d_1, \ldots, d_k \in \mathbb{N}$ with $d_1 + \ldots + d_k = n + 1$,
(iii) $f_{ij} \in \mathbb{C}$ for $j = 0, \ldots, d_i - 1$ and $i = 1, \ldots, k$,
we search for a polynomial p with

$$p^{(j)}(\lambda_i) = f_{ij}, \qquad j = 0, \ldots, d_i - 1, \qquad i = 1, \ldots, k. \tag{2.12}$$

Lemma 2.7. *For any choice of parameters, there exists a unique polynomial of degree at most n that solves the Hermite interpolation problem* (2.12).

Proof. Write $p(z) = \alpha_0 + \alpha_1 z + \frac{\alpha_2}{2!}z^2 + \ldots + \frac{\alpha_n}{n!}z^n$. Then

$$p^{(j)}(z) = \alpha_j + \alpha_{j+1}z + \frac{\alpha_{j+2}}{2!}z^2 + \ldots + \frac{\alpha_{n-1}}{(n-1-j)!}z^{n-1-j} + \frac{\alpha_n}{(n-j)!}z^{n-j}.$$

Thus, (2.12) can be written as

$$\alpha_j + \alpha_{j+1}\lambda_i + \frac{\alpha_{j+2}}{2!}\lambda_i^2 + \ldots + \frac{\alpha_{n-1}}{(n-1-j)!}\lambda_i^{n-1-j} + \frac{\alpha_n}{(n-j)!}\lambda_i^{n-j} = f_{ij},$$

which yields $n + 1$ linear equations in the unknown $\alpha = [\alpha_0, \ldots, \alpha_n]^T \in \mathbb{C}^{n+1}$.

Next, we prove that this linear system is nonsingular by showing that the zero-polynomial is the unique solution of the homogeneous system, i.e., with $f_{ij} \equiv 0$. Let the polynomial $q \neq 0$ with $\deg(q) \leq n$ be a solution of (2.12) for $f_{ij} \equiv 0$. Since

$$0 = q(\lambda_i) = q'(\lambda_i) = \cdots = q^{(d_i-1)}(\lambda_i),$$

the polynomial q has a zero of order (at least) d_i at λ_i for $i = 1, \ldots, k$. Hence $\deg(q) \geq d_1 + \cdots + d_k = n + 1$, which is a contradiction. Therefore, the homogeneous linear system is uniquely solved by the zero-polynomial. Thus, the linear algebraic system is nonsingular and there exists a unique solution for (2.12). □

Remark 2.8. The unique polynomial in Lemma 2.7 of degree at most n can be given explicitly.

(1) *Lagrange-Hermite formula*

$$p(z) = \sum_{i=1}^{k} \Big(\sum_{j=0}^{d_i-1} \frac{\phi_i^{(j)}(\lambda_i)}{j!} (z-\lambda_i)^j \Big) \prod_{\substack{j=1 \\ j\neq i}}^{k} (z-\lambda_j)^{d_j}, \quad \text{where} \quad \phi_i(z) = \frac{f(z)}{\prod_{\substack{j=1 \\ j\neq i}}^{k} (z-\lambda_j)^{d_j}},$$

which reduces to the Lagrange interpolation polynomial for a matrix with distinct eigenvalues ($d_i \equiv 1$).

(2) *Newton divided difference form*
Recall that the *divided differences* are defined by

$$f[\mu_1, \ldots, \mu_{k+1}] \coloneqq \begin{cases} \frac{f[\mu_2,\ldots,\mu_{k+1}] - f[\mu_1,\ldots,\mu_k]}{\mu_{k+1}-\mu_1}, & \mu_1 \neq \mu_{k+1}, \\ \frac{f^{(k)}(\mu_{k+1})}{k!}, & \mu_1 = \mu_{k+1}. \end{cases}$$

The (unique) Hermite interpolation polynomial has the form

$$p(z) = f(\widehat{\lambda}_1) + f[\widehat{\lambda}_1, \widehat{\lambda}_2](z-\widehat{\lambda}_1) + \ldots + f[\widehat{\lambda}_1, \ldots, \widehat{\lambda}_{n+1}](z-\widehat{\lambda}_1) \cdot \ldots \cdot (z-\widehat{\lambda}_n),$$

where in $\widehat{\lambda}_1, \ldots, \widehat{\lambda}_{n+1}$ the interpolation node λ_i occurs d_i times, $i = 1, \ldots, k$.

Definition 2.9. Let $A \in \mathbb{C}^{n\times n}$ have the distinct eigenvalues $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_m$ with corresponding indices $\tilde{d}_1, \ldots, \tilde{d}_m$. Furthermore, let f be defined on the spectrum of A and let p be a polynomial with

$$p^{(j)}(\tilde{\lambda}_i) = f^{(j)}(\tilde{\lambda}_i), \qquad j = 0, \ldots, \tilde{d}_i - 1, \qquad i = 1, \ldots, m.$$

Then we define $f(A) \coloneqq p(A)$.

Example 2.10. Let $f(z) = \sqrt{z}$, i.e., $f'(z) = \frac{1}{2\sqrt{z}}$, and

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 4 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \overbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}}^{X} \overbrace{\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}}^{J} \overbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix}}^{X^{-1}}.$$

(1) Jordan form definition:

$$f(A) = Xf(J)X^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

(2) Hermite interpolation definition:
We need to find a polynomial p with $p(1) = 1$, $p'(1) = \frac{1}{2\sqrt{1}} = \frac{1}{2}$ and $p(4) = 2$. This problem is solved by $p(z) = -\frac{1}{18}z^2 + \frac{11}{18}z + \frac{4}{9}$. Then

$$f(A) = p(A) = -\frac{1}{18}A^2 + \frac{11}{18}A + \frac{4}{9}I_3 = \ldots = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

Remark 2.11. By Definition 2.9, the matrix $f(A)$ is always a polynomial in A, that depends on f *and* A.

Theorem 2.12. *Let $A \in \mathbb{C}^{n\times n}$ and p, q be polynomials. Then the following are equivalent:*

(1) p and q have the same values on the spectrum of A, i.e.,

$$[p^{(j)}(\tilde{\lambda}_i) = q^{(j)}(\tilde{\lambda}_i), \qquad j = 0, \ldots, \tilde{d}_i - 1, \qquad i = 1, \ldots, m.$$

(2) $p(A) = q(A)$

Proof. "(1)⇒(2)": Let p and q have the same values on the spectrum of A. Then $r \coloneqq p - q$ vanishes on the spectrum of A, i.e.,

$$r^{(j)}(\tilde{\lambda}_i) = p^{(j)}(\tilde{\lambda}_i) - q^{(j)}(\tilde{\lambda}_i) = 0, \qquad j = 0, \ldots, \tilde{d}_i - 1, \qquad i = 1, \ldots, m.$$

Then

$$r(z) = \underbrace{\Big(\prod_{i=1}^{m}(z - \tilde{\lambda}_i)^{\tilde{d}_i}\Big)}_{M_A(z)}\tilde{r}(z)$$

for some polynomial $\tilde{r}$. Thus, M_A divides r and therefore

$$p(A) - q(A) = r(A) = M_A(A)\tilde{r}(A) = 0,$$

i.e., $p(A) = q(A)$.

"(2)⇒(1)": Let $p(A) = q(A)$. Then for $r \coloneqq p - q$ we find $r(A) = 0$ and therefore M_A divides r i.e., $r(z) = M_A(z)\tilde{r}(z)$ for some polynomial $\tilde{r}$. Thus, by the Leibniz rule (generalized product rule) we have

$$r^{(j)}(\tilde{\lambda}_i) = (M_A \cdot \tilde{r})^{(j)}(\tilde{\lambda}_i) = \sum_{k=0}^{j}\binom{j}{k}M_A^{(k)}(\tilde{\lambda}_i)\tilde{r}^{(j-k)}(\tilde{\lambda}_i) = 0$$

for $j = 0, \ldots, \tilde{d}_i - 1$ and $i = 1, \ldots, m$. Hence, r vanishes on the spectrum of A, and therefore, p and q coincide on the spectrum of A. □

2.3 Taylor Series Definiton

The next two definitions require some notation from complex analysis. In this course, we will recall the main results related to our matrix functions, mostly without proofs or additional background. For more details we refer to the classical textbooks by Ahlfors [1] and Conway [6], and to the more modern textbook by Wegert [26].

Complex Analysis Sidenote

Analytic functions:
Let $\Omega \subseteq \mathbb{C}$ be open and simply connected. A function $f : \Omega \to \mathbb{C}$ is called *analytic* in $z_0 \in \Omega$, if f has a power series representation at z_0, i.e., there exists $r > 0$ with

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k, \qquad |z - z_0| < r. \tag{2.13}$$

The function f is called analytic in Ω if f is analytic at z_0 for all $z_0 \in \Omega$. Some interesting properties of analytic functions f are:

- f is infinitely often differentiable.
- $a_k = \frac{f^{(k)}(z_0)}{k!}$, i.e., the power series representation is the Taylor series.
- If the zero set of f has a limit point in Ω, then f is the zero function.

Definition 2.13. Let f be analytic in $\Omega \subseteq \mathbb{C}$ with Taylor series around z_0

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k, \qquad z \in \Omega. \tag{2.14}$$

Then for any $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subset \Omega$, we define

$$f(A) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (A - z_0 I_n)^k. \tag{2.15}$$

Next, we prove that the matrix series (2.15) is indeed converging.

Theorem 2.14. *Let f be analytic in $D_r(z_0) := \{z \in \mathbb{C} \mid |z - z_0| < r\}$ and let $A \in \mathbb{C}^{n\times n}$ with $q := \frac{\rho(A - z_0 I_n)}{r} < 1$. Then for any consistent norm on $\mathbb{C}^{n\times n}$*

$$\lim_{m\to\infty} \left\| f(A) - \sum_{k=0}^{m} \frac{f^{(k)}(z_0)}{k!} (A - z_0 I_n)^k \right\| = 0, \tag{2.16}$$

where $f(A)$ is defined according to (2.15).

Proof. From complex analysis, we know that analyticity implies that the derivatives of an analytic function can not grow arbitrarily fast. More precisely, for any function that

is analytic in $D_r(z_0)$ there exists a constant C such that $|f^{(k)}(z_0)| \leq C\frac{k!}{r^k}$. (This is an immediate consequence of the Cauchy integral formula for the k-th derivative below.)

Moreover, for any $A \in \mathbb{C}^{n\times n}$ and any $\varepsilon > 0$ there exist a consistent norm $\|\cdot\|_*$ with $\rho(A) \leq \|A\|_* \leq \rho(A) + \varepsilon$; see reference [16, Lem. 5.6.10]. In particular, there exist a consistent norm $\|\cdot\|_*$ with

$$\frac{\rho(A - z_0 I_n)}{r} \leq \frac{\|A - z_0 I_n\|_*}{r} \leq \frac{\rho(A - z_0 I_n)}{r} + \frac{1-q}{2} =: \tilde{q} < 1.$$

Using the triangle inequality, the consistency of the norm $\|\cdot\|_*$, and the geometric series, we find

$$\begin{aligned}
&\Big\| f(A) - \sum_{k=0}^{m} \frac{f^{(k)}(z_0)}{k!}(A - z_0 I_n)^k \Big\|_* = \Big\| \sum_{k=m+1}^{\infty} \frac{f^{(k)}(z_0)}{k!}(A - z_0 I_n)^k \Big\|_* \\
&\leq \sum_{k=m+1}^{\infty} \frac{|f^{(k)}(z_0)|}{k!}\|A - z_0 I_n\|_*^k \leq \sum_{k=m+1}^{\infty} C\frac{\|A - z_0 I_n\|_*^k}{r^k} \leq \sum_{k=m+1}^{\infty} C\left(\frac{\|A - z_0 I_n\|_*}{r}\right)^k \\
&\leq C \sum_{k=m+1}^{\infty} \left(\frac{\rho(A - z_0 I_n)}{r} + \frac{1-q}{2}\right)^k \leq C \sum_{k=m+1}^{\infty} \tilde{q}^k = C \cdot \frac{\tilde{q}^{m+1}}{1-\tilde{q}} \overset{m\to\infty}{\longrightarrow} 0.
\end{aligned}$$

Since all norms on a finite-dimensional vector space are equivalent, the assertion follows. □

2.4 Cauchy Integral Definition

The next definition of a matrix function is based on the Cauchy integral formula.

Complex Analysis Sidenote

Cauchy integral formula:
A *Jordan curve* $\Gamma : [0,1] \to \mathbb{C}$ is a simple (i.e., non-intersecting), closed curve ($\Gamma(0) = \Gamma(1)$) that is oriented counterclockwise; see, e.g., [26, p. 56]. By $\operatorname{int}(\Gamma)$, we denote the *interior* of Γ, i.e., the bounded component of $\mathbb{C} \setminus \Gamma$. Note that $\operatorname{int}(\Gamma)$ is always left to Γ according to its counterclockwise orientation.

Theorem 2.15 (Cauchy integral formula)**.** *Let Γ be a rectifiable*[1]*. Jordan curve and let f be analytic in* $\operatorname{int}(\Gamma)$ *and continuous on Γ. Then for any $z_0 \in \operatorname{int}(\Gamma)$, we have*

$$f(z_0) = \frac{1}{2\pi\mathbf{i}} \int_\Gamma \frac{f(z)}{z - z_0}\,\mathrm{d}z. \tag{2.17}$$

Moreover, for the k-th derivative of f, we have

$$f^{(k)}(z_0) = \frac{k!}{2\pi\mathbf{i}} \int_\Gamma \frac{f(z)}{(z - z_0)^{k+1}}\,\mathrm{d}z. \tag{2.18}$$

We refer to [26, Sect. 4.3] for a proof and more details.

Definition 2.16. Let Γ be a Jordan curve and let f be analytic in $\operatorname{int}(\Gamma)$ and continuous on Γ. Then for any $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \operatorname{int}(\Gamma)$, we define

$$f(A) \coloneqq \frac{1}{2\pi \mathbf{i}} \int_\Gamma f(z)(zI_n - A)^{-1}\,\mathrm{d}z. \tag{2.19}$$

The Integration in (2.19) is meant entry-wise here, i.e.,

$$\int_\Gamma A(z)\,\mathrm{d}z \coloneqq \Big[\int_\Gamma a_{ij}(z)\,\mathrm{d}z\Big]. \tag{2.20}$$

Note that the resolvent $(zI_n - A)^{-1}$ is always defined in (2.19) since $\Lambda(A) \cap \Gamma = \varnothing$. It can be shown that the value of the integral in this definition does not depend on the particular choice of the Jordan curve.

2.5 Equivalence of the Definitions of $f(A)$

In this section, we prove that our definitions for $f(A)$ are equivalent (modulo analyticity in Definitions 2.13 and 2.16). The four definitions have different strengths, e.g.,

- Jordan form definition: Solving matrix equations, e.g., $X^2 = A$.
- Hermite interpolation definition: Deriving basic properties of matrix functions, e.g., $Af(A) = f(A)A$.
- Taylor series definition: Evaluate $f(A)$ for A with a single eigenvalue cluster.
- Cauchy integral definition: Numerical Computation of $f(A)b$, e.g., with *quadrature rules*

$$f(A)b = \frac{1}{2\pi \mathbf{i}} \int_\Gamma f(z)(zI_n - A)^{-1}b\,\mathrm{d}z \approx \frac{1}{2\pi \mathbf{i}} \sum_{k=0}^{m} f(z_k)(z_k I_n - A)^{-1}b \frac{\operatorname{length}(\Gamma)}{m+1}.$$

Theorem 2.17. *Let f be defined on the spectrum of $A \in \mathbb{C}^{n\times n}$. Then the definitions for $f(A)$ in Definition 2.3 and Definition 2.9 are equivalent.*

Proof. Let A have the Jordan decomposition

$$A = XJX^{-1} \quad \text{with} \quad J = \operatorname{diag}\big(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k)\big)$$

and let p be a polynomial with

$$p^{(j)}(\lambda_i) = f^{(j)}(\lambda_i) \qquad \text{for} \qquad j = 0, \ldots, d_i - 1, \qquad i = 1, \ldots, k.$$

[1] A curve is *rectifiable*, if it has finite length

We will now prove $p(A) = Xf(J)X^{-1}$. Indeed,

$$p(A) = p(XJX^{-1}) \overset{\text{Rem. 2.4 (1)}}{=} Xp(J)X^{-1} = X \operatorname{diag}\big(p(J_{d_1}(\lambda_1)), \ldots, p(J_{d_k}(\lambda_k))\big)X^{-1}.$$

So, it is sufficient to show that $p(J_{d_i}(\lambda_i)) = f(J_{d_i}(\lambda_i))$ for $i = 1, \ldots, k$. We know that p has a finite Taylor series at λ_i, i.e., we have

$$\begin{aligned} p(z) &= p(\lambda_i) + p'(\lambda_i)(z-\lambda_i) + \ldots + \frac{p^{(d_i-1)}(\lambda_i)}{(d_i-1)!}(z-\lambda_i)^{d_i-1} + \sum_{k=d_i}^{m} \frac{p^{(k)}(\lambda_i)}{k!}(z-\lambda_i)^k \\ &= f(\lambda_i) + f'(\lambda_i)(z-\lambda_i) + \ldots + \frac{f^{(d_i-1)}(\lambda_i)}{(d_i-1)!}(z-\lambda_i)^{d_i-1} + \sum_{k=d_i}^{m} \frac{p^{(k)}(\lambda_i)}{k!}(z-\lambda_i)^k \end{aligned}$$

for some $m \in \mathbb{N}$ and all $z \in \mathbb{C}$. Similar to Remark 2.4.2.), we use $J_{d_i}(\lambda_i) = \lambda_i I_{d_i} + N$ where

$$N = \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix} \in \mathbb{C}^{d_i \times d_i}.$$

Inserting this in the above equation, we receive

$$\begin{aligned} p(J_{d_i}(\lambda_i)) = &\sum_{k=0}^{d_i-1} \frac{f^{(k)}(\lambda_i)}{k!}(\lambda_i I_{d_i} + N - \lambda_i I_{d_i})^k + \sum_{k=d_i}^{m} \frac{p^{(k)}(\lambda_i)}{k!}(\lambda_i I_{d_i} + N - \lambda_i I_{d_i})^k \\ &= f(\lambda_i)I_{d_i} + f'(\lambda_i)N + \ldots + \frac{f^{(d_i-1)}(\lambda_i)}{(d_i-1)!}N^{d_i-1} + \sum_{k=d_i}^{m} \frac{p^{(k)}(\lambda_i)}{k!}\underbrace{N^k}_{=0} \\ &= f(J_{d_i}(\lambda_i)). \end{aligned}$$

□

Next, we prove the equivalence of Definition 2.3 and Definition 2.13. For this, we use that the radius of convergence of a power series and its derivative coincide.

Lemma 2.18. *Consider the power series*

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k$$

with radius of convergence $r_f > 0$. Then its derivative

$$f'(z) = \sum_{k=1}^{\infty} k a_k (z - z_0)^{k-1}$$

has the same radius of convergence, i.e., $r_{f'} = r_f$.

Proof. Using $\lim_{k\to\infty} \sqrt[k]{k} = 1$, we find

$$r_f = \limsup_{k\to\infty} \frac{1}{\sqrt[k]{|a_k|}} = \limsup_{k\to\infty} \frac{1}{\sqrt[k]{k}\sqrt[k]{|a_k|}} = \limsup_{k\to\infty} \frac{1}{\sqrt[k]{k|a_k|}} = r_{f'}. \qquad \square$$

Theorem 2.19. *Let $f : \Omega \to \mathbb{C}$ have a Taylor series representation around $z_0 \in \Omega \subseteq \mathbb{C}$ with radius of convergence $r > 0$, i.e.,*

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!}(z - z_0)^k, \quad |z - z_0| < r$$

For all $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \{z \in \mathbb{C} \mid |z - z_0| < r\}$, we have

$$f(A) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!}(A - z_0 I_n)^k.$$

Proof. Let A have the Jordan decomposition

$$A = XJX^{-1} = X \operatorname{diag}(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k))X^{-1}.$$

Moreover, denote by

$$f_m(z) \coloneqq \sum_{k=0}^{m} \frac{f^{(k)}(z_0)}{k!}(z - z_0)^k. \tag{2.21}$$

the Taylor polynomial of f of order m. Its j-th derivative is given by

$$f_m^{(j)}(z) = \sum_{k=j}^{m} \frac{f^{(k)}(z_0)}{k!} k(k-1)\cdot\ldots\cdot(k-j+1)(z - z_0)^{k-j} = \sum_{k=j}^{m} \frac{f^{(k)}(z_0)}{(k-j)!}(z - z_0)^{k-j}.$$

Since f_m is a polynomial, we have

$$\begin{aligned}\lim_{m\to\infty} f_m(A) &= \lim_{m\to\infty} f_m(XJX^{-1}) = \lim_{m\to\infty} X f_m(\operatorname{diag}(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k)))X^{-1} \\ &= X\left(\lim_{m\to\infty} f_m(\operatorname{diag}(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k)))\right)X^{-1}.\end{aligned}$$

Thus, we can restrict to the case of a Jordan block $J_d(\lambda)$ with $\lambda \in \Lambda(A) \subset D_r(z_0)$. We have $J_d(\lambda) = \lambda I_d + N$ with

$$N = \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix}.$$

Hence,

$$\begin{aligned}
f_m\left(J_d(\lambda)\right) &= \sum_{k=0}^{m} \frac{f^{(k)}(z_0)}{k!}\big(J_d(\lambda) - z_0 I_d\big)^k = \sum_{k=0}^{m} \frac{f^{(k)}(z_0)}{k!}\big((\lambda - z_0)I_n + N\big)^k \\
&= \sum_{k=0}^{m} \frac{f^{(k)}(z_0)}{k!} \sum_{j=0}^{k} \binom{k}{j} (\lambda - z_0)^{k-j} N^j \\
&= \sum_{k=0}^{m} \sum_{j=0}^{k} \frac{1}{j!}\frac{f^{(k)}(z_0)}{(k-j)!}(\lambda - z_0)^{k-j} N^j = \sum_{j=0}^{m} \frac{1}{j!} N^j \sum_{k=j}^{m} \frac{f^{(k)}(z_0)}{(k-j)!}(\lambda - z_0)^{k-j} \\
&= \sum_{j=0}^{m} \frac{f_m^{(j)}(\lambda)}{j!} N^j = \sum_{j=0}^{\min\{m,d-1\}} \frac{f_m^{(j)}(\lambda)}{j!} N^j,
\end{aligned}$$

where we used that $\sum\limits_{k=0}^{m}\sum\limits_{j=0}^{k} a_{kj} = \sum\limits_{j=0}^{m}\sum\limits_{k=j}^{m} a_{kj}$ holds for all $m \in \mathbb{N}$ and $a_{kj} \in \mathbb{C}$.

By Lemma 2.18, we have $\lim\limits_{m\to\infty} f_m^{(j)}(\lambda) = f^{(j)}(\lambda)$ for all $j \in \mathbb{N}$. Therefore,

$$\lim_{m\to\infty} f_m(J_d(\lambda)) = \lim_{m\to\infty} \sum_{j=0}^{\min\{m,d-1\}} \frac{f_m^{(j)}(\lambda)}{j!} N^j = \sum_{j=0}^{d-1} \frac{f^{(j)}(\lambda)}{j!} N^j \overset{\text{cf. Rem. 2.4 (2)}}{=} f(J_d(\lambda)). \quad \square$$

Finally, we prove the equivalence of Definition 2.3 and Definition 2.16.

Theorem 2.20. *Let Γ be a Jordan curve, and let f be analytic in $\operatorname{int}(\Gamma)$ and continuous on Γ. Then for all $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \operatorname{int}(\Gamma)$ the definitions for $f(A)$ in Definitions 2.3 and 2.16 are equivalent.*

Proof. Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \operatorname{int}(\Gamma)$ have the Jordan decomposition

$$A = XJX^{-1} = X \operatorname{diag}(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k))X^{-1}.$$

Then

$$\begin{aligned}
&\frac{1}{2\pi\mathbf{i}} \int_\Gamma f(z)(zI_n - A)^{-1}\,\mathrm{d}z \\
&= \frac{1}{2\pi\mathbf{i}} \int_\Gamma f(z)(X(zI_n - J)X^{-1})^{-1}\,\mathrm{d}z \\
&= X \cdot \frac{1}{2\pi\mathbf{i}} \int_\Gamma f(z)(zI_n - J)^{-1}\,\mathrm{d}z \cdot X^{-1} \\
&= X \cdot \frac{1}{2\pi\mathbf{i}} \int_\Gamma f(z)(zI_n - \operatorname{diag}(J_{d_1}(\lambda_1), \ldots, J_{d_k}(\lambda_k))^{-1}\,\mathrm{d}z \cdot X^{-1} \\
&= X \operatorname{diag}\left(\frac{1}{2\pi\mathbf{i}} \int_\Gamma f(z)(zI_{d_1} - J_{d_1}(\lambda_1))^{-1}, \ldots, \frac{1}{2\pi\mathbf{i}} \int_\Gamma f(z)(zI_{d_k} - J_{d_k}(\lambda_k))^{-1}\right) X^{-1},
\end{aligned}$$

where we used $\Gamma \cap \Lambda(A) = \varnothing$. Hence, it is enough to prove the result for a Jordan block $J_d(\lambda) = \lambda I_d + N$. The inverse of $J_d(\lambda)$ is explicitly given by

$$(zI_d - J_d(\lambda_d))^{-1} = \begin{bmatrix} \frac{1}{z-\lambda_d} & \frac{1}{(z-\lambda_d)^2} & \cdots & \frac{1}{(z-\lambda_d)^d} \\ & \frac{1}{z-\lambda_d} & \ddots & \vdots \\ & & \ddots & \frac{1}{(z-\lambda_d)^2} \\ & & & \frac{1}{z-\lambda_d} \end{bmatrix}. \tag{2.22}$$

Moreover, the entry (i, j) of

$$F \coloneqq \frac{1}{2\pi\mathbf{i}} \int_\Gamma f(z)(zI_d - J_d(\lambda))^{-1} \, \mathrm{d}z \in \mathbb{C}^{d\times d}$$

is

$$f_{ij} = \frac{1}{2\pi\mathbf{i}} \int_\Gamma \frac{f(z)}{(z-\lambda)^{(j-i+1)}} \, \mathrm{d}z = \frac{f^{(j-i)}(\lambda)}{(j-i)!} = f(J_d(\lambda))_{ij},$$

and therefore $F = f(J_d(\lambda))$. □

2.6 Primary and Non-Primary Matrix Functions

One application of matrix functions is solving matrix equations, e.g., for $A \in \mathbb{C}^{n\times n}$ we search for a matrix $X \in \mathbb{C}^{n\times n}$ s.t.

$$X^2 = A \qquad \text{or} \qquad \exp(X) = A.$$

The above equations lead to the matrix square root and the matrix logarithm. While primary matrix functions can be written as polynomials in A, this is not the case for all solutions of the above equations, since $z \mapsto \sqrt{z}$ or $z \mapsto \log(z)$ are roughly speaking "multi-valued". Such solutions are called *non-primary matrix functions*. For example, we can find them by choosing different branches of the function for different Jordan blocks corresponding to the same eigenvalue in the Jordan form definition.

Example 2.21. Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \text{and} \qquad f(z) = \sqrt{z},$$

i.e., we search for solutions for $X^2 = A$. By using the two branches of the square root, we find the solutions

$$X_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad X_2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \qquad \text{(primary)}.$$

By choosing different branches for the two diagonal entries, we find

$$X_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{and} \quad X_4 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \text{(non-primary)}.$$

Furthermore, there is a whole family of solutions that are not derived from the Jordan form of A given by Householder reflections:

$$X_\varphi = \begin{bmatrix} \sin(\varphi) & \cos(\varphi) \\ \cos(\varphi) & -\sin(\varphi) \end{bmatrix} \qquad \text{(non-primary)}.$$

Even more generally, any *involutory matrix*, i.e., $X = X^{-1}$ is a solution of $X^2 = A$. Note, that all involutory (2×2)-matrices have the form

$$X_{a,b,c} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

with $a^2 + bc = 1$ or the form $X = \begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix}$.

In the following, we will always consider primary matrix functions, if not stated otherwise.

3 Properties of Matrix Functions

In this section, we study properties of general matrix functions (Section 3.1) and introduce important matrix function, namely the matrix exponential, the matrix logarithm, the matrix square root and the matrix sign function (Section 3.2).

3.1 Properties for General f

First, we consider sums, products and compositions of matrix functions.

Theorem 3.1. *Let $f, g : \Omega \to \mathbb{C}$ be defined on the spectrum of $A \in \mathbb{C}^{n\times n}$. Then*

$$(f+g)(A) = f(A) + g(A) \quad \textit{and} \quad (f \cdot g)(A) = f(A)g(A),$$

where $f+g$ and $f \cdot g$ is the pointwise sum and product.

Proof. Let p interpolate f and q interpolate g on the spectrum of A, i.e.,

$$f^{(j)}(\lambda_i) = p^{(j)}(\lambda_i) \quad \text{and} \quad g^{(j)}(\lambda_i) = q^{(j)}(\lambda_i), \qquad j = 0, \dots, d_i - 1, \qquad i = 1, \dots, k.$$

We then have

$$\begin{aligned}(f+g)^{(j)}(\lambda_i) &= f^{(j)}(\lambda_i) + g^{(j)}(\lambda_i) = p^{(j)}(\lambda_i) + q^{(j)}(\lambda_i) = (p+q)^{(j)}(\lambda_i),\\ (f \cdot g)^{(j)}(\lambda_i) &= \sum_{k=0}^{j} \binom{j}{k} f^{(k)}(\lambda_i) g^{(j-k)}(\lambda_i) = \sum_{k=0}^{j} \binom{j}{k} p^{(k)}(\lambda_i) q^{(j-k)}(\lambda_i) = (p \cdot q)^{(j)}(\lambda_i).\end{aligned}$$

Thus, $p+q$ interpolates $f+g$ and $p \cdot q$ interpolates $f \cdot g$ on the spectrum of A. Hence,

$$\begin{aligned}(f+g)(A) &= (p+q)(A) = p(A) + q(A) = f(A) + g(A),\\ (f \cdot g)(A) &= (p \cdot q)(A) = p(A)q(A) = f(A)g(A),\end{aligned}$$

where the second equality in each equation line is valid because we are considering polynomials in a matrix A. □

Theorem 3.2. *Let $g : \Omega \to \mathbb{C}$ be defined on the spectrum of $A \in \mathbb{C}^{n\times n}$. Moreover let $\tilde{\lambda}_1, \dots, \tilde{\lambda}_m$ be the distinct eigenvalues A with indices $\tilde{d}_1, \dots, \tilde{d}_m$ and let*

$$f^{(j)}(g(\tilde{\lambda}_i)) \quad \textit{exist} \quad \textit{for} \quad j = 0, \dots, \tilde{d}_i - 1, \qquad i = 1, \dots, m.$$

Then $(f \circ g)(A) = f(g(A))$.

Proof. Let $B \coloneqq g(A)$ and $\tilde{\mu}_i \coloneqq g(\tilde{\lambda}_i)$ for $i = 1, \dots, m$. Then the index $\tilde{\delta}_i$ of $\tilde{\mu}_i$ fulfills $\tilde{\delta}_i \leq \tilde{d}_i$. Furthermore, let p be a polynomial with

$$p^{(j)}(\tilde{\mu}_i) = f^{(j)}(\tilde{\mu}_i) = f^{(j)}(g(\tilde{\lambda}_i)), \qquad j = 0, \dots, \tilde{d}_i - 1, \qquad i = 1, \dots, m.$$

Then in particular, p interpolates f on the spectrum of B, so $f(B) = p(B)$. We show that $f \circ g$ and $p \circ g$ coincide on the spectrum of A. For this, we use a generalization of the chain

rule to higher derivatives called *Faà di Bruno's formula*. In this, formula $B_{j,k}$ denote the so-called *Bell polynomials*. The details do not matter here. It is only important that $(f \circ g)^{(j)}(\tilde{\lambda}_i)$ only depends on $f(g(\tilde{\lambda}_i)), \dots, f^{(j)}(g(\tilde{\lambda}_i))$ and $g'(\tilde{\lambda}_i), \dots, g^{(j)}(\tilde{\lambda}_i)$. We have

$$\begin{aligned}(f \circ g)^{(j)}(\tilde{\lambda}_i) &= \sum_{k=1}^{j} f^{(k)}(g(\tilde{\lambda}_i)) \cdot B_{j,k}\big(g'(\tilde{\lambda}_i), \dots, g^{(n-k+1)}(\tilde{\lambda}_i)\big) \\ &= \sum_{k=1}^{j} p^{(k)}(g(\tilde{\lambda}_i)) \cdot B_{j,k}\big(g'(\tilde{\lambda}_i), \dots, g^{(n-k+1)}(\tilde{\lambda}_i)\big) = (p \circ g)^{(j)}(\tilde{\lambda}_i)\end{aligned}$$

for $j = 0, \dots, \tilde{d}_i - 1$ and $i = 1, \dots, m$. Therefore, $(f \circ g)(A) = (p \circ g)(A)$. The function $p \circ g$ only consists of the evaluation of g and then multiplications and additions. Thus, by applying Theorem 3.1 multiple times, we receive

$$(f \circ g)(A) = (p \circ g)(A) \overset{\text{Thm. } 3.1}{=} p(g(A)) = p(B) = f(B) = f(g(A)). \qquad \square$$

To extend well-known results from the scalar case, e.g., trigonometric identities, to matrix functions, we often use the "*matrix function induction*", which works as follows. We start by proving the desired result for (i) diagonal matrices, then for (ii) diagonalizable matrices and finally for (iii) general matrices. This principle is based on the following theorem.

Theorem 3.3. *Let $\Omega \subseteq \mathbb{C}$ be open in $\mathbb{C}$ (or $\Omega \subseteq \mathbb{R}$ be open in $\mathbb{R}$) and define*

$$\begin{aligned}\mathbb{C}^{n\times n}(\Omega) &:= \{A \in \mathbb{C}^{n\times n} \mid \Lambda(A) \subseteq \Omega\}, \\ \mathcal{D}(\Omega) &:= \{A \in \mathbb{C}^{n\times n}(\Omega) \mid A \text{ is diagonalizable}\}.\end{aligned}$$

(1) The set $\mathcal{D}(\Omega)$ is dense in $\mathbb{C}^{n\times n}(\Omega)$, i.e., for any $A \in \mathbb{C}^{n\times n}(\Omega)$ there exists a sequence $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{D}(\Omega)$ with $A = \lim_{n\to\infty} A_n$.

(2) If $f : \Omega \to \mathbb{C}$ is $(n-1)$-times (real) continuously differentiable, then the corresponding matrix function $f : \mathbb{C}^{n\times n}(\Omega) \to \mathbb{C}^{n\times n}$ (according to Section 2) is continuous .

(3) If a map $f : \mathbb{C}^{n\times n}(\Omega) \to \mathbb{C}^{n\times n}$ (does not need to be a matrix function) is continuous and $f(D) = 0$ for all $D \in \mathcal{D}(\Omega)$, then $f(A) = 0$ for all $A \in \mathbb{C}^{n\times n}(\Omega)$.

(4) If the maps $f, g : \mathbb{C}^{n\times n}(\Omega) \to \mathbb{C}^{n\times n}$ are continuous with $f(D) = g(D)$ for all $D \in \mathcal{D}(\Omega)$, then $f(A) = g(A)$ for all $A \in \mathbb{C}^{n\times n}(\Omega)$.

Proof. (1) Exercise.

(2) See [15, Lem. 6.1.28].

(3) Let $A \in \mathbb{C}^{n\times n}(\Omega)$. By 1.), there exists a sequence $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{D}(\Omega)$ with $A = \lim_{n\to\infty} A_n$. Then

$$f(A) = f\big(\lim_{n\to\infty} A_n\big) = \lim_{n\to\infty} f(A_n) = 0.$$

(4) Similarly, $f(A) = \lim_{n\to\infty} f(A_n) = \lim_{n\to\infty} g(A_n) = g(A)$. □

The next corollary is an example of a "*matrix function induction*".

Corollary 3.4. *Let $A \in \mathbb{C}^{n\times n}$. Then*

(1) $\sin(A)^2 + \cos(A)^2 = I_n$

(2) $\sin(2A) = 2\sin(A)\cos(A)$

(3) $\exp(\mathbf{i}A) = \cos(A) + \mathbf{i}\sin(A)$.

Proof. (1) First, recall that $\sin(z)^2 + \cos(z)^2 = 1$ holds for all $z \in \mathbb{C}$. Then for any diagonal matrix $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \in \mathbb{C}^{n\times n}$, we have

$$\begin{aligned}\sin(D)^2 + \cos(D)^2 &= \sin(\operatorname{diag}(\lambda_1, \ldots, \lambda_n))^2 + \cos(\operatorname{diag}(\lambda_1, \ldots, \lambda_n))^2 \\ &= \operatorname{diag}(\sin(\lambda_1)^2, \ldots, \sin(\lambda_n)^2) + \operatorname{diag}(\cos(\lambda_1)^2, \ldots, \cos(\lambda_n)^2) \\ &= \operatorname{diag}(\sin(\lambda_1)^2 + \cos(\lambda_1)^2, \ldots, \sin(\lambda_n)^2 + \cos(\lambda_n)^2) = I_n.\end{aligned}$$

Next, let $A \in \mathbb{C}^{n\times n}$ be diagonalizable, i.e., $A = XDX^{-1}$. Then

$$\begin{aligned}\sin(A)^2 + \cos(A)^2 &= \sin(XDX^{-1})^2 + \cos(XDX^{-1})^2 \\ &\overset{\text{Thm. 3.5 (3)}}{=} X\sin(D)^2X^{-1} + X\cos(D)^2X^{-1} = X(\sin(D)^2 + \cos(D)^2)X^{-1} = XX^{-1} = I_n.\end{aligned}$$

Finally, let $A \in \mathbb{C}^{n\times n}$. Since the diagonalizable matrices are dense in $\mathbb{C}^{n\times n}$ by Theorem 3.3 (1), we find a sequence of diagonalizable matrices A_n with $A_n \overset{n\to\infty}{\longrightarrow} A$. Moreover, since any analytic f yields a continuous matrix function from $\mathbb{C}^{n\times n}$ to $\mathbb{C}^{n\times n}$ by Theorem 3.3 (2), we have for $f(z) = \sin(z)^2 + \cos(z)^2$

$$f(A) = f\left(\lim_{n\to\infty} A_n\right) = \lim_{n\to\infty} f(A_n) = \lim_{n\to\infty} I_n = I_n.$$

The proof of (2) and (3) is analogous. □

Theorem 3.5. *Let f be defined on the spectrum of $A \in \mathbb{C}^{n\times n}$. Then*

(1) $Af(A) = f(A)A$

(2) $f(A^T) = f(A)^T$

(3) For all nonsingular $X \in \mathbb{C}^{n\times n}$ we have $f(XAX^{-1}) = Xf(A)X^{-1}$.

(4) $\Lambda(f(A)) = f(\Lambda(A))$.

(5) For all $X \in \mathbb{C}^{n\times n}$ with $XA = AX$ we have $Xf(A) = f(A)X$.

(6) If A is block-triangular, i.e.,

$$A = \begin{bmatrix} A_{11} & \dots & A_{1m} \\ & \ddots & \vdots \\ & & A_{mm} \end{bmatrix},$$

then $f(A)$ is block-triangular as well, has the same block structure, and it holds $(f(A))_{ii} = f(A_{ii})$.

(7) If $A = \operatorname{diag}(A_{11}, \dots, A_{mm})$, i.e., A is block-diagonal, then $f(A)$ is also block-diagonal with $f(A) = \operatorname{diag}(f(A_{11}), \dots, f(A_{mm}))$.

(8) $f(I_m \otimes A) = I_m \otimes f(A)$

(9) $f(A \otimes I_m) = f(A) \otimes I_m$

Recall that the *Kronecker product* is given by

$$\otimes : \mathbb{C}^{n\times m} \times \mathbb{C}^{k\times \ell} \to \mathbb{C}^{mk \times n\ell}, \quad [a_{ij}] \otimes B \coloneqq \begin{bmatrix} a_{11}B & \cdots & a_{1m}B \\ \vdots & & \vdots \\ a_{n1}B & \cdots & a_{nm}B \end{bmatrix}.$$

Proof. (1), (2), (3), and (5) follow because $f(A)$ is a polynomial in A. For (2), note that A and A^T have the same eigenvalues and eigenvectors.

(4) Follows directly from Definition 2.3

(6) Let p be a polynomial that interpolates f on the spectrum of A. Then from $f(A) = p(A)$, we conclude that $f(A)$ is block-triangular with the same block structure and has diagonal blocks $p(A_{11}), \dots, p(A_{mm})$. Since the spectrum of A_{ii} is a subset on the spectrum of A, the polynomial p also interpolates f on the spectrum of A_{ii}. Hence $p(A_{ii}) = f(A_{ii})$ for $i = 1, \dots, m$.

(7) Same reasoning as in (6).

(8) Using $I_m \otimes A = \operatorname{diag}(A, \dots, A)$, we find

$$f(I_m \otimes A) = f\big(\operatorname{diag}(A, \dots, A)\big) \overset{(7)}{=} \operatorname{diag}\big(f(A), \dots, f(A)\big) = I_m \otimes f(A).$$

(9) Using $A \otimes B = P(B \otimes A)P^T$ for a permutation matrix P that does not depend on A and B, we find

$$f(A \otimes I_m) = f(P(I_m \otimes A)P^T) \overset{(3)}{=} Pf(I_m \otimes A)P^T \overset{(8)}{=} P(I_m \otimes f(A))P^T = f(A) \otimes I_m.$$

□

The permutation matrix P in the proof of (9) is also called a *commutation matrix*. It has a rather simple structure, e.g., for $n = m = 3$, it is given by

$$P = \left[\begin{array}{ccc|ccc|ccc} 1 & & & & & & & & \\ & & & 1 & & & & & \\ & & & & & & 1 & & \\ \hline & 1 & & & & & & & \\ & & & & 1 & & & & \\ & & & & & & & 1 & \\ \hline & & 1 & & & & & & \\ & & & & & 1 & & & \\ & & & & & & & & 1 \end{array}\right] \in \mathbb{R}^{9\times 9}.$$

Remark 3.6. (1) Note, that other reasonable looking identities may not hold, e.g., in general, we have $f(A^H) \neq f(A)^H$. To see this, let $A = I_n$ and $f(z) = \mathbf{i}z$. Then

$$f(A^H) = \mathbf{i}I_n^H = \mathbf{i}I_n \neq -\mathbf{i}I_n = f(A)^H.$$

(2) Properties involving two matrices $A, B \in \mathbb{C}^{n\times n}$, e.g., $\exp(A+B) = \exp(A)\exp(B)$, usually only hold, if A and B commute.

The following theorem answers the question under which condition the identity $f(A^H) = f(A)^H$ holds, and, in particular, when do real matrices A result in real matrices $f(A)$. Recall that for any $\Omega \subseteq \mathbb{C}$, we denote

$$\mathbb{C}^{n\times n}(\Omega) := \{A \in \mathbb{C}^{n\times n} \mid \Lambda(A) \subseteq \Omega\} \quad \text{and} \quad \mathbb{R}^{n\times n}(\Omega) := \{A \in \mathbb{R}^{n\times n} \mid \Lambda(A) \subseteq \Omega\}.$$

Theorem 3.7 ([14, Thm. 3.2])**.** *Let $\Omega \subseteq \mathbb{C}$ be open and such that each connected component is closed under complex conjugation. Moreover, let f be analytic on Ω. Then the following are equivalent.*

(1) $f(A^H) = f(A)^H$ for all $A \in \mathbb{C}^{n\times n}(\Omega)$;

(2) $f(\overline{A}) = \overline{f(A)}$ for all $A \in \mathbb{C}^{n\times n}(\Omega)$;

(3) $f(\mathbb{R}^{n\times n}(\Omega)) \subseteq \mathbb{R}^{n\times n}$;

(4) $f(\mathbb{R} \cap \Omega) \subseteq \mathbb{R}$.

Proof. "(1)⇔(2)": We have

$$f(A^H) = f(A)^H \quad \Leftrightarrow \quad f(\overline{A}) = f((A^H)^T) \overset{\text{Thm. 3.5 (2)}}{=} f(A^H)^T = \big(f(A)^H\big)^T = \overline{f(A)}.$$

"(2)⇒(3)": Let $A \in \mathbb{R}^{n\times n}(\Omega)$. Then we have $f(A) = f(\overline{A}) \overset{(ii)}{=} \overline{f(A)}$, and hence, $f(A) \in \mathbb{R}^{n\times n}$.

"(3)⇒(4)": Let $a \in \mathbb{R} \cap \Omega$. Then $aI_n \in \mathbb{R}^{n\times n}(\Omega)$, and therefore, we have $f(aI_n) = \operatorname{diag}(f(a), \ldots, f(a)) \overset{(3)}{\in} \mathbb{R}^{n\times n}$ by assumption. In particular, $f(a) \in \mathbb{R}$.

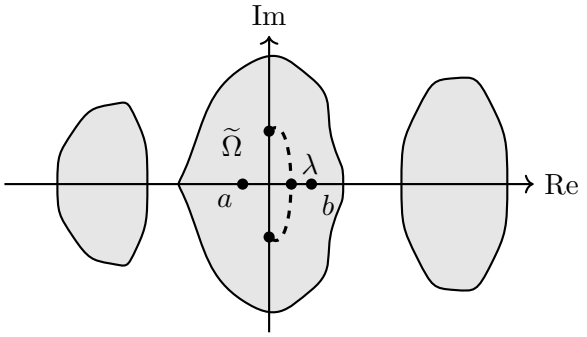


Figure 2: Construction in the proof of Theorem 3.7

Complex Analysis Sidenote

Identity theorem ([26, Thm. 3.4.1]): Let $\Omega \subseteq \mathbb{C}$ be a domain (open & connected) and $f, g : \Omega \to \mathbb{C}$ be analytic such that $\{z \in \Omega \mid f(z) = g(z)\}$ contains a limit point. Then $f \equiv g$ on Ω.

Schwarz reflection principle ([26, Thm. 6.6.1]): Let $\Omega \subseteq \mathbb{C}$ be a domain and let $\overline{\Omega} \coloneqq \{\overline{z} \mid z \in \Omega\}$. If $f : \Omega \to \mathbb{C}$ is analytic, then $g : \overline{\Omega} \to \mathbb{C}$, $g(z) = \overline{f(\overline{z})}$ is also analytic.

"(4)⇒(2)": Let $\widetilde{\Omega}$ be a connected component of Ω. Since $\widetilde{\Omega}$ is path connected and closed under conjugation, it contains some $\lambda \in \mathbb{R}$. Since $\widetilde{\Omega}$ is open, it moreover, contains a real interval $]a, b[$ with $\lambda \in]a, b[$; see Figure 2.

We define $g(z) \coloneqq \overline{f(\overline{z})}$ for all $z \in \widetilde{\Omega}$. Then g is analytic by the Schwarz reflection principle.
By (4), we have $f(]a, b[) \subseteq \mathbb{R}$, i.e., $f(z) = \overline{f(\overline{z})} = g(z)$ for all $z \in]a, b[$. Since $]a, b[$ has a limit point, we find $f(z) = g(z) = \overline{f(\overline{z})}$ for all $z \in \widetilde{\Omega}$ by the identity theorem.
The same holds for all connected components of Ω, i.e., $f(z) = \overline{f(\overline{z})}$ for all $z \in \Omega$ which implies $f(z) = \overline{f(z)}$. Thus, $f(\overline{D}) = \overline{f(D)}$ holds for all complex diagonal matrices. For diagonalizable $A \in \mathbb{C}^{n\times n}(\Omega)$ we find

$$\begin{aligned} f\big(\overline{A}\big) &= f\big(\overline{XDX^{-1}}\big) = f\big(\overline{X} \cdot \overline{D} \cdot \overline{X}^{-1}\big) = \overline{X} \cdot f\big(\overline{D}\big) \cdot \overline{X}^{-1} \\ &= \overline{X} \cdot \overline{f(D)} \cdot \overline{X^{-1}} = \overline{Xf(D)X^{-1}} = \overline{f(A)}. \end{aligned}$$

By the matrix function induction principle (see Theorem 3.3), we deduce $f(\overline{A}) = \overline{f(A)}$ for all $A \in \mathbb{C}^{n\times n}(\Omega)$. □

Next, we study matrix functions of upper triangular matrices. This is of particular interest for the numerical computation of matrix function since every matrix $A \in \mathbb{C}^{n\times n}$ has a Schur decomposition $A = QRQ^H$ and we can therefore reduce the computation of $f(A)$ to the computation of $f(R)$ via

$$f(A) = f(QRQ^H) = Qf(R)Q^H.$$

Proposition 3.8. *Let $T \in \mathbb{C}^{2\times 2}$ be upper triangular and f defined on the spectrum of T. Then*

$$f(T) = f\left(\begin{bmatrix} \lambda_1 & t_{12} \\ 0 & \lambda_2 \end{bmatrix}\right) = \begin{bmatrix} f(\lambda_1) & t_{12} f[\lambda_1, \lambda_2] \\ 0 & f(\lambda_2) \end{bmatrix}. \tag{3.1}$$

Recall that the divided difference of the first order is defined by

$$f[\lambda_1, \lambda_2] = \begin{cases} \dfrac{f(\lambda_2) - f(\lambda_1)}{\lambda_2 - \lambda_1}, & \lambda_1 \neq \lambda_2, \\ f'(\lambda_1), & \lambda_1 = \lambda_2. \end{cases}$$

Proof. By Theorem 3.5, we know that $f(T)$ is upper triangular and has the diagonal elements $f(\lambda_1)$ and $f(\lambda_2)$, so we only need to find the entry $(1,2)$ of $f(T)$. We use the identity $Tf(T) = f(T)T$, i.e.,

$$\begin{bmatrix} \lambda_1 & t_{12} \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} f(\lambda_1) & x \\ 0 & f(\lambda_2) \end{bmatrix} = \begin{bmatrix} f(\lambda_1) & x \\ 0 & f(\lambda_2) \end{bmatrix} \begin{bmatrix} \lambda_1 & t_{12} \\ 0 & \lambda_2 \end{bmatrix}. \tag{3.2}$$

For $\lambda_1 \neq \lambda_2$, considering the entry $(1,2)$ in (3.2) yields

$$0 = \lambda_1 x + t_{12} f(\lambda_2) - f(\lambda_1) t_{12} - x\lambda_2 = t_{12}\left(f(\lambda_2) - f(\lambda_1)\right) + x(\lambda_1 - \lambda_2),$$

and therefore,

$$x = t_{12} \frac{f(\lambda_2) - f(\lambda_1)}{\lambda_2 - \lambda_1} = t_{12} f[\lambda_1, \lambda_2].$$

For $\lambda_1 = \lambda_2$, we use that the matrix function is continuous for sufficiently smooth f by Theorem 3.3 2.) and consider the limit for $\lambda_2 \to \lambda_1$. We find

$$\begin{aligned} f(T) &= f\left(\lim_{\lambda_2 \to \lambda_1} \begin{bmatrix} \lambda_1 & t_{12} \\ 0 & \lambda_2 \end{bmatrix}\right) = \lim_{\lambda_2 \to \lambda_1} f\left(\begin{bmatrix} \lambda_1 & t_{12} \\ 0 & \lambda_2 \end{bmatrix}\right) \\ &= \lim_{\lambda_2 \to \lambda_1} \left(\begin{bmatrix} f(\lambda_1) & t_{12} f[\lambda_1, \lambda_2] \\ 0 & f(\lambda_2) \end{bmatrix}\right) = \begin{bmatrix} f(\lambda_1) & t_{12} f'(\lambda_1) \\ 0 & f(\lambda_1) \end{bmatrix}. \end{aligned}$$

□

Note that Proposition 3.8 can als be proven without using the continuity of the matrix function. Next, we generalize Proposition 3.8 from (2×2) matrices to $(n \times n)$ matrices.

Theorem 3.9 ([12, Thm. 4.11])**.** *Let f be defined on the spectrum of an upper triangular matrix $T = [t_{ij}] \in \mathbb{C}^{n\times n}$. Then $[f_{ij}] := f(T)$ is upper triangular as well with $f_{ii} = f(t_{ii})$ for $i = 1, \ldots, n$ and*

$$f_{ij} = \sum_{(s_0, \ldots, s_k) \in S_{ij}} t_{s_0,s_1} t_{s_1,s_2} \cdot \ldots \cdot t_{s_{k-1},s_k} f[t_{s_0,s_0}, \ldots, t_{s_k,s_k}], \qquad \text{for } 1 \leq i < j \leq n.$$

Here, S_{ij} denotes the set of strictly increasing integer sequences of any length from i to j, e.g., $S_{03} = \{(0,1,2,3), (0,1,3), (0,2,3), (0,3)\}$.

Remark 3.10. (1) We verify Theorem 3.9 for $T \coloneqq [t_{ij}] \coloneqq J_n(\lambda)$. We have

$$t_{ij} = \begin{cases} \lambda, & i = j, \\ 1, & j = i+1, \\ 0, & \text{otherwise.} \end{cases}$$

The only non-zero entries of T in the strict upper triangular part are $t_{\ell,\ell+1}$. For $1 \leq i < j \leq n$, we have

$$\begin{aligned} f_{ij} &= \sum_{(s_0,\ldots,s_k)\in\mathcal{S}_{ij}} t_{s_0 s_1} \cdot \ldots \cdot t_{s_{k-1} s_k} \cdot \ldots \cdot f[t_{s_0 s_0}, \ldots, t_{s_k s_k}] \\ &= t_{i,i+1} \cdot \ldots \cdot t_{j-1,j} \cdot f[\underbrace{\lambda, \ldots, \lambda}_{(j-i+1) \text{ times}}] \\ &= 1 \cdot \ldots \cdot 1 \cdot \frac{f^{(j-i)}(\lambda)}{(j-i)!} = \frac{f^{(j-i)}(\lambda)}{(j-i)!}, \end{aligned}$$

which indeed coincides with the entry (i, j) of $f(J_n(\lambda))$.

(2) To compute $f(T)$ as proposed in Theorem 3.9, we require $\mathcal{O}(2^n)$ floating point operations (flops), which is too expensive for practical computations. Instead, we can, just like in the (2×2) case, use the identity $Tf(T) = f(T)T$ and $\big(f(T)\big)_{ii} = f(t_{ii})$ to deduce the off-diagonal elements. This leads to the so-called *Parlett recurrence*, *block Parlett recurrence* and the *Schur-Parlett Algorithm*; see Section 5.

3.2 Properties for Special f

In this section, we study basic properties of the matrix square root, the matrix exponential, the matrix logarithm, and the matrix sign function.

3.2.1 Visualizing Analytic Functions, Complex Square Root & Logarithm

We start with an overview on the visualization of analytic functions; see the textbook [26] or the survey [27] for a comprehensive presentation.

Complex Analysis Sidenote

Phase plots - visualizing complex functions: Real functions $f : D \to \mathbb{R}$, $D \subseteq \mathbb{R}$, are usually visualized via their *graph*

$$\Gamma_f = \{(x, f(x)) \mid x \in D\} \subseteq \mathbb{R} \times \mathbb{R}.$$

For $f : \Omega \to \mathbb{C}$, $\Omega \subseteq \mathbb{C}$, the graph Γ_f is a subset of $\mathbb{C} \times \mathbb{C}$ which is a 4-dimensional space over $\mathbb{R}$. Hence, f can not be visualized in the same manner. Instead, we consider a *phase plot* of f, where the complex plane is colored according to the

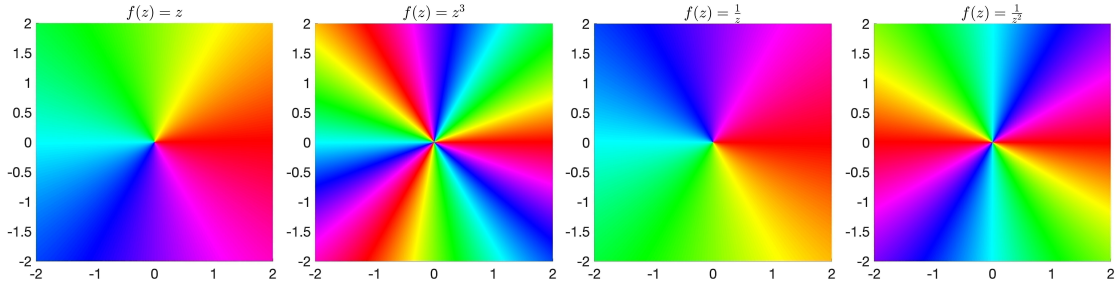


Figure 3: The order of zeros and poles of analytic functions in phase plots.

phase of f

$$z \mapsto \frac{f(z)}{|f(z)|}. \tag{3.3}$$

In particular, phase plots do not depend on the absolute value, but only on the argument of $f(z)$. Note that the phase determines an analytic function up to a multiplicative constant. Many characteristics of analytic functions can be read off from a phase plot, e.g.,

- *zeros* - points z_0 where on a small circle (counter-clockwise oriented) around z_0 all colors appear in the order red, yellow, green. The number of runs through the color wheel determines the order of the zero.

  In a small neighborhood around a zero z_0 of f of order k, the phase plot of f looks like the phase plot of $(z - z_0)^k$.

- *poles* - points z_0 where on a small circle (counter-clockwise oriented) around z_0 all colors appear in the order green, yellow, red. The number of runs through the color wheel determines the order of the pole.

  In a small neighborhood around a pole z_0 of f of order k, the phase plot of f looks like the phase plot of $(z - z_0)^{-k}$.

- *critical points* ($f'(z_0) = f''(z_0) = \ldots = f^k(z_0) = 0$) - intersection points of $k + 1$ isochromatic lines.

Example 3.11. In Figure 3, we see the functions

$$f_1(z) = z, \quad f_2(z) = z^3, \quad f_3(z) = \frac{1}{z}, \quad f_4(z) = \frac{1}{z^3}.$$

On a small circle about 0 the color wheel is traveled around $+1$, $+3$, -1, -2 times, respectively. Thus, f_1 has a a zero of order 1 at z_0, f_2 has a a zero of order 3 at z_0, f_3 has a a pole of order 1 at z_0, f_4 has a a zero of order 2 at z_0, respectively.

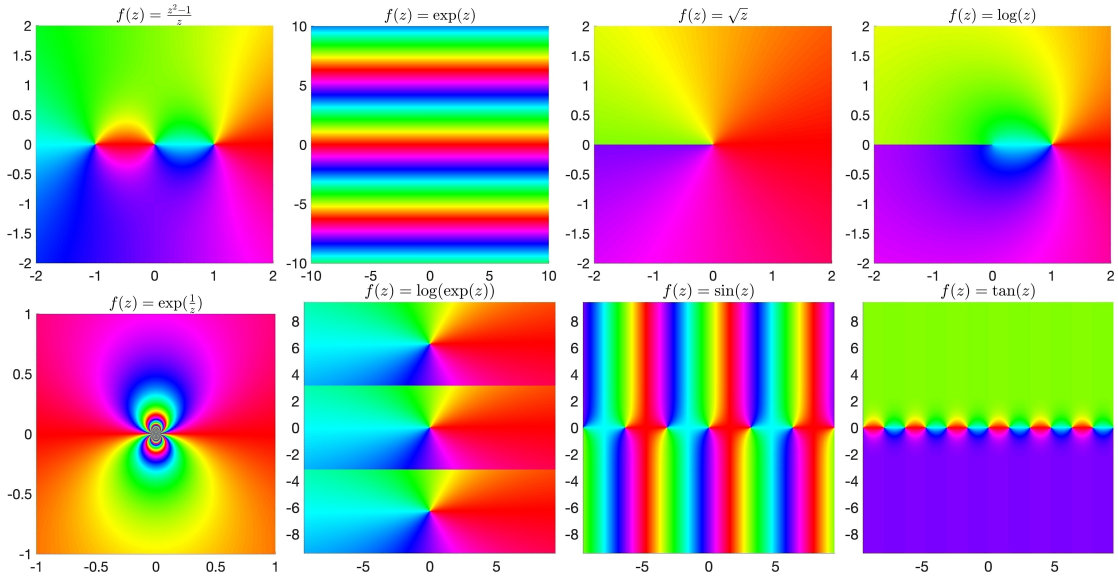


Figure 4: Phase plots for several analytic functions.

Complex Analysis Sidenote

Complex square root and complex logarithm: Recall that any $z \in \mathbb{C} \setminus \{0\}$ has a unique polar form

$$z = r\mathrm{e}^{\mathbf{i}\varphi}$$

where $r > 0$ and $\varphi \in]-\pi, \pi[$.

- Complex square root: Any $w \in \mathbb{C}$ with $w^2 = z$ is called a square root of z. For $z = r\mathrm{e}^{\mathbf{i}\varphi} \in \mathbb{C} \setminus \{0\}$, there are the two square roots

$$\sqrt{r}\mathrm{e}^{\mathbf{i}\frac{\varphi}{2}} \qquad \text{and} \qquad \sqrt{r}\mathrm{e}^{\mathbf{i}(\frac{\varphi}{2}+\pi)}. \tag{3.4}$$

  The first branch, i.e., the map

$$\sqrt{\cdot} : \mathbb{C} \setminus \mathbb{R}_{\leq 0} \to \mathbb{C}, \quad r\mathrm{e}^{\mathbf{i}\varphi} \mapsto \sqrt{r}\mathrm{e}^{\mathbf{i}\frac{\varphi}{2}} \tag{3.5}$$

  is called *principal square root*.

  Both branches can not be continuously extended to the *branch cut* $\mathbb{R}_{\leq 0}$. However, glueing them together at the branch cut yields a *Riemann surface* on which f is continuous; see Figure 5. More details and illuminating illustrations can be found in [26, Ch. 7]

- Complex logarithm: Any $w \in \mathbb{C}$ with $\mathrm{e}^w = z$ is called a *logarithm* of z. Assuming that the well-known logarithm identities also apply to complex

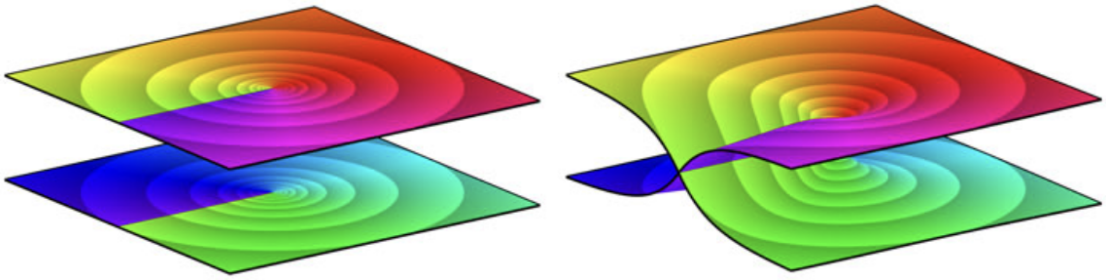


Figure 5: Left: the two branches of the complex square root (top principal branch); right: the Riemann surface of the square root [26, Fig. 7.2].

numbers, we receive the infinite family of solutions of $\mathrm{e}^w = z$

$$\log(z) = \log(r\mathrm{e}^{\mathbf{i}\varphi}) = \ln(r) + \log(\mathrm{e}^{\mathbf{i}\varphi}) = \ln(r) + \mathbf{i}\varphi + 2\pi k\mathbf{i}, \qquad k \in \mathbb{Z}$$

where we write ln to distinguish the logarithm for real numbers from the logarithm for complex numbers. Every $k \in \mathbb{Z}$ gives a branch of the complex logarithm. Glueing them together at the branch cut $\mathbb{R}_{\leq 0}$ yields a Riemann surface as above; see Figure 6. The branch with $k = 0$, i.e.,

$$\log : \mathbb{C} \setminus \mathbb{R}_{\leq 0} \to \{z \in \mathbb{C} \mid \mathrm{Im}(z) \in \,]-\pi, \pi[\}, \quad r\mathrm{e}^{\mathbf{i}\varphi} \mapsto \ln(r) + i\varphi \tag{3.6}$$

is the *principal logarithm*. For all $z = r\mathrm{e}^{\mathbf{i}\varphi} \in \mathbb{C} \setminus \mathbb{R}_{\leq 0}$, we indeed have

$$\mathrm{e}^{\log(z)} = \mathrm{e}^{\ln(r)+i\varphi} = \mathrm{e}^{\ln(r)}\mathrm{e}^{\mathbf{i}\varphi} = r\mathrm{e}^{\mathbf{i}\varphi} = z.$$

Also, for any $z = x + \mathbf{i}y \in \{z \in \mathbb{C} \mid \mathrm{Im}(z) \in \,]-\pi, \pi[\}$, we receive

$$\log(\mathrm{e}^z) = \log(\mathrm{e}^{x+\mathbf{i}y}) = \log(\mathrm{e}^x\mathrm{e}^{\mathbf{i}y}) \overset{y\in]-\pi,\pi[}{=} \ln(\mathrm{e}^x) + \mathbf{i}y = x + \mathbf{i}y = z.$$

As for the real logarithm, we have $\log'(z) = \frac{1}{z}$. However, many well-known identities for the real logarithm only hold modulo $2\pi\mathbf{i}$ in the complex case.

Example 3.12. We consider the phase plots in Figure 4.

(1) The first plot shows $f(z) = \frac{z^2-1}{z}$ with zeros at -1 and 1, and a pole at 0.

(2) The next plot shows $f(z) = \mathrm{e}^z = \mathrm{e}^{x+\mathbf{i}y} = \mathrm{e}^x\mathrm{e}^{\mathbf{i}y}$, which has no zeros and a $2\pi\mathbf{i}$-periodic phase.

(3) In the third phase plot, we see the principlal branch of the square root, particularly the branch cut $\mathbb{C} \setminus \mathbb{R}_{\leq 0}$, where $z \mapsto \sqrt{z}$ is not continuous. Also $\sqrt{z} \in \{z \in \mathbb{C} \mid \mathrm{Re}(z) > 0\}$, i.e., $\arg(\sqrt{z}) = \frac{\varphi}{2} \in \,]\frac{\pi}{2}, \frac{\pi}{2}[$ is apparent in the plot.

(4) Also $f(z) = \log(z)$ has the branch cut $\mathbb{C}\setminus\mathbb{R}_{\leq 0}$, where f is not continuous. Moreover, we see the only zero of f at $z_0 = 1$.

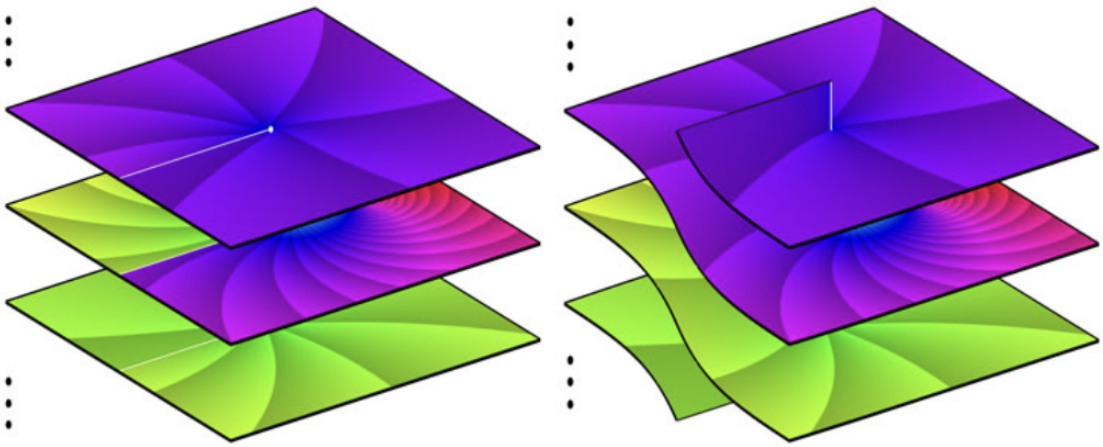


Figure 6: Left: three branches of the complex logarithm root including the principal branch (mid); right: Riemann surface of the logarithm with three sheets. The dots indicate the infinite number of branches and sheets [26, Fig. 7.6].

(5) The function $f(z) = \mathrm{e}^{\frac{1}{z}}$ has a rather wild behavior near the point $z_0 = 0$. In fact this point is an *essential singularity* and the behavior of f near such a point is described in the *Great Picard's Theorem*; see [6, Sect. XII §4].

If an analytic function f has an essential singularity at a point z_0, then in each neighborhood of z_0, $f(z)$ assumes each complex number, with at most a single exception, infinitely often.

(6) We see that $\log(\mathrm{e}^z) = z$ only holds on the strip $\{z \in \mathbb{C} \mid \mathrm{Im}(z) \in]-\pi, \pi[\}$ and that $f(z) = \log(\mathrm{e}^z)$ has a $2\pi\mathbf{i}$-periodic phase.

(7) Phase plot 7 shows the sine function including its zeros $\{\pi k \mid k \in \mathbb{Z}\}$ and critical points (zeros of the cosine) that appear as color saddles at $\{\frac{\pi}{2} + \pi k \mid k \in \mathbb{Z}\}$.

(8) The tangent with zeros at $\{\pi k \mid k \in \mathbb{Z}\}$ and poles at $\{\frac{\pi}{2} + \pi k \mid k \in \mathbb{Z}\}$ is displayed.

3.2.2 Matrix Square Root

First, we define the unique principal square root of a matrix with eigenvalues in $\mathbb{C} \setminus \mathbb{R}_{\leq 0}$.

Theorem 3.13. *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$.*

(1) There exists a unique matrix $X \in \mathbb{C}^{n\times n}$ with $X^2 = A$ and

$$\Lambda(X) \subseteq \{z \in \mathbb{C} \mid \mathrm{Re}(z) > 0\}, \tag{3.7}$$

which we call the principal square root *of A and we write $A^{\frac{1}{2}} := X$.*

(2) If $A \in \mathbb{R}^{n\times n}$, then $A^{\frac{1}{2}} \in \mathbb{R}^{n\times n}$.

Proof. (1) Let $X^2 = A$, i.e., X is a square root of A. If X is a non-primary square root, then X has eigenvalues $\mu_1, \mu_2 \in \mathbb{C}$ with $\mu_1 \neq \mu_2$ and $\mu_1^2 = \mu_2^2$. Thus, either $\operatorname{Re}(\mu_1) \leq 0$ or $\operatorname{Re}(\mu_2) \leq 0$, i.e., (3.7) is violated. Since A has no eigenvalues in $\mathbb{R}_{\leq 0}$, according to Definition 2.3, there exists only one (primary) square root with eigenvalues in $\{z \in \mathbb{C} \mid \operatorname{Re}(z) > 0\}$.

(2) Follows by Theorem 3.7 since $\sqrt{\cdot}$ is analytic on $\mathbb{C} \setminus \mathbb{R}_{\leq 0} = \overline{\mathbb{C} \setminus \mathbb{R}_{\leq 0}}$ and $f(a) \in \mathbb{R}$ holds for all $a \in (\mathbb{C} \setminus \mathbb{R}_{\leq 0}) \cap \mathbb{R} = \mathbb{R}_{>0}$. □

Corollary 3.14. *If $A \in \mathbb{C}^{n\times n}$ is Hermitian positive definite (HPD), then $A^{\frac{1}{2}}$ is also HPD.*

Proof. Since $\Lambda(A) \subseteq \mathbb{R}_{>0} \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$, the principal square root $A^{\frac{1}{2}}$ is defined. By Theorem 3.7,

$$(A^{\frac{1}{2}})^H = (A^H)^{\frac{1}{2}} = A^{\frac{1}{2}}.$$

Hence, A is Hermitian. Moreover,

$$\Lambda(A^{\frac{1}{2}}) = \left\{\sqrt{\lambda} \mid \lambda \in \Lambda(A)\right\} \subseteq \mathbb{R}_{>0}$$

by the spectral mapping theorem (Theorem 3.5 (4)). □

3.2.3 Matrix Exponential and Matrix Logarithm

The matrix exponential is not only the most studied matrix function, but it has motivated the field of matrix functions in the first place.

We start with the matrix function pendent of the functional equation of the exponential.

Theorem 3.15. *Let $A, B \in \mathbb{C}^{n\times n}$. The following are equivalent.*

(1) $\exp((A+B)t) = \exp(At)\exp(Bt)$ *holds for all $t \in \mathbb{C}$.*

(2) $AB = BA$

Proof. The proof requires theory about commuting matrices which is beyond the scope of this course. We refer to the lecture "Matrix Theory". □

The next result can be interpreted as the matrix exponential version of (i) $|z| = 1 \Leftrightarrow z = \mathrm{e}^{\mathbf{i}\varphi}, \varphi \in \mathbb{R}$, and (ii) $e^0 = 1$.

Theorem 3.16. *(1) The matrix $A \in \mathbb{C}^{n\times n}$ is unitary if and only if there exists a Hermitian matrix $B \in \mathbb{C}^{n\times n}$ with $A = \exp(\mathbf{i}B)$.*

(2) The matrix $A \in \mathbb{C}^{n\times n}$ is real and orthogonal with $\det(A) = 1$ if and only if there exists a skew-symmetric matrix $B \in \mathbb{R}^{n\times n}$ with $A = \exp(B)$.

Proof. (1) "⇒": Let A be unitary and denote its eigenvalues by $\lambda_1, \ldots, \lambda_n$. We then have $|\lambda_1| = \ldots = |\lambda_n| = 1$, i.e., all eigenvalues of A have the form

$$\lambda_j = \mathrm{e}^{\mathbf{i}\varphi_j}, \qquad \varphi_j \in \mathbb{R}, \qquad j = 1, \ldots, n.$$

Since A is normal, we find a unitary matrix $U \in \mathbb{C}^{n\times n}$ such that

$$A = U \operatorname{diag}(\lambda_1, \ldots, \lambda_n)\, U^H,$$

and therefore

$$\begin{aligned} A &= U \operatorname{diag}(\lambda_1, \ldots, \lambda_n)\, U^H = U \operatorname{diag}\left(\mathrm{e}^{\mathbf{i}\varphi_1}, \ldots, \mathrm{e}^{\mathbf{i}\varphi_n}\right) U^H \\ &= U \exp(\operatorname{diag}(\mathbf{i}\varphi_1, \cdots, \mathbf{i}\varphi_n))\, U^H = U \exp(\mathbf{i} \operatorname{diag}(\varphi_1, \ldots, \varphi_n))\, U^H \\ &\overset{\text{Thm.3.3.}}{=} \exp(\mathbf{i} \underbrace{U \operatorname{diag}(\varphi_1, \ldots, \varphi_n)\, U^H}_{=:B}). \end{aligned}$$

Moreover, B is Hermitian since $\varphi_j \in \mathbb{R}$ for $j = 1, \ldots, n$.

"⇐": Let $B \in \mathbb{C}^{n\times n}$ be Hermitian and $A = \exp(\mathbf{i}B)$. Then

$$\begin{aligned} AA^H &= \exp(\mathbf{i}B) \exp(\mathbf{i}B)^H \overset{\text{Thm. 3.7}}{=} \exp(\mathbf{i}B) \exp\left((\mathbf{i}B)^H\right) = \exp(\mathbf{i}B) \exp(-\mathbf{i}B) \\ &\overset{\text{Thm. 3.15}}{=} \exp(\mathbf{i}B - \mathbf{i}B) = \exp(0) = I_n. \end{aligned}$$

Hence, A is unitary.

(2) "⇒": Let A be orthogonal with $\det(A) = 1$. Then there exists a real block eigendecomposition

$$A = QDQ^T,$$

with orthogonal $Q \in \mathbb{R}^{n\times n}$ and block-diagonal $D = \operatorname{diag}(D_{11}, \ldots, D_{kk})$, where

$$D_{jj} = [1], \quad D_{jj} = [-1], \quad \text{or} \quad D_{jj} = \begin{bmatrix} \cos(\varphi_j) & \sin(\varphi_j) \\ -\sin(\varphi_j) & \cos(\varphi_j) \end{bmatrix}, \varphi_j \in \mathbb{R}$$

for $j = 1, \ldots, k$.

Note that in the last case, we have $\det(D_{jj}) = \cos(\varphi_j)^2 + \sin(\varphi_j)^2 = 1$. We conclude from $\det(A) = 1$, that the second type ($D_{jj} = [-1]$) occurs an even number of times on the diagonal. W.l.o.g. we assume that these blocks occur in pairs and can therefore be interpreted as (2×2) blocks of the third type with $\varphi_j = \pi$.

Define $\Delta \coloneqq \operatorname{diag}(\Delta_{11}, \ldots, \Delta_{kk})$ with

$$\Delta_{jj} \coloneqq \begin{cases} [0], & \text{if } D_{jj} = [1], \\ \begin{bmatrix} 0 & \varphi_j \\ -\varphi_j & 0 \end{bmatrix}, & \text{if } D_{jj} = \begin{bmatrix} \cos(\varphi_j) & \sin(\varphi_j) \\ -\sin(\varphi) & \cos(\varphi_j) \end{bmatrix}, \end{cases}$$

for $j = 1, \ldots, k$. Then Δ is skew-symmetric and a somewhat lengthy computation gives $\exp(\Delta_{jj}) = D_{jj}$ for $j = 1, \ldots, n$. We receive

$$A = QDQ^T = Q\exp(\Delta)Q^T = \exp(\underbrace{Q\Delta Q^T}_{=:B}).$$

"$\Leftarrow$": Let $B \in \mathbb{R}^{n\times n}$ be skew-symmetric and $A = \exp(B)$. Then by Theorem 3.7, $A \in \mathbb{R}^{n\times n}$. Furthermore,

$$\begin{aligned} AA^T &= \exp(B)\exp(B)^T = \exp(B)\exp(B^T) = \exp(B + B^T) \\ &= \exp(B - B) = \exp(0) = I_n. \end{aligned}$$

Hence, A is orthogonal. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of B. Since $\mathrm{Re}(\lambda_j) = 0$ for $j = 1, \ldots, n$ and eigenvalues of real matrices occur in complex conjugate pairs, we have $\sum_{j=1}^n \lambda_j = 0$. Thus,

$$\det(A) = \det(\exp(B)) = \prod_{j=1}^n \mathrm{e}^{\lambda_j} = \mathrm{e}^{\sum_{j=1}^n \lambda_j} = \mathrm{e}^0 = 1. \qquad \square$$

The following well-known result is a consequence of the scalar functional equation.

Theorem 3.17 (Jacobi's formula)**.** *Let $A \in \mathbb{C}^{n\times n}$. Then* $\det(\exp(A)) = \mathrm{e}^{\mathrm{tr}(A)}$.

Proof. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. Then $\mathrm{e}^\lambda_1, \ldots, \mathrm{e}^\lambda_n$ are the eigenvalues of $\exp(A)$ by the spectral mapping theorem (Theorem 3.5 (4)), and therefore

$$\det(\exp(A)) = \prod_{j=1}^n \mathrm{e}^{\lambda_j} = \mathrm{e}^{\sum_{j=1}^n \lambda_j} = \mathrm{e}^{\mathrm{tr}(A)}. \qquad \square$$

Theorem 3.18. *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$.*

(1) There exists a unique matrix $X \in \mathbb{C}^{n\times n}$ with $\exp(X) = A$ and

$$\Lambda(X) \subseteq \{z \in \mathbb{C} \mid \mathrm{Im}(z) \in \,]-\pi, \pi[\}, \tag{3.8}$$

which we call the principal logarithm of A *and we write* $\log(A) \coloneqq X$.

(2) If $A \in \mathbb{R}^{n\times n}$, then $\log(A) \in \mathbb{R}^{n\times n}$.

Proof. Similar to the proof of Theorem 3.13. We omit the details. $\square$

Theorem 3.19. *Let $A \in \mathbb{C}^{n\times n}$.*

(1) If $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$ then $\exp(\log(A)) = A$.

(2) If $\Lambda(A) \subseteq \{z \in \mathbb{C} \mid \mathrm{Im}(z) \in \,]-\pi, \pi[\}$ then $\log(\exp(A)) = A$.

Proof. The statements follow from

$$\begin{aligned} \mathrm{e}^{\log(z)} &= z \qquad \text{for all } z \in \mathbb{C} \setminus \mathbb{R}_{\leq 0}, \\ \log(\mathrm{e}^z) &= z \qquad \text{for all } z \in \mathbb{C} \text{ with } \operatorname{Im}(z) \in \,]-\pi, \pi[\end{aligned}$$

by the matrix function induction principle; see Theorem 3.3. □

We close this section with *Richter's integral formula* for the matrix logarithm, that allows us to compute $\log(A)b$ via quadrature rules; see Section 6.2. For the proof of the integral formula, we need the following lemma.

Lemma 3.20. *Let be $[a, b] \subseteq \mathbb{R}$ with $a \neq b$, $U \subseteq \mathbb{C}^{n\times n}$ open and $f : [a, b] \times U \to \mathbb{C}^{n\times n}$ continuous. Then the map*

$$\Phi : U \to \mathbb{C}^{n\times n}, \qquad A \mapsto \int_a^b f(t, A)\,\mathrm{d}t,$$

is continuous.

Proof. Let $A \in U$ and $\varepsilon > 0$. Since U is open, there exists $\delta_1 > 0$ with

$$S \coloneqq \{B \in \mathbb{C}^{n\times n} \mid \|A - B\| \leq \delta_1\} \subseteq U,$$

where $\|\cdot\|$ is an arbitrary norm on $\mathbb{C}^{n\times n}$. Note that the set $[a, b] \times S$ is compact. By the Heine-Cantor theorem, the restricted map $f|_{[a,b]\times S}$ is uniformly continuous. Thus, there exists $\delta_2 > 0$ such that

$$\|f(t_1, B_1) - f(t_2, B_2)\| < \frac{\varepsilon}{b - a}$$

holds for all $(t_1, B_1), (t_2, B_2) \in [a, b] \times S$ with $\max\{|t_1 - t_2|, \|B_1 - B_2\|\} < \delta_2$.

Finally, for all $B \in U$ with $\|A - B\| < \min\{\delta_1, \delta_2\}$, we have

$$\begin{aligned} \|\Phi(A) - \Phi(B)\| &= \left\| \int_a^b f(t, A)\,\mathrm{d}t - \int_a^b f(t, B)\,\mathrm{d}t \right\| = \left\| \int_a^b \big(f(t, A) - f(t, B)\big)\,\mathrm{d}t \right\| \\ &\leq \int_a^b \|f(t, A) - f(t, B)\|\,\mathrm{d}t \leq \int_a^b \frac{\varepsilon}{b - a}\,\mathrm{d}t = \varepsilon. \end{aligned}$$

□

Theorem 3.21 ([12, Thm. 11.1])**.** *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Then*

$$\log(A) = \int_0^1 (A - I_n)\big(t(A - I_n) + I_n\big)^{-1}\,\mathrm{d}t. \tag{3.9}$$

Proof. For $z \in \mathbb{C} \setminus \mathbb{R}_{\leq 0}$, we have

$$\int_0^1 \frac{z - 1}{t(z - 1) + 1}\,\mathrm{d}t = \log\big(t(z - 1) + 1\big)\Big|_{t=0}^1 = \log(z) - \log(1) = \log(z).$$

Thus, (3.9) holds for diagonal matrices. Let A be diagonalizable, i.e., $A = XDX^{-1}$ with a diagonal matrix $D \in \mathbb{C}^{n\times n}$ and $\Lambda(D) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Then the assertion holds for A since

$$\begin{aligned}\log(A) = \log(XDX^{-1}) &= X\log(D)X^{-1} = X\Big(\int_0^1 (D - I_n)\big(t(D-I_n)+I_n\big)^{-1}\,\mathrm{d}t\Big)X^{-1}\\ &= \int_0^1 (XDX^{-1} - XX^{-1})\big(t(XDX^{-1}-XX^{-1})+XX^{-1}\big)^{-1}\,\mathrm{d}t\\ &= \int_0^1 (A - I_n)\big(t(A-I_n)+I_n\big)^{-1}\,\mathrm{d}t.\end{aligned}$$

Now, recall that $\log(z)$ is analytic. Hence, by Theorem 3.3 (2), the matrix logarithm is continuous. Moreover, the right-hand side of (3.9) is continuous by Lemma 3.20. Thus, by the density of diagonalizable matrices (Theorem 3.3 (1)), the assertion follows for all $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. □

3.2.4 Matrix Sign Function

Finally, we study the matrix sign function, which has surprising applications in solving matrix equations, for example.

Definition 3.22. The *sign function* is defined as

$$\operatorname{sign} : \mathbb{C} \setminus \mathbf{i}\mathbb{R} \to \mathbb{C}, \qquad \operatorname{sign}(z) \coloneqq \begin{cases} 1, & \operatorname{Re}(z) > 0,\\ -1, & \operatorname{Re}(z) < 0,\end{cases}$$

where we write $\mathbf{i}\mathbb{R} \coloneqq \{z \in \mathbb{C} \mid \operatorname{Re}(z) = 0\}$.

The sign function is analytic and we have $\operatorname{sign}^{(k)}(z) = 0$ for all $z \in \mathbb{C} \setminus \mathbf{i}\mathbb{R}$ and $k \geq 1$. Hence, for all $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbf{i}\mathbb{R}$, the *matrix sign function* $\operatorname{sign}(A)$ is defined.

Remark 3.23. (1) Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C}\setminus\mathbf{i}\mathbb{R}$ and let $A = XJX^{-1}$ be a Jordan decomposition of A where $J = \operatorname{diag}(J_+, J_-)$ is ordered such that any eigenvalue λ of $J_+ \in \mathbb{C}^{m\times m}$ fulfills $\operatorname{Re}(\lambda) > 0$ and any eigenvalue λ of $J_- \in \mathbb{C}^{(n-m)\times(n-m)}$ fulfills $\operatorname{Re}(\lambda) < 0$. Then

$$\begin{aligned}\operatorname{sign}(A) &= X\operatorname{sign}(J)X^{-1} = X\operatorname{diag}\big(\operatorname{sign}(J_+), \operatorname{sign}(J_-)\big)X^{-1}\\ &= X\operatorname{diag}(I_m, -I_{n-m})X^{-1} = X\begin{bmatrix} I_m & 0\\ 0 & -I_{n-m}\end{bmatrix}X^{-1}.\end{aligned}$$

In particular, $\operatorname{sign}(A)$ is diagonalizable with eigenvalues ± 1.

(2) For $z \in \mathbb{C}$ with $\operatorname{Re}(z) > 0$, we have

$$\frac{z}{\sqrt{z^2}} = \frac{z}{z} = 1 = \operatorname{sign}(z),$$

and for $z \in \mathbb{C}$ with $\operatorname{Re}(z) < 0$, we have

$$\frac{z}{\sqrt{z^2}} = \frac{z}{-z} = -1 = \operatorname{sign}(z).$$

Thus, the identity $\operatorname{sign}(z) = \dfrac{z}{\sqrt{z^2}}$ holds for all $z \in \mathbb{C} \setminus \mathbf{i}\mathbb{R}$. By the matrix function induction principle (Theorem 3.3), we can extend this identity to the matrix sign function and receive

$$\operatorname{sign}(A) = A(A^2)^{-\frac{1}{2}} \tag{3.10}$$

for all $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbf{i}\mathbb{R}$.

Theorem 3.24. *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbf{i}\mathbb{R}$.*

(1) $\operatorname{sign}(A)^2 = I_n$, *i.e.,* $\operatorname{sign}(A)$ *is self-inverse.*

(2) If $A \in \mathbb{R}^{n\times n}$, then $\operatorname{sign}(A) \in \mathbb{R}^{n\times n}$.

(3) The matrices $P \coloneqq \frac{1}{2}(I_n + \operatorname{sign}(A))$ and $Q \coloneqq \frac{1}{2}(I_n - \operatorname{sign}(A))$ are projections onto the A-invariant subspaces corresponding to the eigenvalues in the right and left half-plane, respectively.

Proof. (1) Using the same notation as in Remark 3.23 (1), we have

$$\operatorname{sign}(A)^2 = X \begin{bmatrix} I_m & 0 \\ 0 & -I_{n-m} \end{bmatrix} X^{-1} X \begin{bmatrix} I_m & 0 \\ 0 & -I_{n-m} \end{bmatrix} X^{-1} = I_n.$$

(2) This follows from Theorem 3.7 since $\operatorname{sign}(z)$ is analytic, the connected components of $\mathbb{C} \setminus \mathbf{i}\mathbb{R}$ are closed under complex conjugation, and $\operatorname{sign}(z) \in \mathbb{R}$ for all $z \in \mathbb{C}$.

(3) We have

$$P^2 = \frac{1}{2}\big(I_n + \operatorname{sign}(A)\big) \cdot \frac{1}{2}\big(I_n + \operatorname{sign}(A)\big) = \frac{1}{4}\big(I_n + 2\operatorname{sign}(A) + \underbrace{\operatorname{sign}(A)^2}_{=I_n}\big) = P.$$

Hence, P is a projection. Furthermore, using the notation from Remark 3.23 (1) again, we receive

$$\begin{aligned} P &= \frac{1}{2}\big(I_n + \operatorname{sign}(A)\big) = \frac{1}{2}\Big(I_n + X \begin{bmatrix} I_m & 0 \\ 0 & -I_{n-m} \end{bmatrix} X^{-1}\Big) \\ &= \frac{1}{2} X \begin{bmatrix} 2I_m & 0 \\ 0 & 0 \end{bmatrix} X^{-1} = X \begin{bmatrix} I_m & 0 \\ 0 & 0 \end{bmatrix} X^{-1}. \end{aligned}$$

The proof for Q is analogous. □

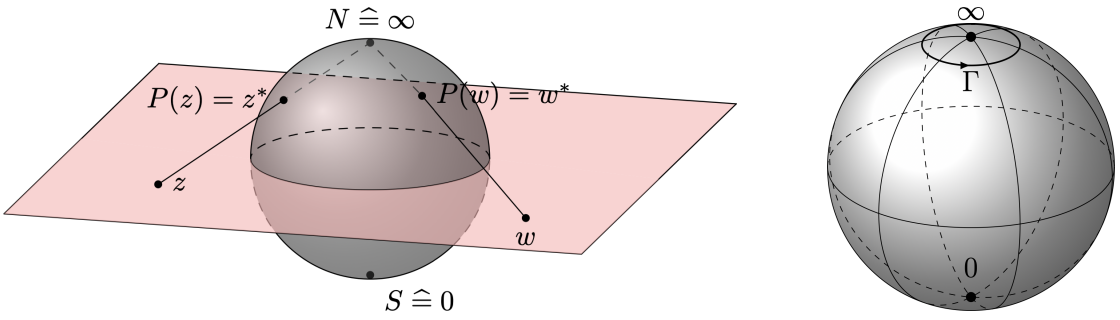


Figure 7: Left: contructions of the Riemann sphere $\widehat{\mathbb{C}}$. Right: a curve Γ surrounding the point $\infty \in \widehat{\mathbb{C}}$.

Next, we prove an integral formula for $\operatorname{sign}(A)$ that, similar to Theorem 3.21, allows us to compute $\operatorname{sign}(A)b$ via quadrature formulas. For this, we use the Cauchy integral formula but on the *Riemann sphere* $\widehat{\mathbb{C}} := \mathbb{C} \cup \{0\}$ instead of $\mathbb{C}$.

We briefly recall the *stereographic projection*. Let $\mathbb{S}^2$ be the sphere with radius 1 centered at the origin (in $\mathbb{C}$). The *stereographic image* $z \in \mathbb{C}$ of a point z^* is the intersection point of the line from the north pole N of $\mathbb{S}^2$ to z; see Fig 7 (left). Moreover, we define the stereographic image of N as ∞. In this construction, the point ∞ is just a usual point that close to very large numbers, as the point 0 is close to very small numbers. Together, $\widehat{\mathbb{C}}$ can be identified with the whole sphere $\mathbb{S}^2$. We refer to [26, pp. 20–22] for further information.

Theorem 3.25. *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbf{i}\mathbb{R}$. Then*

$$\operatorname{sign}(A) = \frac{2}{\pi} A \int_0^\infty (t^2 I_n + A^2)^{-1} \,\mathrm{d}t. \tag{3.11}$$

Sketch of the proof. We use the Cauchy integral formula (Theorem 2.15), but for a *Hankel contour* instead of a Jordan curve, which is a rectifiable (in $\widehat{\mathbb{C}}$), simple and closed curve through ∞ about the origin. One example for a Hankel contour is the oriented elliptic triangle

$$\Gamma := \big(\infty,\, \varepsilon + \varepsilon\mathbf{i}\,,\, \varepsilon - \varepsilon\mathbf{i}\,,\, \infty\big)$$

on $\widehat{\mathbb{C}}$, which we will use in this proof. The key idea is to integrate $f(z) = z^{-\frac{1}{2}}$ along this contour and then consider the limit as ε goes to 0.

By the Cauchy integral definition of matrix functions, we have

$$\begin{aligned}(A^2)^{-\frac{1}{2}} = f(A^2) &= \frac{1}{2\pi\mathbf{i}} \int_\Gamma z^{-\frac{1}{2}} (zI_n - A^2)^{-1} \,\mathrm{d}z \\ &= \frac{1}{2\pi\mathbf{i}} \int_\infty^{\varepsilon+\varepsilon\mathbf{i}} z^{-\frac{1}{2}} (zI_n - A^2)^{-1} \,\mathrm{d}z + \frac{1}{2\pi\mathbf{i}} \underbrace{\int_{\varepsilon+\varepsilon\mathbf{i}}^{\varepsilon-\varepsilon\mathbf{i}} z^{-\frac{1}{2}} (zI_n - A^2)^{-1} \,\mathrm{d}z}_{\xrightarrow{\varepsilon\to 0} 0} \\ &\quad + \frac{1}{2\pi\mathbf{i}} \int_{\varepsilon-\varepsilon\mathbf{i}}^\infty z^{-\frac{1}{2}} (zI_n - A^2)^{-1} \,\mathrm{d}z.\end{aligned}$$

In both integrals, we use the substitution $t = \mathbf{i}z^{\frac{1}{2}}$, i.e., $\mathrm{d}t = \frac{1}{2}\mathbf{i}z^{-\frac{1}{2}}\,\mathrm{d}z$ and $z = -t^2$. Then, the part of Γ, that is slightly above the negative real axis, stays there (first integral), and the part, that is slightly below the negative real axis, is mapped to a curve that is slightly above the positive real axis (second integral). We find for the first integral

$$\begin{aligned}\frac{1}{2\pi\mathbf{i}}\int_{\infty}^{\varepsilon+\varepsilon\mathbf{i}} z^{-\frac{1}{2}}(zI_n - A^2)^{-1}\,\mathrm{d}z &= \frac{1}{\pi}\int_{\infty}^{\varepsilon+\varepsilon\mathbf{i}} (-zI_n + A^2)^{-1}\cdot\frac{1}{2}\mathbf{i}z^{-\frac{1}{2}}\,\mathrm{d}z\\ \overset{\text{subst.}}{=} \frac{1}{\pi}\int_{\infty}^{\mathcal{O}(|\varepsilon|^{\frac{1}{2}})} (t^2I_n + A^2)^{-1}\,\mathrm{d}t &\xrightarrow{\varepsilon\to 0} \frac{1}{\pi}\int_{-\infty}^{0}(t^2I_n + A^2)^{-1}\,\mathrm{d}t = \frac{1}{\pi}\int_0^{+\infty}(t^2I_n + A^2)^{-1}\,\mathrm{d}t,\end{aligned}$$

where the contour integral on $\widehat{\mathbb{C}}$ converges to an improper real integral on $]-\infty, 0]$ for $\varepsilon \to 0$. Similarly, for the second integral, we find

$$\begin{aligned}&\frac{1}{2\pi\mathbf{i}}\int_{\varepsilon-\varepsilon\mathbf{i}}^{\infty} z^{-\frac{1}{2}}(zI_n - A^2)^{-1}\,\mathrm{d}z \overset{\text{subst.}}{=} \frac{1}{\pi}\int_{\mathcal{O}(|\varepsilon|^{\frac{1}{2}})}^{\infty}(t^2I_n + A^2)^{-1}\,\mathrm{d}t\\ \xrightarrow{\varepsilon\to 0}&\frac{1}{\pi}\int_0^{+\infty}(t^2I_n + A^2)^{-1}\,\mathrm{d}t,\end{aligned}$$

where the contour integral on $\widehat{\mathbb{C}}$ now converges to an improper real integral on $[0, \infty[$. We denoted complex infinity by ∞ and the real infinities with $+\infty$ and $-\infty$, repectively. Together, we find

$$(A^2)^{-\frac{1}{2}} = \frac{2}{\pi}\int_0^{\infty}(t^2I_n + A^2)^{-1}\,\mathrm{d}t.$$

Finally, using $\operatorname{sign}(A) = A(A^2)^{-\frac{1}{2}}$, we recieve the claimed formula. □

Theorem 3.26. *Let $A, B \in \mathbb{C}^{n\times n}$ with $\Lambda(AB) \subseteq \mathbb{C}\setminus\mathbb{R}_{\leq 0}$ and define $C \coloneqq A(BA)^{-\frac{1}{2}}$. Then*

$$\operatorname{sign}\left(\begin{bmatrix}0 & A\\ B & 0\end{bmatrix}\right) = \begin{bmatrix}0 & C\\ C^{-1} & 0\end{bmatrix}. \tag{3.12}$$

Proof. Define $P \coloneqq \begin{bmatrix}0 & A\\ B & 0\end{bmatrix}$. Then $P^2 = \begin{bmatrix}AB & 0\\ 0 & BA\end{bmatrix}$. Note that $\Lambda(AB) = \Lambda(BA)$ from which we receive $\Lambda(P^2) = \Lambda(AB) = \Lambda(BA) \subseteq \mathbb{C}\setminus\mathbb{R}_{\leq 0}$. This implies $\Lambda(P) \subseteq \mathbb{C}\setminus\mathbf{i}\mathbb{R}$, i.e., sign is defined on the spectrum of P. Moreover,

$$\begin{aligned}\operatorname{sign}(P) \overset{\text{Rem. 3.23 (2)}}{=} P(P^2)^{-\frac{1}{2}} &= \begin{bmatrix}0 & A\\ B & 0\end{bmatrix}\begin{bmatrix}AB & 0\\ 0 & BA\end{bmatrix}^{-\frac{1}{2}} = \begin{bmatrix}0 & A\\ B & 0\end{bmatrix}\begin{bmatrix}(AB)^{-\frac{1}{2}} & 0\\ 0 & (BA)^{-\frac{1}{2}}\end{bmatrix}\\ &= \begin{bmatrix}0 & A(BA)^{-\frac{1}{2}}\\ B(AB)^{-\frac{1}{2}} & 0\end{bmatrix} =: \begin{bmatrix}0 & C\\ D & 0\end{bmatrix}.\end{aligned}$$

Theorem 3.24 (1) yields

$$I_{2n} = \begin{bmatrix}I_n & 0\\ 0 & I_n\end{bmatrix} = \operatorname{sign}(P)^2 = \begin{bmatrix}0 & C\\ D & 0\end{bmatrix}^2 = \begin{bmatrix}CD & 0\\ 0 & DC\end{bmatrix},$$

i.e., $D = C^{-1}$. □

Remark 3.27. An important special case of Theorem 3.26 is that

$$\operatorname{sign}\left(\begin{bmatrix} 0 & A \\ I_n & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} & 0 \end{bmatrix} \tag{3.13}$$

holds for all $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Consequently, any (numerical) method for computing $\operatorname{sign}(A)$ yields a method for computing $A^{\frac{1}{2}}$.

Another application of the special case in Remark 3.27 leads to an integral formula for the matrix square root, that relies on Theorem 3.25.

Theorem 3.28. *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Then*

$$A^{\frac{1}{2}} = \frac{2}{\pi} A \int_0^\infty (t^2 I_n + A)^{-1} \,\mathrm{d}t. \tag{3.14}$$

Proof. We apply Theorem 3.25 to $\tilde{A} \coloneqq \begin{bmatrix} 0 & A \\ I_n & 0 \end{bmatrix} \in \mathbb{C}^{(2n)\times(2n)}$. Then $\tilde{A}^2 = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix}$ and we receive

$$\begin{aligned}
\begin{bmatrix} 0 & A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} & 0 \end{bmatrix} &\overset{\text{Rem. 3.27}}{=} \operatorname{sign}(\tilde{A}) \overset{\text{Thm. 3.25}}{=} \frac{2}{\pi} \tilde{A} \int_0^\infty (t^2 I_{2n} + \tilde{A}^2)^{-1} \,\mathrm{d}t \\
&= \frac{2}{\pi} \begin{bmatrix} 0 & A \\ I_n & 0 \end{bmatrix} \int_0^\infty \begin{bmatrix} t^2 I_n + A & 0 \\ 0 & t^2 I_n + A \end{bmatrix}^{-1} \mathrm{d}t \\
&= \frac{2}{\pi} \begin{bmatrix} 0 & A \\ I_n & 0 \end{bmatrix} \int_0^\infty \begin{bmatrix} (t^2 I_n + A)^{-1} & 0 \\ 0 & (t^2 I_n + A)^{-1} \end{bmatrix} \mathrm{d}t \\
&= \frac{2}{\pi} \begin{bmatrix} 0 & A \\ I_n & 0 \end{bmatrix} \begin{bmatrix} \int_0^\infty (t^2 I_n + A)^{-1} \,\mathrm{d}t & 0 \\ 0 & \int_0^\infty (t^2 I_n + A)^{-1} \,\mathrm{d}t \end{bmatrix} \\
&= \begin{bmatrix} 0 & \frac{2}{\pi} A \int_0^\infty (t^2 I_n + A)^{-1} \,\mathrm{d}t \\ \frac{2}{\pi} \int_0^\infty (t^2 I_n + A)^{-1} \,\mathrm{d}t & 0 \end{bmatrix}.
\end{aligned}$$

Comparing the top right block entries finishes the proof. □

Applications of the Matrix Sign Function for Solving Matrix Equations

Matrix equations are everywhere systems and control and systems in which time-dependent models used to predict, simulate and control real-world phenomena can often be represented as linear (or nonlinear) dynamical systems. We refer to the lecture "Control Theory" for further information.

(1) Sylvester equation. Given $A \in \mathbb{C}^{n\times n}$, $B \in \mathbb{C}^{m\times m}$ and $C \in \mathbb{C}^{n\times m}$, we search for a matrix $X \in \mathbb{C}^{n\times m}$ with

$$AX - XB = C. \tag{3.15}$$

Theorem 3.29. *Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. Then the following are equivalent.*

(1) The Sylvester equation $AX - XB = C$ is uniquely solvable for all $C \in \mathbb{C}^{n\times m}$.

(2) $\Lambda(A) \cap \Lambda(B) = \varnothing$

Proof. Note that (3.15) can be rewritten as a linear system with $n \cdot m$ equations and $n \cdot m$ unknowns. Thus, (3.15) is uniquely solvable for all $C \in \mathbb{C}^{n\times m}$ if and only if the homogeneous equation $AX - XB = 0$ has only the trivial solution $X = 0$.

"(1)⇒(2)": We prove this by contraposition. Let $\lambda \in \mathbb{C}$ be a common eigenvalue of A and of B. Moreover let $v \in \mathbb{C}^n \setminus \{0\}$ be a (right-)eigenvector of A corresponding to λ and $u \in \mathbb{C}^m \setminus \{0\}$ be a left-eigenvector of B corresponding to λ, i.e., $u^H B = \lambda u^H$. Define $X \coloneqq vu^H \in \mathbb{C}^{n\times m} \setminus \{0\}$. Then

$$AX - XB = \underbrace{Av}_{=\lambda v} u^H - v \underbrace{u^H B}_{=\lambda u^H} = \lambda v u^H - \lambda v u^H = 0,$$

i.e., $AX - XB = 0$ has a non-zero solution.

"(2)⇒(1)": Let $\Lambda(A) \cap \Lambda(B) = \varnothing$. Moreover, let $X \in \mathbb{C}^{n\times m}$ satisfy $AX - XB = 0$. We inductively receive $A^k X = XB^k$ for $k \in \mathbb{N}$. Thus, $p(A)X = Xp(B)$ for any polynomial p. Using the Cayley-Hamilton theorem, we receive

$$0 = P_A(A)X = XP_A(B),$$

where P_A denotes the characteristic polynomial of A. By $\Lambda(P_A(B)) = P_A(\Lambda(B))$ (Theorem 3.5 (4)) and since no eigenvalue of B is a zero of P_A, we know $0 \notin \Lambda(P_A(B))$, i.e., $P_A(B)$ is nonsingular. Thus, we conclude $X = 0$. □

Next, we discuss how the matrix sign function can be used to solve a Sylvester equation. Assume

$$\Lambda(A) \subseteq \{z \in \mathbb{C} \mid \operatorname{Re}(z) > 0\} \qquad \text{and} \qquad \Lambda(B) \subseteq \{z \in \mathbb{C} \mid \operatorname{Re}(z) < 0\}. \tag{3.16}$$

Then by the above theorem, there exists a unique solution of (3.15). Note, that $AX - XB = C$ is equivalent to

$$\begin{bmatrix} A & C \\ 0 & B \end{bmatrix} = \begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix}^{-1}, \tag{3.17}$$

which can easily be seen from

$$\begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix}^{-1} = \begin{bmatrix} A & -XB \\ 0 & B \end{bmatrix} \begin{bmatrix} I_n & X \\ 0 & I_m \end{bmatrix} = \begin{bmatrix} A & AX - XB \\ 0 & B \end{bmatrix}.$$

Now let $X \in \mathbb{C}^{n\times m}$ be a solution of (3.15). From (3.16), we receive $\operatorname{sign}(A) = I_n$ and $\operatorname{sign}(B) = -I_m$. Therefore,

$$\begin{aligned}\operatorname{sign}\left(\begin{bmatrix} A & C \\ 0 & B \end{bmatrix}\right) &= \operatorname{sign}\left(\begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix}\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}\begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix}^{-1}\right) \\ &= \begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix}\begin{bmatrix} \operatorname{sign}(A) & 0 \\ 0 & \operatorname{sign}(B) \end{bmatrix}\begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix}^{-1} \\ &= \begin{bmatrix} I_n & -X \\ 0 & I_m \end{bmatrix}\begin{bmatrix} I_n & 0 \\ 0 & -I_m \end{bmatrix}\begin{bmatrix} I_n & X \\ 0 & I_m \end{bmatrix} = \begin{bmatrix} I_n & 2X \\ 0 & -I_m \end{bmatrix},\end{aligned}$$

i.e., the solution X can be read off from the top right block.

Note that there exist specialized and more sophisticated methods to solve (3.15), especially the Bartels-Stewart algorithm. We refer to the survey [24, Sect. 4] for further informations and references.

(2) Lyapunov equation. We consider the special case of the Sylvester equation where $B = -A^H$, i.e., given $A, C \in \mathbb{C}^{n\times n}$, we search for a matrix $X \in \mathbb{C}^{n\times n}$ with

$$AX + XA^H = C. \tag{3.18}$$

Since this is a special case of the Sylvester equation, we can apply the above strategy of using the matrix sign function.

(3) Algebraic Riccati equation. Given $A \in \mathbb{C}^{n\times n}$ and Hermitian matrices $F, G \in \mathbb{C}^{n\times n}$, we search for $X \in \mathbb{C}^{n\times n}$ with

$$XFX - A^H X - XA = G. \tag{3.19}$$

Under somewhat involved conditions, see e.g. [4, Sect. 2.1], we find that this equation has a unique Hermitian solution $X \in \mathbb{C}^{n\times n}$ with

$$\Lambda(A - FX) \subseteq \{z \in \mathbb{C} \mid \operatorname{Re}(z) < 0\}. \tag{3.20}$$

Equation (3.19) is equivalent to

$$\begin{bmatrix} A^H & G \\ F & -A \end{bmatrix} = \begin{bmatrix} X & -I_n \\ I_n & 0 \end{bmatrix}\begin{bmatrix} -(A-FX) & -F \\ 0 & (A-FX)^H \end{bmatrix}\begin{bmatrix} X & -I_n \\ I_n & 0 \end{bmatrix}^{-1},$$

which can be seen from

$$\begin{aligned}&\begin{bmatrix} X & -I_n \\ I_n & 0 \end{bmatrix}\begin{bmatrix} -(A-FX) & -F \\ 0 & (A-FX)^H \end{bmatrix}\begin{bmatrix} X & -I_n \\ I_n & 0 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} -XA + XFX & -XF - A^H + XF \\ -A + FX & -F \end{bmatrix}\begin{bmatrix} 0 & I_n \\ -I_n & X \end{bmatrix} \\ &= \begin{bmatrix} A^H & XFX - A^H X - XA \\ F & -A \end{bmatrix}.\end{aligned}$$

We receive

$$
\begin{aligned}
S &\coloneqq \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \coloneqq \operatorname{sign}\left(\begin{bmatrix} A^H & G \\ F & -A \end{bmatrix} \right) \\
&\overset{\substack{\text{Thm.} \\ 3.5\,(3)}}{=} \begin{bmatrix} X & -I_n \\ I_n & 0 \end{bmatrix} \operatorname{sign}\left(\begin{bmatrix} -(A - FX) & -F \\ 0 & (A - FX)^H \end{bmatrix} \right) \begin{bmatrix} X & -I_n \\ I_n & 0 \end{bmatrix}^{-1} \\
&\overset{(3.20)}{=} \begin{bmatrix} X & -I_n \\ I_n & 0 \end{bmatrix} \begin{bmatrix} I_n & Z \\ 0 & -I_n \end{bmatrix} \begin{bmatrix} X & -I_n \\ I_n & 0 \end{bmatrix}^{-1} \\
&= \begin{bmatrix} X & XZ + I_n \\ I_n & Z \end{bmatrix} \begin{bmatrix} 0 & I_n \\ -I_n & IX \end{bmatrix} \\
&= \begin{bmatrix} -XZ - I_n & XZX + 2X \\ -Z & ZX + I_n \end{bmatrix}
\end{aligned}
$$

for some matrix $Z \in \mathbb{C}^{n \times n}$. We obtain

$$
\begin{aligned}
S_{11} = -XZ - I_n \quad &\implies (S_{11} - I_n)X = -XZX - 2X = -S_{12}, \\
S_{22} = ZX - I_n \quad &\implies -S_{22} + I_n = -ZX = S_{21}X.
\end{aligned}
$$

Which yields the overdetermined linear system

$$
\begin{cases} (S_{11} - I_n)X = -S_{12}, \\ S_{21}X = -S_{22} + I_n. \end{cases} \tag{3.21}
$$

The system matrix $\begin{bmatrix} S_{11} - I_n \\ S_{21} \end{bmatrix}$ of (3.21) has full rank since

$$
\begin{bmatrix} I_n & -X \end{bmatrix} \begin{bmatrix} S_{11} - I_n \\ S_{21} \end{bmatrix} = S_{11} - I_n - XS_{21} = -XZ - I_n - I_n + XZ = \underbrace{-2I_n}_{\text{rank } n}.
$$

Overall, we have to compute the sign function of a $2n \times 2n$ matrix and we have solve a linear system with $2n^2$ equations and n^2 unknowns, e.g., with a QR decomposition to get the (unique) solution of (3.19).

4 Conditioning of Matrix Functions

Let $f : \mathcal{X} \to \mathcal{Y}$ be a map between two (finite-dimensional) normed spaces $(\mathcal{X}, \|\cdot\|_{\mathcal{X}})$ and $(\mathcal{Y}, \|\cdot\|_{\mathcal{Y}})$ with $f(x) = y$ and $f(x + \Delta x)\widehat{y}$. We define the *(relative) forward error* and the *(relative) backward error* of f at x by

$$\|\Delta x\|_{\mathcal{X}}, \quad \frac{\|\Delta x\|_{\mathcal{X}}}{\|x\|_{\mathcal{X}}}, \quad \|\widehat{y} - y\|_{\mathcal{Y}}, \quad \frac{\|\widehat{y} - y\|_{\mathcal{Y}}}{\|y\|_{\mathcal{Y}}},$$

respectively; see Figure 8. Although y is the exact output for the input x, due to perturbations, measurements, or inexact computations, we may get only the inexact output $\widehat{y}$, which itself is the exact output for a perturbed input $x + \Delta x$. The size of this perturbation in x is referred as the backward error. In general, we are interested in the forward error, but it is usually not accessible since we do not know the exact solution. Therefore, we try to estimate or bound via forward error by the backward error.

In the framework of matrix functions, we have $\mathcal{X} = \mathcal{Y} = \mathbb{C}^{n\times n}$ usually equipped with the same norm.

4.1 Condition Number and Fréchet Derivative

Forward and backward error are related via the condition number.

Definition 4.1. Let $U \subseteq \mathbb{C}^{n\times n}$, $\|\cdot\|$ a norm on $\mathbb{C}^{n\times n}$ and $f : U \to \mathbb{C}^{n\times n}$. We define

(1) the *absolute condition number* of f at A as

$$\kappa_f^{\text{abs}}(A) \coloneqq \lim_{\varepsilon\to 0} \sup_{\|\Delta A\|\le\varepsilon} \frac{\|f(A + \Delta A) - f(A)\|}{\|\Delta A\|}, \tag{4.1}$$

(2) the *relative condition number* of f at A as

$$\kappa_f^{\text{rel}}(A) \coloneqq \lim_{\varepsilon\to 0} \sup_{\|\Delta A\|\le\varepsilon} \frac{\|f(A + \Delta A) - f(A)\|}{\|f(A)\|} \cdot \frac{\|A\|}{\|\Delta A\|}. \tag{4.2}$$

If the condition number is small, then small changes in the input data only lead to small changes in the output data. In this case, we call f *well-conditioned* and *ill-conditioned* otherwise.

Remark 4.2. Definition 4.1 gives rise to the *rule of thumb*.

$$\text{(relative) forward error} \lesssim \text{(relative) condition number} \cdot \text{(relative) backward error}$$

Consequently, if we know the condition number and can approximate or bound the backward error, we receive a (approximate) bound for the forward error.

The condition number of f at A can be expressed using the Fréchet derivative of f at A.

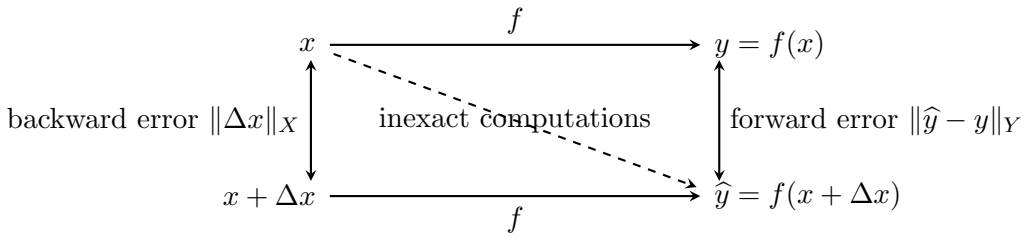


Figure 8: Forward and backward errors.

Definition 4.3. Let $U \subseteq \mathbb{C}^{n\times n}$ be open. A map $f : U \to \mathbb{C}^{n\times n}$ is *Fréchet differentiable* at $A \in U$, if there exists a linear map

$$L_f(A) : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}, \quad X \mapsto L_f(A, X) \tag{4.3}$$

with

$$f(A + X) - f(A) - L_f(A, X) \in o(\|X\|), \tag{4.4}$$

i.e.,

$$\lim_{\varepsilon\to 0} \sup_{\|X\|\le\varepsilon} \frac{\|f(A + X) - f(A) - L_f(A, X)\|}{\|X\|} = 0, \tag{4.5}$$

where $\|\cdot\|$ is an arbitrary norm on $\mathbb{C}^{n\times n}$. We call $L_f(A)$ the *Fréchet derivative* of f at A.

The Fréchet derivative is unique if it exists.

Theorem 4.4. *Let $f : U \to \mathbb{C}^{n\times n}$ be Fréchet differentiable at $A \in U$, and let $L_f(A)$ and $\tilde{L}_f(A)$ be two linear maps fulfilling* (4.4)*. Then $L_f(A) = \tilde{L}_f(A)$.*

Proof. We define the remainders

$$R(X) \coloneqq f(A + X) - f(A) - L_f(A, X),$$
$$\tilde{R}(X) \coloneqq f(A + X) - f(A) - \tilde{L}_f(A, X).$$

Then $R, \tilde{R} \in o(\|X\|)$ by assumption, and

$$L_f(A, X) - \tilde{L}_f(A, X) = \tilde{R}(X) - R(X) \qquad \text{for all } X \in \mathbb{C}^{n\times n}.$$

Now fix $X \in \mathbb{C}^{n\times n}$ and define $X_t \coloneqq tX$ for $t > 0$. Using the linearity of $L_f(A)$ and $\tilde{L}_f(A)$, we receive

$$\begin{aligned} L_f(A, X) - \tilde{L}_f(A, X) =& \frac{1}{t}\left(L_f(A, X_t) - \tilde{L}_f(A, X_t)\right) = \|X\| \cdot \frac{L_f(A, X_t) - \tilde{L}_f(A, X_t)}{\|X_t\|} \\ =& \|X\|\left(\frac{\tilde{R}(X_t)}{\|X_t\|} - \frac{R(X_t)}{\|X_t\|}\right) \xrightarrow{t\to 0} 0. \end{aligned}$$

Since the left-hand side is independent of t, we conclude $L_f(A, X) = \tilde{L}_f(A, X)$. □

Theorem 4.5. *(1) Let $f, g : U \to \mathbb{C}^{n\times n}$ be Fréchet differentiable at $A \in U$. Then $f + g$ and $f \cdot g$ are also Fréchet differentiable at A with*

$$L_{f+g}(A, X) = L_f(A, X) + L_g(A, X), \qquad \text{"sum rule"}$$

and

$$L_{f\cdot g}(A, X) = L_f(A, X)g(A) + f(A)L_g(A, X). \qquad \text{"product rule"}$$

(2) Let g and f be Fréchet differentiable at $A \in \mathbb{C}^{n\times n}$ and $g(A) \in \mathbb{C}^{n\times n}$, respectively. Then $f \circ g$ is Fréchet differentiable at A with

$$L_{f\circ g}(A, X) = L_f(g(A), L_g(A, X)). \qquad \text{"chain rule"}$$

(3) Let $f : U \to f(U)$ be bijective and Fréchet differentiable, and let $A \in U$. If $L_f(A)$ is nonsingular, then $L_{f^{-1}}(f(A))$ exists with

$$L_{f^{-1}}(f(A), X) = L_f^{-1}(A, X). \qquad \text{"derivative of the inverse"}$$

In particular, we have for all $X \in \mathbb{C}^{n\times n}$

$$L_f(A, L_{f^{-1}}(f(A), X)) = X.$$

Proof. Exercise, or see [12, Thms. 3.2–3.5]. □

We continue with some examples that will be of particular interest in what follows.

Example 4.6. (1) Let $f(A) = A$, i.e., $f = \mathrm{id}$. Then

$$f(A + X) - f(A) = A + X - A = X,$$

and therefore, $L_f(A, X) = X$.

(2) Let $f(A) = A^2$. Then

$$f(A + X) - f(A) = (A + X)^2 - A^2 = AX + XA + \underbrace{X^2}_{\in o(\|X\|)},$$

i.e., $L_f(A, X) = AX + XA$.

(3) Let $A \in \mathbb{C}^{n\times n}$ be nonsingular and let $f(A) = A^{-1}$. Then $(\mathrm{id} \cdot f)(A) = A \cdot A^{-1} = I_n$. Hence, by the product rule for the Fréchet derivative (Theorem 4.5 (1)), we find

$$0 = L_{\mathrm{id}\cdot f}(A, X) = L_{\mathrm{id}}(A, X)f(A) + \mathrm{id}(A)L_f(A, X) \overset{(1)}{=} XA^{-1} + AL_f(A, X).$$

Reordering yields $L_f(A, X) = -A^{-1}XA^{-1}$.

(4) Let $f(A) = \sum_{k=0}^{m} \alpha_k A^k$. Then

$$\begin{aligned} f(A+X) - f(A) &= \sum_{k=0}^{m} \alpha_k (A+X)^k - \sum_{k=0}^{m} \alpha_k A^k \\ &= \sum_{k=1}^{m} \alpha_k \sum_{j=1}^{k} A^{j-1} X A^{k-j} + o(\|X\|), \end{aligned}$$

where all terms with more than one X in the second sum is incorporated in $o(\|X\|)$. Hence, we find

$$L_f(A,X) = \sum_{k=1}^{m} \alpha_k \sum_{j=1}^{k} A^{j-1} X A^{k-j}.$$

Next, we prove that the condition number is just the operator norm of the Fréchet derivative, which is given by

$$\|L_f(A)\| = \sup_{X \neq 0} \frac{\|L_f(A,X)\|}{\|X\|} = \sup_{\|X\|=1} \|L_f(A,X)\| = \sup_{\|X\| \leq 1} \|L_f(A,X)\|.$$

Theorem 4.7. *Let $U \subseteq \mathbb{C}^{n\times n}$, $\|\cdot\|$ a norm on $\mathbb{C}^{n\times n}$ and $f : U \to \mathbb{C}^{n\times n}$ Fréchet differentiable at $A \in U$. Then*

$$\kappa_f^{\text{abs}}(A) = \|L_f(A)\| \qquad \text{and} \qquad \kappa_f^{\text{rel}}(A) = \|L_f(A)\| \frac{\|A\|}{\|f(A)\|}$$

where the norm in $\|L_f(A)\|$ is the induced norm of the norm $\|\cdot\|$.

Proof. Using the linearity of the Fréchet derivative, we recieve

$$\begin{aligned} \kappa_f^{\text{abs}}(A) &= \lim_{\varepsilon \to 0} \sup_{\|\Delta A\| \leq \varepsilon} \frac{\|f(A+\Delta A) - f(A)\|}{\|\Delta A\|} = \lim_{\varepsilon \to 0} \sup_{\|\Delta A\| \leq \varepsilon} \frac{\|L_f(A,\Delta A) + o(\|\Delta A\|)\|}{\|\Delta A\|} \\ &= \lim_{\varepsilon \to 0} \sup_{\|\Delta A\| \leq \varepsilon} \left\| L_f\big(A, \tfrac{\Delta A}{\|\Delta A\|}\big) + \underbrace{\frac{o(\|\Delta A\|)}{\|\Delta A\|}}_{\xrightarrow{\varepsilon \to 0} 0} \right\| = \sup_{\|\widetilde{A}\|=1} \|L_f(A,\widetilde{A})\| = \|L_f(A)\|. \end{aligned}$$

The other assertion follows from

$$\kappa_f^{\text{rel}}(A) = \kappa_f^{\text{abs}}(A) \frac{\|A\|}{\|f(A)\|}.$$ □

In Theorem 3.3 (2), we saw that any sufficiently smooth f induces a continuous matrix function. The next result is the analog for a Fréchet differentiable matrix function.

Theorem 4.8. *Let $\Omega \subseteq \mathbb{C}$ and $f : \Omega \to \mathbb{C}$ be $2n-1$ times (real) continuously differentiable. Then the corresponding primary matrix function $f : \mathbb{C}^{n\times n}(\Omega) \to \mathbb{C}^{n\times n}$ is Fréchet differentiable at every $A \in \mathbb{C}^{n\times n}(\Omega)$. Moreover, the Fréchet derivative $L_f(A,X)$ is continuous both in $A \in \mathbb{C}^{n\times n}(\Omega)$ and in $X \in \mathbb{C}^{n\times n}$.*

Proof. The proof relies on Gateaux derivatives (pendent of directional derivatives in general spaces). We omit the details here and refer to [12, Thm. 3.8]. □

4.1.1 Connection of Matrix Functions and Fréchet Derivatives

The Fréchet derivative can be computed using matrix functions.

Theorem 4.9. *Let $\Omega \subseteq \mathbb{C}$, $f : \Omega \to \mathbb{C}$ be $2n-1$ times (real) continuously differentiable, $A \in \mathbb{C}^{n\times n}(\Omega)$, and $X \in \mathbb{C}^{n\times n}$. Then*

$$f\left(\begin{bmatrix} A & X \\ 0 & A \end{bmatrix}\right) = \begin{bmatrix} f(A) & L_f(A,X) \\ 0 & f(A) \end{bmatrix}. \tag{4.6}$$

Proof. See [12, Thm. 3.8]. □

Remark 4.10. (1) By Theorem 4.9 we can compute $L_f(A, X)$ by computing $f(\tilde{A})$ for

$$\tilde{A} = \begin{bmatrix} A & X \\ 0 & A \end{bmatrix}.$$

Note that one should exploit the special structure of $\tilde{A}$ when computing $f(\tilde{A})$.

(2) For $n = 1$, equation (4.6) coincides with the expression for (2×2) upper triangular matrices in (3.1).

Using the *Hadamard product*, which is the entry-wise matrix product, i.e., the map

$$\odot : \mathbb{C}^{n\times m} \times \mathbb{C}^{n\times m} \to \mathbb{C}^{n\times m}, \qquad [a_{ij}] \odot [b_{ij}] \coloneqq [a_{ij} \cdot b_{ij}], \tag{4.7}$$

we give an explicit expression for the Fréchet derivative of f at A, i.e., $L_f(A, X)$.

Theorem 4.11. *Let $\Omega \subseteq \mathbb{C}$, $f : \Omega \to \mathbb{C}$ be $2n-1$ times (real) continuously differentiable, and let $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \in \mathbb{C}^{n\times n}(\Omega)$. Then*

$$L_f(D, X) = \big[f[\lambda_i, \lambda_j] \cdot x_{ij}\big]_{ij} = \big[f[\lambda_i, \lambda_j]\big]_{ij} \odot X$$

holds for all $X = [x_{ij}] \in \mathbb{C}^{n\times n}$.

Sketch of the proof. First, let D have distinct eigenvalues. By Theorem 4.9,

$$f\left(\begin{bmatrix} D & X \\ 0 & D \end{bmatrix}\right) = \begin{bmatrix} f(D) & L_f(D,X) \\ 0 & f(D) \end{bmatrix}.$$

Using Theorem 3.5 (1), i.e., $Af(A) = f(A)A$, yields

$$\begin{bmatrix} D & X \\ 0 & D \end{bmatrix} \begin{bmatrix} f(D) & L_f(D,X) \\ 0 & f(D) \end{bmatrix} = \begin{bmatrix} f(D) & L_f(D,X) \\ 0 & f(D) \end{bmatrix} \begin{bmatrix} D & X \\ 0 & D \end{bmatrix}.$$

The top right block entry of this equation is equivalent to

$$Xf(D) - f(D)X = L_f(D,X)D - D\,L_f(D,X).$$

Since D is diagonal, the entry (i,j), $i \neq j$, reads

$$\begin{aligned} & f(\lambda_j)x_{ij} - f(\lambda_i)x_{ij} = (\lambda_j - \lambda_i)\big(L_f(D,X)\big)_{ij} \\ \overset{\lambda_i \neq \lambda_j}{\Longrightarrow} \quad & \big(L_f(D,X)\big)_{ij} = \frac{f(\lambda_j) - f(\lambda_i)}{\lambda_j - \lambda_i} x_{ij} = f[\lambda_i, \lambda_j] \cdot x_{ij}. \end{aligned}$$

Since matrices with distinct eigenvalues are dense in $\mathbb{C}^{n\times n}(\Omega)$ and $L_f(D,X)$ is continuous, we obtain the result for general D. □

Note that the case $i = j$ is not covered in above sketch of the proof.

4.1.2 Kronecker Form of the Fréchet Derivative

Using the *vectorization* operator

$$\operatorname{vec} : \mathbb{C}^{n\times m} \to \mathbb{C}^{nm}, \quad \operatorname{vec}([x_1, \ldots, x_m]) = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}, \tag{4.8}$$

we represent the linear map $L_f(A)$ by a matrix, which will be very convenient in what follows. The unique matrix $\mathcal{K}_f(A) \in \mathbb{C}^{n^2\times n^2}$, that fulfills

$$\operatorname{vec}(L_f(A,X)) = \mathcal{K}_f(A)\operatorname{vec}(X). \tag{4.9}$$

is called the *Kronecker form* of $L_f(A)$. In particular, X is an eigenvector of $L_f(A)$ corresponding to λ if and only if $\operatorname{vec}(X)$ is an eigenvector of $\mathcal{K}_f(A)$ corresponding to λ.

Next, we prove that the operator Frobenius norm of the Fréchet derivative and the spectral norm of its Kronecker form coincide.

Theorem 4.12. *Let f be Fréchet differentiable on $U \subseteq \mathbb{C}^{n\times n}$ and $A \in U$. Then*

$$\|L_f(A)\|_F = \|\mathcal{K}_f(A)\|_2. \tag{4.10}$$

Proof. Using the obvious identity $\|X\|_F = \|\operatorname{vec}(X)\|_2$ twice, we find

$$\begin{aligned} \|L_f(A)\|_F &= \sup_{\|X\|_F=1} \|L_f(A,X)\|_F = \sup_{\|\operatorname{vec}(X)\|_2=1} \|\operatorname{vec}\big(L_f(A,X)\big)\|_2 \\ &= \sup_{\|\operatorname{vec}(X)\|_2=1} \|\mathcal{K}_f(A)\operatorname{vec}(X)\|_2 = \sup_{y\in\mathbb{C}^{n^2},\|y\|_2=1} \|\mathcal{K}_f(A)y\|_2 = \|\mathcal{K}_f(A)\|_2. \end{aligned}$$ □

4.2 Bounding the Condition Number

The main goal of this subsection is to derive bounds for $\kappa_f^{\text{abs}}(A)$ and $\kappa_f^{\text{rel}}(A)$ by bounding the *spectral radius* of $\mathcal{K}_f(A)$, i.e.,

$$\rho(\mathcal{K}_f(A)) = \max\{\lambda \mid \lambda \in \Lambda(\mathcal{K}_f(A))\}. \tag{4.11}$$

We start with basic identities involving the Kronecker product and the Theorem of Stephanos, which can be interpreted as a spectral mapping theorem for bivariate polynomials.

Lemma 4.13. *(1) Let $A, C \in \mathbb{C}^{n\times n}$ and $B, D \in \mathbb{C}^{m\times m}$. Then*

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD). \tag{4.12}$$

(2) Let $A \in \mathbb{C}^{n\times n}, B \in \mathbb{C}^{m\times m}$ and $C \in \mathbb{C}^{n\times m}$. Then

$$\mathrm{vec}(ACB) = (B^T \otimes A)\mathrm{vec}(C). \tag{4.13}$$

Proof. (1) First, let $F = [F_{ij}] := (A \times B)(C \otimes D)$. Then the block entry F_{ij} of F is

$$F_{ij} = \sum_{k=1}^{n} (a_{ik}B)(c_{kj}D) = \left(\sum_{k=1}^{n} a_{ik}c_{kj}\right) BD.$$

Moreover, let $H := AC$. The block entry (i, j) of $G = [G_{ij}] = (AC) \otimes (BD)$ is

$$G_{ij} = h_{ij}BD = \left(\sum_{k=1}^{n} a_{ik}c_{kj}\right) BD.$$

(2) Let $j \in \{1, \ldots, m\}$. The jth column of ACB is

$$\begin{aligned}(ACB)e_j = (AC)(Be_j) = (AC)b_j = \sum_{k=1}^{n}(AC)e_k b_{kj} = \sum_{k=1}^{n} b_{kj}ACe_k \\ = \begin{bmatrix} b_{1j}A & b_{2j}A & \ldots & b_{mj}A \end{bmatrix} \mathrm{vec}(C).\end{aligned}$$

Thus, $\mathrm{vec}(ACB) = (B^T \otimes A)\mathrm{vec}(C)$. □

A *bivariate polynomial* is a polynomial in commuting variables t_1 and t_2, i.e., it is of the form

$$p(t_1, t_2) = \sum_{i,j=0}^{k} \alpha_{ij} t_1^i t_2^j \in \mathbb{C}[t_1, t_2]. \tag{4.14}$$

If such a polynomial $p \in \mathbb{C}[t_1, t_2]$ is given, we define for $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$

$$p(A, B) = \sum_{i,j=0}^{k} \alpha_{ij} A^i \otimes B^j. \tag{4.15}$$

Note that we have $A^i \otimes B^j \neq B^j \otimes A^i$ in general.

Theorem 4.14 (Theorem of Stephanos)**.** *Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. Moreover, let $p \in \mathbb{C}[t_1, t_2]$ be a bivariate polynomial. Then*

$$\Lambda(p(A, B)) = \{p(\lambda, \mu) \mid \lambda \in \Lambda(A), \mu \in \Lambda(B)\}. \tag{4.16}$$

Proof. Let

$$A = XJ_AX^{-1}, \qquad B = YJ_BY^{-1}$$

be Jordan decompositions of A and B. Then by Lemma 4.13 (1), we have

$$\begin{aligned}
p(A,B) &= \sum_{i,j=0}^{k} \alpha_{ij} A^i \otimes B^j = \sum_{i,j=0}^{k} \alpha_{ij} (XJ_AX^{-1})^i \otimes (YJ_BY^{-1})^j \\
&= \sum_{i,j=0}^{k} \alpha_{ij} (XJ_A^iX^{-1}) \otimes (YJ_B^jY^{-1}) \\
&= \sum_{i,j=0}^{k} \alpha_{ij} ((XJ_A^i) \otimes (YJ_B^j)(X^{-1} \otimes Y^{-1}) \\
&= \sum_{i,j=0}^{k} \alpha_{ij} (X \otimes Y)(J_A^i \otimes J_B^j)(X^{-1} \otimes Y^{-1}) \\
&= (X \otimes Y) \left(\sum_{i,j=0}^{k} \alpha_{ij} J_A^i \otimes J_B^j \right) (X^{-1} \otimes Y^{-1}) \\
&= (X \otimes Y) p(J_A, J_B)(X^{-1} \otimes Y^{-1}) = (X \otimes Y) p(J_A, J_B)(X \otimes Y)^{-1}.
\end{aligned}$$

In particular, we used $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$, which is an immediate consequence of Lemma 4.13 (1).

Since J_A^i, J_B^j and $J_A^i \otimes J_B^j$ are upper triangular matrices, the eigenvalues of $p(J_A, J_B)$ are on the diagonal, where we find $p(\lambda, \mu)$ for $\lambda \in \Lambda(A)$ and $\mu \in \Lambda(B)$. □

The next result characterizes the spectrum of $L_f(A)$ and therefore also for $\mathcal{K}_f(A)$. Recall, that for any eigenvalue λ of $\mathcal{K}_f(A)$ and any consistent norm $\|\cdot\|$ on $\mathbb{C}^{n\times n}$, we have

$$|\lambda| \leq \rho(\mathcal{K}_f(A)) \leq \|\mathcal{K}_f(A)\|, \tag{4.17}$$

where $\rho(\mathcal{K}_f(A))$ denotes the spectral radius of $\mathcal{K}_f(A)$ (see Equation (1.6)).

Theorem 4.15. *Let $\Omega \subseteq \mathbb{C}$, $f : \Omega \to \mathbb{C}$ be $2n-1$ times (real) continuously differentiable, and let $A \in \mathbb{C}^{n\times n}(\Omega)$. Then*

$$\Lambda(L_f(A)) = \Lambda(\mathcal{K}_f(A)) = \{f[\lambda, \mu] \mid \lambda, \mu \in \Lambda(A)\}. \tag{4.18}$$

Sketch of the proof. We only prove the result for f being a polynomial. Let

$$f(z) = \sum_{k=0}^{m} \alpha_k z^k.$$

Then by Example 4.6 (4), we get

$$L_f(A, X) = \sum_{k=1}^{m} a_k \sum_{j=1}^{k} A^{j-1} X A^{k-j}.$$

Using the identity $\operatorname{vec}(ABC) = (C^T \otimes A)\operatorname{vec}(B)$ (Lemma 4.13 (2)), we find that the Kronecker form of $L_f(A)$ is given by

$$\mathcal{K}_f(A) = \sum_{k=1}^{m} a_k \sum_{j=1}^{k} (A^{k-j})^T \otimes A^{j-1}.$$

By the Theorem of Stephanos (Theorem 4.14), the eigenvalues of $\mathcal{K}_f(A)$ are

$$\sum_{k=1}^{m} a_k \sum_{j=1}^{k} \lambda^{k-j}\mu^{j-1}, \qquad \lambda, \mu \in \Lambda(A).$$

For $\lambda = \mu$, we find

$$\sum_{k=1}^{m} a_k \sum_{j=1}^{k} \lambda^{k-j}\lambda^{j-1} = \sum_{k=1}^{m} a_k k \lambda^{k-1} = f'(\lambda) = f[\lambda, \mu].$$

For $\lambda \neq \mu$, we have for all $k \in \mathbb{N}$

$$\begin{aligned}\left(\sum_{j=1}^{k} \lambda^{k-j}\mu^{j-1}\right)(\lambda - \mu) &= \sum_{j=1}^{k} \lambda^{k-j+1}\mu^{j-1} - \sum_{j=1}^{k} \lambda^{k-j}\mu^{j} \\ &= \sum_{j=0}^{k-1} \lambda^{k-j}\mu^{j} - \sum_{j=1}^{k} \lambda^{k-j}\mu^{j} = \lambda^k - \mu^k,\end{aligned}$$

and therefore

$$\sum_{j=1}^{k} \lambda^{k-j}\mu^{j-1} = \frac{\lambda^k - \mu^k}{\lambda - \mu}.$$

This yields

$$\begin{aligned}\sum_{k=1}^{m} a_k \sum_{j=1}^{k} \lambda^{k-j}\mu^{j-1} &= \sum_{k=1}^{m} a_k \frac{\lambda^k - \mu^k}{\lambda - \mu} = \frac{1}{\lambda - \mu}\left(\sum_{k=1}^{m} a_k \lambda^k - \sum_{k=1}^{m} a_k \mu^k\right) \\ &= \frac{f(\lambda) - f(\mu)}{\lambda - \mu} = f[\lambda, \mu].\end{aligned}$$

□

For a diagonal matrix D, the Frobenius norm of $L_f(D)$ coincides with the spectral radius of $L_f(D)$.

Theorem 4.16. *Let $\Omega \subseteq \mathbb{C}$, $f : \Omega \to \mathbb{C}$ be $2n-1$ times (real) continuously differentiable, and let $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \in \mathbb{C}^{n\times n}(\Omega)$. Then*

$$\|L_f(D)\|_F = \max_{\sigma \in \Lambda(L_f(D))} |\sigma| = \max_{1 \le i,j \le n} |f[\lambda_i, \lambda_j]|. \tag{4.19}$$

Proof. We write

$$\operatorname{diag}_{ij}(f[\lambda_i,\lambda_j]) = \operatorname{diag}\left(\begin{bmatrix} f[\lambda_1,\lambda_1] & & \\ & \ddots & \\ & & f[\lambda_n,\lambda_1] \end{bmatrix}, \dots, \begin{bmatrix} f[\lambda_1,\lambda_n] & & \\ & \ddots & \\ & & f[\lambda_n,\lambda_n] \end{bmatrix}\right).$$

Then

$$\begin{aligned}\|L_f(D)\|_F &= \sup_{X\neq 0} \frac{\|L_f(D,X)\|_F}{\|X\|_F} \overset{\text{Thm. 4.11}}{=} \sup_{X\neq 0} \frac{\|\left[f[\lambda_i,\lambda_j]\right]_{ij} \odot X\|_F}{\|X\|_F} \\ &= \sup_{X\neq 0} \frac{\|\operatorname{vec}\left(\left[f[\lambda_i,\lambda_j]\right]_{ij} \odot X\right)\|_2}{\|\operatorname{vec}(X)\|_2} = \sup_{\operatorname{vec}(X)\neq 0} \frac{\|\operatorname{diag}_{ij}(f[\lambda_i,\lambda_j])\operatorname{vec}(X)\|_2}{\|\operatorname{vec}(X)\|_2} \\ &= \sup_{y\neq 0} \frac{\|\operatorname{diag}_{ij}(f[\lambda_i,\lambda_j])y\|_2}{\|y\|_2} = \|\operatorname{diag}_{ij}(f[\lambda_i,\lambda_j])\|_2 = \max_{1\leq i,j\leq n} |f[\lambda_i,\lambda_j]|. \square\end{aligned}$$

The last theorem in this chapter requires two additional lemmas.

Lemma 4.17. *Let $A \in \mathbb{C}^{n\times m}$ and $B \in \mathbb{C}^{m\times k}$. Then*

(1) $\|AB\|_F \leq \|A\|_2\|B\|_F$,

(2) $\|AB\|_F \leq \|A\|_F\|B\|_2$.

Proof. (1) We have

$$\begin{aligned}\|AB\|_F^2 =& \|A[b_1,\dots,b_k]\|_F^2 = \|[Ab_1,\dots,Ab_k]\|_F^2 = \sum_{j=1}^k \|Ab_j\|_2^2 \\ \leq& \sum_{j=1}^k \|A\|_2^2\|b_j\|_2^2 = \|A\|_2^2 \sum_{j=1}^k \|b_j\|_2^2 = \|A\|_2^2\|B\|_F^2.\end{aligned}$$

(2) The statement follows from (1) by applying the transpose:

$$\|AB\|_F = \|(AB)^T\|_F = \|B^TA^T\|_F \leq \|B^T\|_2\|A^T\|_F = \|A\|_F\|B\|_2. \qquad \square$$

Lemma 4.18. *Let $\Omega \subseteq \mathbb{C}$, $f : \Omega \to \mathbb{C}$ be $2n-1$ times (real) continuously differentiable. Moreover, let $A \in \mathbb{C}^{n\times n}(\Omega)$ be diagonalizable, i.e., $A = ZDZ^{-1}$ for some diagonal matrix $D \in \mathbb{C}^{n\times n}$, and let $X \in \mathbb{C}^{n\times n}$. Define $\widetilde{X} \coloneqq Z^{-1}XZ$. Then*

$$L_f(A,X) = ZL_f(D,\widetilde{X})Z^{-1}. \tag{4.20}$$

Proof. We have

$$\begin{aligned}ZL_f(D,\widetilde{X})Z^{-1} =& Z\big(f(D+\widetilde{X}) - f(D) - o(\|\widetilde{X}\|)\big)Z^{-1} \\ =& Zf(D+\tilde{X})Z^{-1} - Zf(D)Z^{-1} - Z\,o(\|\tilde{X}\|)\,Z^{-1} \\ \overset{\text{Thm. 3.5}}{=}& f\left(ZDZ^{-1} + ZZ^{-1}XZZ^{-1}\right) - f(ZDZ^{-1}) - Z\,o(\|\widetilde{X}\|)\,Z^{-1} \\ =& f(A+X) - f(A) - o(\|X\|_\sim)\end{aligned}$$

with $\|Y\|_\sim := \|Z^{-1}YZ\|$. Since $\|\cdot\|$ and $\|\cdot\|_\sim$ are equivalent, we have $o(\|X\|_\sim) = o(\|X\|)$. It follows that

$$ZL_f(D, \widetilde{X})Z^{-1} = f(A+X) - f(A) - o(\|X\|) = L_f(A, X). \qquad \square$$

Finally, we combine the previous results in this chapter to obtain bounds for the absolute condition number $\kappa_f^{\text{abs}}(A)$, wich of course, also yield similar bounds on the relative condition number $\kappa_f^{\text{rel}}(A)$.

Theorem 4.19. *Let $\Omega \subseteq \mathbb{C}$, $f : \Omega \to \mathbb{C}$ be $2n-1$ times (real) continuously differentiable, and let $A \in \mathbb{C}^{n\times n}(\Omega)$.*

(1) Then we have the lower bound

$$\max_{\lambda,\mu\in\Lambda(A)} \big|f[\lambda,\mu]\big| \le \kappa_f^{\text{abs}}(A), \tag{4.21}$$

where $\kappa_f^{\text{abs}}(A)$ is be defined with respect to an arbitrary norm on $\mathbb{C}^{n\times n}$.

(2) If A is diagonalizable, i.e., $A = ZDZ^{-1}$ for a diagonal matrix $D \in \mathbb{C}^{n\times n}$, then we have the upper bound

$$\kappa_f^{\text{abs}}(A) \le \kappa_2(Z)^2 \max_{\lambda,\mu\in\Lambda(A)} \big|f[\lambda,\mu]\big|, \tag{4.22}$$

where $\kappa_f^{\text{abs}}(A)$ is meant with respect to $\|\cdot\|_F$.

(3) If A is normal, then

$$\kappa_f^{\text{abs}}(A) = \max_{\lambda,\mu\in\Lambda(A)} \big|f[\lambda,\mu]\big|, \tag{4.23}$$

where $\kappa_f^{\text{abs}}(A)$ is meant with respect to $\|\cdot\|_F$.

Proof. (1) We have

$$\kappa_f^{\text{abs}}(A) \overset{\text{Thm. 4.7}}{=} \|L_f(A)\| \overset{\substack{\text{induced}\\ \text{norm}}}{\ge} \rho(L_f(A)) \overset{\text{Thm. 4.15}}{=} \max_{\lambda,\mu\in\Lambda(A)} \big|f[\lambda,\mu]\big|.$$

(2) It holds

$$\begin{aligned}
\kappa_f^{\text{abs}}(A) =& \|L_f(A)\|_F \overset{\text{Lem. 4.18}}{=} \max_{\|X\|_F=1} \|ZL_f(D, Z^{-1}XZ)Z^{-1}\|_F \\
\overset{\text{Lem. 4.17}}{\le}& \sup_{\|X\|_F=1} \|Z\|_2\|Z^{-1}\|_2\|L_f(D, Z^{-1}XZ)\|_F \\
\le& \kappa_2(Z) \sup_{\|X\|_F=1} \|L_f(D)\|_F\|Z^{-1}XZ\|_F \\
\overset{\text{Lem. 4.17}}{\le}& \kappa_2(Z)\|L_f(D)\|_F \sup_{\|X\|_F=1} \|Z^{-1}\|_2\|Z\|_2\|X\|_F \\
=& \kappa_2(Z)^2\|L_f(D)\|_F \overset{\text{Thm. 4.16}}{=} \kappa_2(Z)^2 \max_{\lambda,\mu\in\Lambda(A)} \big|f[\lambda,\mu]\big|.
\end{aligned}$$

(3) We have $A = UDU^H$ for a unitary matrix $U \in \mathbb{C}^{n\times n}$. Since $\kappa_2(U) = 1$, we receive

$$\max_{\lambda,\mu\in\Lambda(A)} \big|f[\lambda,\mu]\big| \overset{(1)}{\leq} \kappa_f^{\text{abs}}(A) \overset{(2)}{\leq} \max_{\lambda,\mu\in\Lambda(A)} \big|f[\lambda,\mu]\big|. \qquad \square$$

Remark 4.20. For a diagonalizable matrix A, the condition number $\kappa_f^{\text{abs}}(A)$ depends on the first order divided differences $f[\lambda,\mu]$ for $\lambda,\mu \in \Lambda(A)$ and the condition number of the eigenvector matrix Z.

Example 4.21. (1) For $f(z) = z^2$ and $\lambda \neq \mu$, we have

$$f[\lambda,\mu] = \frac{f(\lambda) - f(\mu)}{\lambda - \mu} = \frac{\lambda^2 - \mu^2}{\lambda - \mu} = \lambda + \mu,$$

which will only be large for large λ or μ, i.e., f is well-conditioned at A if the eigenvalues of A are not to large.

(2) For $f(z) = \sqrt{z}$ and $\lambda \neq \mu$, we have

$$f[\lambda,\mu] = \frac{f(\lambda) - f(\mu)}{\lambda - \mu} = \frac{\sqrt{\lambda} - \sqrt{\mu}}{\lambda - \mu} = \frac{1}{\sqrt{\lambda} + \sqrt{\mu}},$$

which will be very large for small λ and μ, i.e., f is ill-conditioned at A if A is close to singular.

5 Computation of $f(A)$

This chapter is devoted to study popular methods for the numerical computation of $f(A)$. To choose the right method (also in a more general setting), we should always ask ourselves the following questions.

- Which operations can be performed?

  (E.g., multiplication with A, multiplication with A^H, solving a linear system with system matrix A, matrix factorizations.)

- What is the size of the matrix?

- What is the desired accuracy?

In what follows, we assume that A is small/medium sized, i.e., we can use matrix decompositions and direct methods for solving linear systems.

5.1 Techniques to Compute $f(A)$

We consider three approaches to compute $f(A)$.

(1) *Decomposition approach*

- For normal A, we compute an eigendecomposition $A = UDU^H$ and can then compute $f(A) = Uf(D)U^H$.
- For a non-normal matrix A, we use the Schur decomposition $A = UTU^H$ and then calculate $f(A) = Uf(T)U^H$. For this, we need a method to compute matrix functions of upper triangular matrices. This leads to the Schur-Parlett algorithm that is discussed in Section 5.2.

(2) *Iteration approach*

We construct a map $g : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ and build a matrix sequence by setting

$$X_{k+1} \coloneqq g(X_k), \qquad k \geq 0, \tag{5.1}$$

for some $X_0 \in \mathbb{C}^{n\times n}$, such that

$$\lim_{k\to\infty} X_k = f(A) \tag{5.2}$$

holds. Important examples of this approach are "Newton-like methods" which we study in Section 5.3.

(3) *Approximation approach*

We approximate f by a polynomial p or a rational function r, e.g., by Taylor approximation or Padé approximation, and then compute $p(A)$ or $r(A)$ as an approximation for $f(A)$, respectively. This leads, for example, to the "Scaling and Squaring methods" that we will see in Section 5.4.

Note, that the first approach leads to a general purpose method, while the latter approaches are usually used for specific f.

One may wonder why the above approaches differ so much from the definitions in Chapter 2, where we defined $f(A)$ using the Jordan form, the Hermite interpolation polynomial, the Taylor series, and the Cauchy integral formula. In fact, the first three definitons do not immediately lead to well-suited methods for computing $f(A)$, while the last one can be used for the computation of $f(A)b$.

(1) *Jordan form definition*

As discussed in Remark 2.1, computing a Jordan form of A is not stable, and, in particular, not continuous in the entries of A.

(2) *Hermite interpolation definition*

We are interested in methods for computing $f(A)$ that use $\mathcal{O}(n^3)$ flops. Since the multiplication of two (full) $(n \times n)$ matrices requires $2n^3$ flops, the evaluation of a polynomial of degree n in A requires $\mathcal{O}(n^4)$ flops, which is too expensive. Note that for any non-derogatory matrix, the Hermite interpolation polynomial is of degree n.

(3) *Taylor series definition*

Inserting A in a truncated Taylor series, i.e., in a Taylor polynomial, has the same performance problems as the Hermite interplation approach, since a good approximation of f typically requires a Taylor polynomial of degree $\mathcal{O}(n)$, even if all eigenvalues of A are located in a tight cluster.

(4) *Cauchy integral definition*

The Cauchy integral formula can be used in combination with quadrature formulas for calculating $f(A)b$; see Chapter 6.

5.2 Schur-Parlett Algorithm

The Schur-Parlett algorithm, which was introduced by Davies and Higham in 2003 [7], is considered to be the "state of the art" approach for the computation of $f(A)$ for general f. It consists of three main steps.

(1) Compute a (block) Schur decomposition $A = UTU^H$; see Subsection 5.2.3.

(2) Evaluate $f(T)$; see Subsections 5.2.1 and 5.2.2.

(3) Return $Uf(T)U^H$.

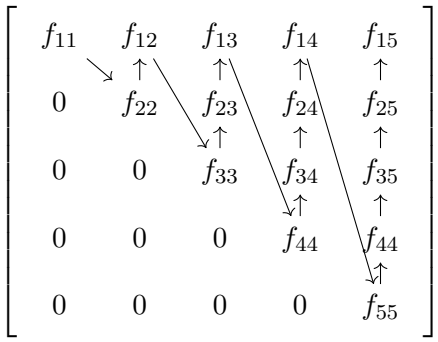


Figure 9: Traversing of the entries f_{ij} of $F = f(T)$ in the Parlett recurrence.

5.2.1 Parlett Recurrence

In Proposition 3.8, we proved the following formula for $f(T)$ if $T = [t_{ij}] \in \mathbb{C}^{2\times 2}$ is upper triangular

$$f\left(\begin{bmatrix} t_{11} & t_{12} \\ 0 & t_{22} \end{bmatrix}\right) = \begin{bmatrix} f(t_{11}) & t_{12} f[t_{11}, t_{22}] \\ 0 & f(t_{22}) \end{bmatrix},$$

by using $Tf(T) = f(T)T$. For an upper triangular $T = [t_{ij}] \in \mathbb{C}^{n\times n}$, we can do the same. Denote $f(T) = [f_{ij}]$. Then

$$\underbrace{\begin{bmatrix} t_{11} & t_{12} & \cdots & t_{1n} \\ & t_{22} & \cdots & t_{2n} \\ & & \ddots & \vdots \\ & & & t_{nn} \end{bmatrix}}_{T} \underbrace{\begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ & f_{22} & \cdots & f_{2n} \\ & & \ddots & \vdots \\ & & & f_{nn} \end{bmatrix}}_{f(T)} = \underbrace{\begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ & f_{22} & \cdots & f_{2n} \\ & & \ddots & \vdots \\ & & & f_{nn} \end{bmatrix}}_{f(T)} \underbrace{\begin{bmatrix} t_{11} & t_{12} & \cdots & t_{1n} \\ & t_{22} & \cdots & t_{2n} \\ & & \ddots & \vdots \\ & & & t_{nn} \end{bmatrix}}_{T}.$$

Note, that the diagonal entries of $f(T)$ can be easily computed because $f_{ii} = f(t_{ii})$ for $i = 1, \ldots, n$ by Theorem 3.5 (6). Let $1 \le i < j \le n$ and assume $t_{ii} \neq t_{jj}$. Considering the entry (i, j) in the above equation yields

$$\begin{aligned} & t_{ii} f_{ij} + t_{ij} f_{jj} + \sum_{k=i+1}^{j-1} t_{ik} f_{kj} = f_{ii} t_{ij} + f_{ij} t_{jj} + \sum_{k=i+1}^{j-1} f_{ik} t_{kj} \\ \Leftrightarrow \qquad & f_{ij}(t_{ii} - t_{jj}) = t_{ij}(f_{ii} - f_{jj}) + \sum_{k=i+1}^{j-1} f_{ik} t_{kj} - \sum_{k=i+1}^{j-1} t_{ik} f_{kj}. \end{aligned}$$

For $t_{ii} \neq t_{jj}$, the above equation is equivalent to

$$f_{ij} = t_{ij} \cdot \frac{f_{ii} - f_{jj}}{t_{ii} - t_{jj}} + \sum_{k=i+1}^{j-1} \frac{f_{ik} t_{kj} - t_{ik} f_{kj}}{t_{ii} - t_{jj}}. \tag{5.3}$$

Algorithm 1 Parlett recurrence

Input: $T = [t_{ij}] \in \mathbb{C}^{n\times n}$ upper triangular with simple eigenvalues, f defined on the spectrum of T

Output: $F = [f_{ij}] = f(T)$

```
for i = 1, ..., n do
    f_ii = f(t_ii)
end for
for j = 2, ..., n do
    for i = j − 1, ..., 1 do
        f_ij = t_ij · (f_ii − f_jj)/(t_ii − t_jj) + Σ_{k=i+1}^{j−1} (f_ik t_kj − t_ik f_kj)/(t_ii − t_jj)
    end for
end for
```

This formula for f_{ij} only depends on the diagonal entries of $f(T)$, the entries f_{ik} with $i < k$, and the entries f_{kj} with $j > k$. Thus, we can calculate $f(T)$ by traversing its entries in an appropriate order; see Figure 9. This gives rise to the *Parlett recurrence*.

Remark 5.1. (1) Algorithm 1 requires $\frac{2}{3}n^3 + \mathcal{O}(n^2)$ floating points operations (flops).

(2) Computing a Schur decomposition with the QR algorithm and applying Algorithm 1 was the first implementation of the `MATLAB` routine `funm` and was in use until 2003. However, this method has two major drawbacks.

- Eigenvalues of A have to be distinct.
- Clustered eigenvalues may lead to a significant loss of accuracy due to the phenomenon of cancellation; see Example 5.2 or [12, pp. 87-89].

Example 5.2. Consider the matrix

$$A = \begin{bmatrix} 1 & -1 & \dots & -1 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ 0 & \dots & 0 & 1 \end{bmatrix} \in \mathbb{R}^{10\times 10},$$

which only has the eigenvalue 1. Thus, the method proposed in Remark 5.1 (2) will break down since $t_{ii} = t_{jj}$. To avoid this, we consider two perturbations of A.

(1) $A_{1,k} = A + \Delta A_{1,k}$, where $\Delta A_{1,k} \in \mathbb{C}^{10\times 10}$ is *diagonal* with normally distributed entries with mean 0 and variance 10^{-k}.

(2) $A_{2,k} = A + \Delta A_{2,k}$, where $\Delta A_{2,k} \in \mathbb{C}^{10\times 10}$ is *upper triangular* with normally distributed entries with mean 0 and variance 10^{-k}.

k	1	2	3	4	5	6
`sqrtm` for $A_{1,k}$	1.0e–15	1.6e–15	3.3e–15	1.9e–15	3.0e–15	2.6e–15
Parlett recurrence for $A_{1,k}$	9.9e–16	1.6e–15	4.5e–15	6.5e–14	3.2e–13	8.1e–13
`sqrtm` for $A_{2,k}$	1.6e–16	1.6e–16	2.2e–16	1.8e–16	1.5e–16	2.2e–16
Parlett recurrence for $A_{2,k}$	1.3e–09	2.0e-01	1.8e+08	1.8e+17	7.8e+26	1.4e+36

Table 1: Relative residuals in Example 5.2.

Computing $A_{j,k}^{\frac{1}{2}}$ with the `MATLAB` built-in routine `sqrtm` and the method proposed in Remark 5.1 (2) yields computed outputs $X_{j,k}$. The corresponding residuals

$$\frac{\|X_{j,k}^2 - A_{j,k}\|_F}{\|A_{j,k}^2 - A_{j,k}\|_F}$$

are displayed in Table 1. For a diagonal perturbation $(A_{1,k})$, the Parlett recurrence yields reasonable results, although the eigenvalues become more and more clustered. For a triangular perturbation $(A_{2,k})$, the Parlett recurrence performs already for $k = 2$ extremely poor, despite the fact that A has very small size. Thus, eigenvalue cluster can lead to numerical difficulties. However, we also observe that the residual of `sqrtm` is always in the order of the machine precision, i.e. these difficulties can be resolved.

To overcome a breakdown of Algorithm 1 if $t_{ii} = t_{jj}$, i.e., if T has a non-simple eigenvalue, we replace T by a block triangular matrix

$$T = \begin{bmatrix} T_{11} & \cdots & T_{1m} \\ & \ddots & \vdots \\ & & T_{mm} \end{bmatrix},$$

where T_{ii} are square matrices for $i = 1, \ldots, m$, such that $\Lambda(T_{ii}) \cap \Lambda(T_{jj}) = \varnothing$ for $i \neq j$. This requires an appropriate reordering of the Schur form. We denote the blocks of $F = f(T)$ according to the blocks T_{ij} of T by F_{ij}. Then the analog of (5.3) is

$$T_{ii}F_{ij} - F_{ij}T_{jj} = F_{ii}T_{ij} - T_{ij}F_{jj} + \sum_{k=i+1}^{j-1} \big(F_{ik}T_{kj} - T_{ik}F_{kj}\big), \tag{5.4}$$

which is a Sylvester equation in the unknown F_{ij} for $1 \leq i < j \leq n$. In particular, there exists a unique solution F_{ij} if and only if $\Lambda(T_{ii}) \cap \Lambda(T_{jj}) = \varnothing$ by Theorem 3.29.

The Sylvester equations (5.4) in Algorithm 2 can be solved with the Bartels-Stewart algorithm. However, there are still two remaining problems.

(1) How can we evaluate $f(T_{ii})$, $i = 1, \ldots, m$, accurately and efficiently? Is this as complicated as computing $f(T)$? → Subsection 5.2.2

(2) How can we find a Schur form with adequate blocking? → Subsection 5.2.3

Here, the key idea is to find a block structure, such that the spectrum of each diagonal block is a "cluster", and such that the spectra of different diagonal blocks are well separated. Such blocks are called *atomic blocks*.

Algorithm 2 Block Parlett recurrence

Input: $T = [T_{ij}] \in \mathbb{C}^{n\times n}$ block upper triangular with $\Lambda(T_{ii}) \cap \Lambda(T_{jj}) = \varnothing$, f defined on the spectrum of T

Output: $F = [F_{ij}] = f(T)$

for $i = 1, \ldots, m$ **do**

$F_{ii} = f(T_{ii})$

end for

for $j = 2, \ldots, m$ **do**

for $i = j-1, \ldots, 1$ **do**

$$\text{solve } T_{ii}F_{ij} - F_{ij}T_{jj} = F_{ii}T_{ij} - T_{ij}F_{jj} + \sum_{k=i+1}^{j-1} \left(F_{ik}T_{kj} - T_{ik}F_{kj}\right) \text{ for } F_{ij}$$

end for

end for

5.2.2 Evaluating Atomic Blocks

As mentioned in Chapter 2, a primary matrix function $f(A)$ can be defined by the Taylor series of f. The truncated Taylor series around z_0, i.e., the Taylor polynomial of order m

$$f_m(z) = \sum_{k=0}^{m} \frac{f^{(k)}(z_0)}{k!}(z - z_0)^k \tag{5.5}$$

can be used to compute $f(A)$ approximately provided that the truncation error $\|f(A) - f_m(A)\|$ is small enough. Recall, that approximating $f(A)$ by the Taylor series is not competitive with other numerical methods, since it is too expensive ($\mathcal{O}(mn^3)$) and the convergence is too slow. However, if A is triangular with a clustered spectrum and of small size, e.g., for atomic blocks, the Taylor series approach can be used.

Assume that $T \in \mathbb{C}^{n\times n}$ is upper triangular and its eigenvalues form a single cluster. Define

$$\sigma \coloneqq \frac{1}{n} \sum_{\lambda\in\Lambda(T)} \lambda \qquad \text{and} \qquad M \coloneqq T - \sigma I_n. \tag{5.6}$$

Then σ is the mean of the eigenvalues of T. Furthermore, we define the following decomposition of T

$$T = \sigma I_n + M. \tag{5.7}$$

If f has a Taylor series around σ, i.e., there exists $r > 0$ with

$$f(\sigma + z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\sigma)}{k!} z^k \ , \qquad |z - \sigma| < r, \tag{5.8}$$

and $\Lambda(T) \subseteq \{z \in \mathbb{C} \mid |z - \sigma| < r\}$, then by Theorem 2.19 we have

$$f(T) = \sum_{k=0}^{\infty} \frac{f^{(k)}(\sigma)}{k!} M^k. \tag{5.9}$$

Next, we decompose M into its diagonal part D and its strictly upper triangular part N (which is nilponent), i.e.,

$$M = D + N. \tag{5.10}$$

If the entries d_{ii} of D are sufficiently small, i.e., all the diagonal entries of T are close to σ, then M^k becomes small enough to obtain a good approximation for $f(T)$ by computing $f_m(T)$ if $m \approx n$.

Theorem 5.3. *Let $D = \operatorname{diag}(d_{11}, \ldots, d_{nn}) \in \mathbb{C}^{n\times n}$ with $|d_{ii}| \leq r$ for $i = 1, \ldots, n$, and let $N \in \mathbb{C}^{n\times n}$ be strictly upper triangular. Then for any induced matrix norm with an underlying absolute vector norm (i.e., a norm $\|\cdot\|$ with $\|\,|x|\,\| = \|x\|$, where $|x|$ is the vector with the entry-wise absolute values of x), we have*

$$\left\|(D+N)^k\right\| \leq \sum_{j=0}^{\min\{k,n-1\}} \binom{k}{j} r^{k-j}\|N\|^j. \tag{5.11}$$

Proof. Using the binomial identity for matrices, we find

$$\begin{aligned}\left\|(D+N)^k\right\| &= \left\| \sum_{x\in\{0,1\}^k} \prod_{i=1}^{k} N^{x_i} D^{1-x_i} \right\| \leq \sum_{\substack{x\in\{0,1\}^k \\ \|x\|_1 \leq n-1}} \left\| \prod_{i=1}^{k} N^{x_i} D^{1-x_i} \right\| \\ &\leq \sum_{\substack{x\in\{0,1\}^k \\ \|x\|_1 \leq n-1}} \prod_{i=1}^{k} \|N\|^{x_i}\|D\|^{1-x_i} \leq \sum_{j=0}^{\min\{k,n-1\}} \binom{k}{j} r^{k-j}\|N\|^j,\end{aligned}$$

where the second inequality uses that the product of m upper triangular matrices is equal to zero if n factors are strictly upper triangular. The last inequality follows from

$$\|D\| = \max_{1\leq i\leq n} |d_{ii}| \leq r,$$

which is true since $\|\cdot\|$ is induced by an absolute vector norm. □

Remark 5.4. If r is sufficiently small, then (5.9) will converge rapidly since $\|M^k\|$ decays fast. More precisely, for $r \ll \|N\|$ and $r < 1$, Theorem 5.3 yields

$$\left\|M^k\right\| \in \mathcal{O}(r^{k+1-n}\|N\|^{n-1}). \tag{5.12}$$

If the Taylor coefficients $\frac{f^{(k)}(\sigma)}{k!}$ of f are not too large (e.g., for analytic f, see [26, Eq. (4.59)]), we can truncate (5.9) to approximately n summands.

A detailed derivation of a reliable criterion for truncating (5.9) is presented in [12, p. 223] and is based on the following bound for the truncation error

$$\left\| f(\sigma I_n + M) - \sum_{k=0}^{m-1} \frac{f^{(k)}(\sigma)}{k!} M^k \right\| \leq \frac{1}{m!} \max_{t\in[0,1]} \|M^m f^{(m)}(\sigma I_n + tM)\|. \tag{5.13}$$

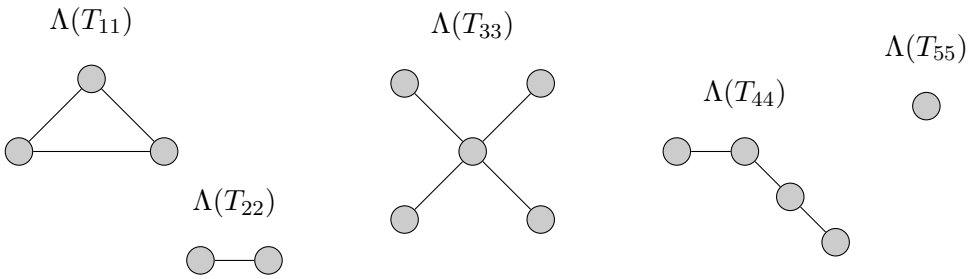


Figure 10: Spectrum $\Lambda(A) = \bigcup_{i=1}^{5} \Lambda(T_{ii})$ interpreted as an (undirected) graph.

5.2.3 Ordering of the Schur Form

We want to compute (or reorder) a block Schur decomposition $A = UTU^H$ with a block triangular

$$T = \begin{bmatrix} T_{11} & \cdots & T_{1m} \\ & \ddots & \vdots \\ & & T_{mm} \end{bmatrix} \in \mathbb{C}^{n \times n}, \tag{5.14}$$

such that the eigenvalues of each diagonal block T_{ii} are close to each other, and such that the eigenvalues of different diagonal blocks T_{ii} and T_{jj} are far away from each other. More precisely, for a given *blocking parameter* $\delta > 0$, we want the following to hold for $1 \leq i < j \leq m$:

(1) $\min\{|\lambda - \mu| \mid \lambda \in \Lambda(T_{ii}),\, \mu \in \Lambda(T_{jj})\} > \delta$

(2) If $|\Lambda(T_{ii})| \geq 2$, then for all $\lambda \in \Lambda(T_{ii})$ there exists $\mu \in \Lambda(T_{ii}) \setminus \{\lambda\}$ with $|\lambda - \mu| \leq \delta$.

Note that (2) implies

$$\max_{\lambda,\mu \in \Lambda(T_{ii})} |\lambda - \mu| \leq (m_i - 1)\delta, \tag{5.15}$$

where m_i is the size of the matrix block T_{ii}. This bound is sharp, e.g., for a T_{ii} with $\Lambda(T_{ii}) = \{\delta, 2\delta, 3\delta, \ldots, m_i\delta\}$.

Graph theoretic interpretation The above construction has the following interpretation in graph theory. Consider the (undirected) graph $G = (V, E)$ with

$$V = \{\lambda_1, \ldots, \lambda_n\}, \qquad E = \{(\lambda_i, \lambda_j) \mid |\lambda_i - \lambda_j| \leq \delta\}.$$

Then each connected component of G corresponds to the eigenvalues a diagonal block T_{ii} in the block Schur form (5.14). An illustration can be found in Figure 10.

A blocking with the above properties can always be computed in a backward stable way. In particular, there exist an algorithm in `LAPACK` written in `Fortran` in the 1970s We omit the details and refer to [12, Sect. 9.3] and references therein. However, note that the above graph is explicitly built in the algorithm.

5.2.4 Algorithm and Remarks

Combining the above parts yields the *Schur-Parlett algorithm* of Davies and Higham.

Algorithm 3 Schur-Parlett algorithm [7]

Input: $f : \Omega \to \mathbb{C}$, $A \in \mathbb{C}^{n\times n}(\Omega)$ analytic, blocking parameter $\delta > 0$
Output: $f(A)$

1: compute a Schur decomposition $A = UTU^H$ (with the QR algorithm)
2: **if** T is diagonal **then** ▷ A is normal
3: **return** $U \operatorname{diag}(f(t_{11}), \ldots, f(t_{mm}))\, U^H$
4: **end if**
5: find $\tilde{U} \in \mathbb{C}^{n\times n}$ unitary such that

$$\tilde{U}^H A \tilde{U} = \begin{bmatrix} T_{11} & \cdots & T_{1m} \\ & \ddots & \vdots \\ & & T_{mm} \end{bmatrix}$$

fulfills (1) and (2) in Sect. 5.2.3
6: evaluate the atomic blocks $F_{ii} = f(T_{ii})$ (using truncated a Taylor series)
7: determine F_{ij} for $1 \le i < j \le m$ with Algorithm 2 (using for example the Bartels-Stewart algorithm to solve Sylvester equations)
8: **return** $\tilde{U} F \tilde{U}^H$

Remark 5.5. (1) `MATLAB`'s `funm` is an implementation of Algorithm 3.

(2) Algorithm 3 requires derivatives of f for computing $f(T_{ii})$, although they may not be necessary to define $f(A)$, e.g., for simple eigenvalues.

(3) A common choice for the blocking parameter is $\delta = 0.1$ *(heuristic)*.

(4) The cost of Algorithm 3 is "roughly between $28n^3$ and $\frac{n^4}{3}$" [12, p. 228]. The worst case applies when the blocks T_{ii} are of size $\mathcal{O}(n)$. Note that the cost for computing the Schur decomposition is already approximately $25n^3$. Hence, almost all effort to perform the algorithm goes into this *initial step*.

(5) The QR algorithm tends to order the diagonal by the modulus, which is good for the blocking procedure.

(6) For a sum of functions

$$f(z) = \sum_{i=0}^{k} f_i(z),$$

the Schur form only needs to be computed once. Similar, for a composition $f = g \circ h$ the matrix $h(T)$ is a Schur form of $h(A)$.

(7) In practice, Algorithm 3 behaves mostly forward stable, i.e.,

$$\frac{\|f(A+\Delta A)-f(A)\|}{\|f(A)\|} \in \mathcal{O}(\kappa_f^{\text{rel}}(A)up(n)),$$

where u is the unit roundoff and $p(n)$ a low degree polynomial that depends on the considered matrix norm. In case of instabilities, one should increase the blocking parameter δ.

(8) Unfortunately, the Schur-Parlett algorithm does not easily extend to real arithmetic, even though there exists a *real Schur decomposition* $A = URU^T$ with the *real Schur form*

$$R = \begin{bmatrix} R_{11} & \cdots & R_{1k} \\ & \ddots & \vdots \\ & & R_{kk} \end{bmatrix} \in \mathbb{R}^{n\times n},$$

where

$$R_{ii} = [\lambda] \in \mathbb{R}^{1\times 1} \quad \text{or} \quad R_{ii} = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \in \mathbb{R}^{2\times 2}$$

for $i = 1, \ldots, n$. The eigenvalues of R_{ii} are λ or $a \pm bi$. The problem in the latter case is that complex-conjugate eigenvalues may not belong to the same eigenvalue cluster with respect to the blocking parameter δ. Hence, we cannot find *real* atomic blocks.

5.3 Newton-like Methods for Computing $\operatorname{sign}(A)$ and $A^{\frac{1}{2}}$

This section is devoted to present iterative methods to compute $\operatorname{sign}(A)$ and $A^{\frac{1}{2}}$ based on Newton's method. Note that there exist several other competitive approaches for the computation of $\operatorname{sign}(A)$ and $A^{\frac{1}{2}}$.

Recall that Newton's method is an iterative method to compute a zero of a (Fréchet) differentiable function f. Given a "good" initial guess x_0, the method converges (at least) quadratically.

Newton's method (scalar version) Given a differentiable function $f : \Omega \to \mathbb{C}$ and an iterate $z_k \in \Omega$, we have

$$f(z) = f(z_k) + f'(z_k)(z - z_k) + o(|z - z_k|). \tag{5.16}$$

Hence, we can approximate f around z_k by the affine linear function

$$z \mapsto f(z_k) + f'(z_k)(z - z_k). \tag{5.17}$$

If $f'(z_k) \neq 0$, then the next iterate z_{k+1} is the zero of the above affine linear function. This leads to the *scalar Newton iteration*

$$z_{k+1} = z_k - \frac{f(z_k)}{f'(z_k)}, \qquad k \geq 0. \tag{5.18}$$

Newton's method (matrix-valued version) Given a Fréchet differentiable function $f : U \to \mathbb{C}^{n\times n}$ and the iterate $X_k \in U$, we have

$$f(X) = f(X_k) + L_f(X_k, X - X_k) + o\left(\|X - X_k\|\right). \tag{5.19}$$

Similar, to the scalar case, the next iterate X_{k+1} has to fulfill

$$0 = f(X_k) + L_f(X_k, X_{k+1} - X_k). \tag{5.20}$$

If $L_f(X_k)$ is bijective, then (5.20) is equivalent to

$$0 = L_f^{-1}(X_k, f(X_k)) + X_{k+1} - X_k, \tag{5.21}$$

where $L_f^{-1}(X_k)$ denotes the inverse of the linear map $L_f(X_k)$. Now we obtain the *matrix-valued Newton iteration*

$$X_{k+1} = X_k - L_f^{-1}\big(X_k, f(X_k)\big), \qquad k \geq 0. \tag{5.22}$$

5.3.1 Newton Sign Iteration

Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subset \mathbb{C} \setminus \mathbf{i}\mathbb{R}$. The key idea of the Newton sign iteration is rather simple. We apply Newton's method to the function

$$f(X) = X^2 - I_n. \tag{5.23}$$

Recall that $\operatorname{sign}(A)^2 = I_n$ (Theorem 3.24 (1)), i.e., $\operatorname{sign}(A)$ is a zero of f, although it is not the only one. Using (5.20) under the assumption that the iterates commute, i.e., $X_kX_{k+1} = X_{k+1}X_k$, yields

$$\begin{aligned} 0 &= f(X_k) + L_f(X_k, X_{k+1} - X_k) \\ &= X_k^2 - I_n + X_k(X_{k+1} - X_k) + (X_{k+1} - X_k)X_k \\ &= X_k^2 - I_n + 2X_k(X_{k+1} - X_k) = -X_k^2 - I_n + 2X_kX_{k+1} \\ \Leftrightarrow \quad X_{k+1} &= \frac{1}{2}X_k^{-1}(X_k^2 + I_n) = \frac{1}{2}(X_k + X_k^{-1}), \end{aligned}$$

where we used that X_k is nonsingular. Hence, for $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subset \mathbb{C} \setminus \mathbf{i}\mathbb{R}$, we define the *Newton sign iteration* by

$$\begin{cases} X_0 = A, \\ X_{k+1} = \dfrac{1}{2}\left(X_k + X_k^{-1}\right), \quad k \geq 0. \end{cases} \tag{5.24}$$

Remark 5.6. For any $\lambda \in \mathbb{C}$ in the open right/left half-plane, the value $\frac{1}{2}(\lambda + \lambda^{-1})$ is also in the right/left half-plane, respectively. Thus, (5.24) is indeed well-defined since for every eigenvalue λ of X_k the corresponding eigenvalue on X_{k+1} is given by $\frac{1}{2}(\lambda + \lambda^{-1})$. In particular, we have $\operatorname{sign}(A) = \operatorname{sign}(X_k)$ for all $k \geq 0$.

Next, we study the convergence of the Newton-sign iteration.

Theorem 5.7. *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbf{i}\mathbb{R}$. Then the sequence $(X_k)_{k\in\mathbb{N}_0}$ defined in (5.24) converges to $S := \operatorname{sign}(A)$ with*

$$\|X_{k+1} - S\| \leq \frac{1}{2}\|X_k^{-1}\|\|X_k - S\|^2 \tag{5.25}$$

for any consistent norm $\|\cdot\|$.

Proof. Since the matrices X_k are rational functions in A for any $k \geq 0$, they commute mutually. Moreover, they commute with $S = \operatorname{sign}(A)$ (see Theorem 3.5). Hence, we receive

$$\begin{aligned} X_{k+1} \pm S &= \frac{1}{2}(X_k + X_k^{-1} \pm 2S) = \frac{1}{2}X_k^{-1}(X_k^2 + I_n \pm 2X_kS) \\ &= \frac{1}{2}X_k^{-1}(X_k^2 \pm X_kS \pm SX_k + S^2) = \frac{1}{2}X_k^{-1}(X_k \pm S)^2, \end{aligned} \tag{5.26}$$

and in particular,

$$\|X_{k+1} - S\| \leq \frac{1}{2}\|X_k^{-1}\|\|X_k - S\|^2,$$

which establishes the inequality.

It remains to show that $(X_k)_{k\in\mathbb{N}_0}$ actually converges. We define

$$G_k := (X_k - S)(X_k + S)^{-1}.$$

Then

$$X_k = (I_n - G_k)^{-1}(I_n + G_k)S, \tag{5.27}$$

and

$$\begin{aligned} G_{k+1} =& (X_{k+1} - S)(X_{k+1} + S)^{-1} \overset{(5.26)}{=} \frac{1}{2}X_k^{-1}(X_k - S)^2\Big(\frac{1}{2}X_k^{-1}(X_k + S)^2\Big)^{-1} \\ =& \big((X_k - S)(X_k + S)^{-1}\big)^2 = G_k^2. \end{aligned}$$

Thus, we receive $G_{k+1} = G_k^2 = \ldots = G_0^{2^{k+1}}$. Furthermore, from

$$\Lambda(G_0) = \Lambda\big((A - S)(A + S)^{-1}\big) = \left\{\frac{\lambda - \operatorname{sign}(\lambda)}{\lambda + \operatorname{sign}(\lambda)} \,\middle|\, \lambda \in \Lambda(A)\right\}$$

we conclude that $\rho(G_0) < 1$, i.e., $\lim\limits_{k\to\infty} G_k = \lim\limits_{k\to\infty} G_0^{2^k} = 0$. Hence,

$$\lim_{k\to\infty} X_k \overset{(5.27)}{=} \lim_{k\to\infty} (I_n - \underbrace{G_k}_{\to 0})^{-1}(I_n + \underbrace{G_k}_{\to 0})S = S. \qquad \square$$

Remark 5.8. (1) In the above proof, we saw that the sequence $(X_k)_{k\in\mathbb{N}_0}$ converges slowly if the sequence $(G_k)_{k\in\mathbb{N}_0}$ converges slowly. This is the case if A has very small or very large eigenvalues since in both cases

$$\rho(G_0) = \max_{\lambda\in\Lambda(A)} \left|\frac{\lambda - \operatorname{sign}(\lambda)}{\lambda + \operatorname{sign}(\lambda)}\right| \approx 1. \tag{5.28}$$

(2) In (5.24), we have to invert X_k in every iteration. Usually, we try to avoid this in practical computations and use inverse-free iterations; see, e.g., the Newton-Schulz sign iteration below.

(3) The Newton sign iteration is locally quadratically convergent. However, depending on A, the initial convergence might be very slow. One way to overcome this problem are *scaled iterations*.

5.3.2 Adaptions of the Newton Sign Iteration

Newton-Schulz sign iteration The key idea is to replace X_k^{-1} with one step of Newton's method for an appropriate function f. For $B \in \mathbb{C}^{n\times n}$ nonsingular, we define

$$f(Y) = Y^{-1} - B. \tag{5.29}$$

Then $f(Y) = 0$ is equivalent to $Y = B^{-1}$, and we have $L_f(Y, Z) = -Y^{-1}ZY^{-1}$; see Example 4.6 (3). Furthermore, we have

$$\begin{aligned} L_f^{-1}(Y, Z) &= L_{f^{-1}}(f(Y), Z) = -(f(Y) + B)^{-1}Z(f(Y) + B)^{-1} \\ &= -(Y^{-1} - B + B)^{-1}Z(Y^{-1} - B + B)^{-1} = -YZY, \end{aligned}$$

where we used $f^{-1}(Y) = (Y + B)^{-1}$, $L_{f^{-1}}(Y, Z) = -(Y + B)^{-1}Z(Y + B)^{-1}$, and Theorem 4.5 (3). Thus, the matrix-valued Newton iteration for f is

$$\begin{aligned} Y_{k+1} &= Y_k - L_f^{-1}(Y_k, f(Y_k)) = Y_k + Y_k f(Y_k)Y_k = Y_k + Y_k(Y_k^{-1} - B)Y_k \\ &= 2Y_k - Y_k B Y_k = Y_k(2I_n - BY_k). \end{aligned}$$

Hence, one step of Newton's method for f with initial matrix B yields the approximation $B(2I_n - B^2)$ for B^{-1}.

By replacing X_k^{-1} in the Newton sign iteration (5.24) with $X_k(2I_n - X_k^2)$ we obtain the *Newton-Schulz sign iteration*

$$\begin{cases} X_0 = A, \\ X_{k+1} = \frac{1}{2}(X_k + X_k(2I_n - X_k^2)) = \frac{1}{2}X_k(3I_n - X_k^2), \quad k \geq 0. \end{cases} \tag{5.30}$$

This iteration converges locally quadratic as well, but this requires somewhat stronger assumptions on the spectrum of A.

Theorem 5.9. *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C}\setminus \mathbf{i}\mathbb{R}$ and $\rho(I_n - A^2) < 1$. Then the sequence $(X_k)_{k\in\mathbb{N}_0}$ defined in (5.30) converges to $S := \operatorname{sign}(A)$ with*

$$\|X_{k+1} - S\| \leq \frac{1}{2}\|X_k + 2S\|\|X_k - S\|^2 \tag{5.31}$$

for any consistent norm $\|\cdot\|$.

Proof. Exercise (similar to the proof of Theorem 5.7). □

Scaled Newton sign iteration Let $a \in \mathbb{C} \setminus \mathbf{i}\mathbb{R}$ and consider the scalar Newton sign iteration (i.e., for $n = 1$)

$$\begin{cases} z_0 = a, \\ z_{k+1} = \frac{1}{2}(z_k + \frac{1}{z_k}), \quad k \geq 0. \end{cases}$$

For $|a| \gg 1$ we have $z_{k+1} \approx \frac{1}{2} z_k$ in the first steps (linear convergence). For $|a| \ll 1$, we have $|z_1| \approx |\frac{1}{2a}| = |\frac{1}{2z_0}| \gg 1$ and again $z_{k+1} \approx \frac{1}{2} z_k$, in the next couple of steps.

To avoid this, consider a scaled Newton sign iteration

$$\begin{cases} X_0 = A, \\ X_{k+1} = \frac{1}{2}(\mu_k X_k + \frac{1}{\mu_k} X_k^{-1}), \quad k \geq 0. \end{cases} \tag{5.32}$$

Typical choices for μ_k are

- *determinantal scaling*: $\mu_k = |\det(X_k)|^{-\frac{1}{n}}$,
- *spectral scaling:* $\mu_k = \sqrt{\dfrac{\rho(X_k^{-1})}{\rho(X_k)}}$,
- *norm scaling:* $\mu_k = \sqrt{\dfrac{\|X_k^{-1}\|}{\|X_k\|}}$.

Example 5.10. We illustrate the effect of a scaling by considering the following random matrices.

(1) $A_1 \in \mathbb{C}^{200\times 200}$ normal with $\Lambda(A_1)$ uniformly distributed $(\mathrm{Re}(\lambda), \mathrm{Im}(\lambda) \in [0, 1])$

(2) $A_2 \in \mathbb{C}^{200\times 200}$ normal with $\Lambda(A_2)$ uniformly distributed $(\mathrm{Re}(\lambda), \mathrm{Im}(\lambda) \in [0, 30])$

(3) $A_3 \in \mathbb{C}^{200\times 200}$ normal with $\Lambda(A_3)$ normally distributed $(\mu = 0,\ \sigma = \sqrt{2})$

(4) $A_4 \in \mathbb{C}^{200\times 200}$ normal with $\Lambda(A_4)$ normally distributed $(\mu = 0,\ \sigma = 20)$

(5) $A_5 \in \mathbb{C}^{200\times 200}$ normal with $\Lambda(A_5)$ normally distributed $(\mu = 5,\ \sigma = \sqrt{2})$

(6) $A_6 \in \mathbb{C}^{200\times 200}$normal with $\Lambda(A_6)$ normally distributed $(\mu = 20,\ \sigma = \sqrt{2})$

The relative residuals

$$\frac{\|X_k^2 - I_n\|_F}{\|A^2 - I_n\|_F}$$

of the scaled Newton sign iterations are displayed in Figure 11. First, we observe that the required number of iterations is fairly small compared to the matrix size ($n = 200$). Depending on the matrix, different scaling techniques are better than others, e.g., in the case of a normal matrix (diagonalizable with perfectly conditioned eigenvectors), determinantal scaling seems to be most beneficial. Moreover, the larger the eigenvalues the larger is the effect of the scaling.

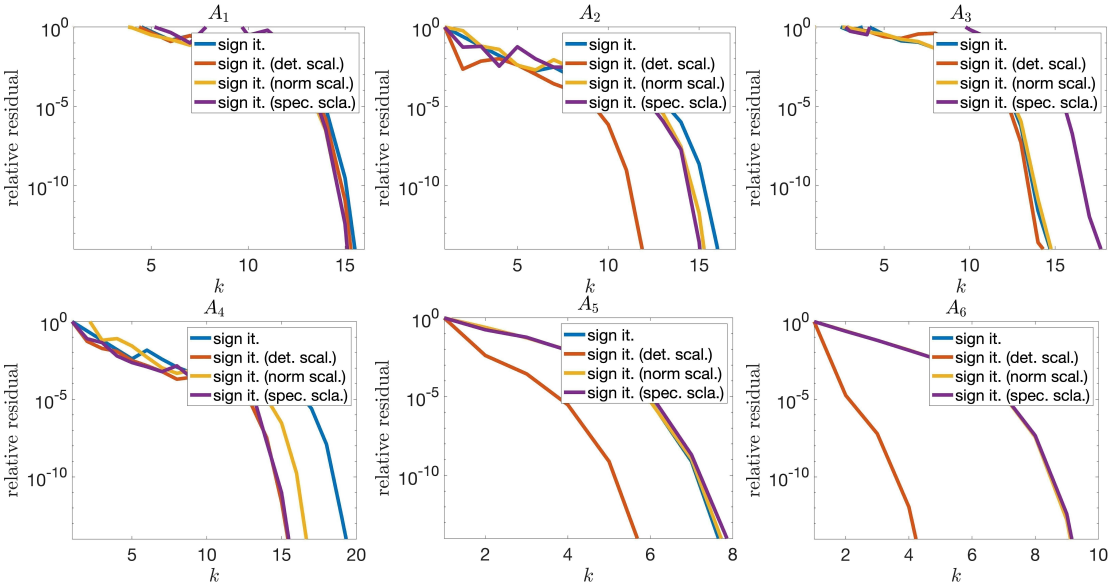


Figure 11: Relative residuals for the Newton sign iteration applied to the matrices $A_1, \ldots, A_6$ in Example 5.10.

5.3.3 Newton Square Root Iteration

Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Similar, to the Newton sign iteration, we use Newton's method for the map

$$f(X) = X^2 - A. \tag{5.33}$$

Then $A^{\frac{1}{2}}$ is a zero of f. Moreover, we have $L_f(X, Y) = XY + YX$; Example 4.6 (2). Again, we assume that all the iterates commute with each other. (Note that this assumption lead to numerical issues discussed below.) Then (5.20) becomes

$$\begin{aligned} 0 &= f(X_k) + L_f(X_k, X_{k+1} - X_k) \\ &= X_k^2 - A + X_k(X_{k+1} - X_k) + (X_{k+1} - X_k)X_k \\ &= -X_k^2 - A + 2X_kX_{k+1} \\ \Leftrightarrow \quad X_{k+1} &= \frac{1}{2}X_k^{-1}(X_k^2 + A) = \frac{1}{2}(X_k + X_k^{-1}A), \end{aligned}$$

and we define the *Newton square root iteration* by

$$\begin{cases} X_0 = A, \\ X_{k+1} = \frac{1}{2}(X_k + X_k^{-1}A), \quad k \geq 0. \end{cases} \tag{5.34}$$

The iteration (5.34) is equivalent to the Newton sign iteration applied to $A^{\frac{1}{2}}$, which leads to a simple convergence proof.

Theorem 5.11. *Let $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Then the sequence $(X_k)_{k\in\mathbb{N}_0}$ defined in* (5.34) *converges to $A^{\frac{1}{2}}$. In particular,*

$$\|X_{k+1} - A^{\frac{1}{2}}\| \leq \frac{1}{2}\|X_k^{-1}\|\|X_k - A^{\frac{1}{2}}\|^2 \tag{5.35}$$

for any consistent norm $\|\cdot\|$.

Proof. Since all eigenvalues of $A^{\frac{1}{2}}$ lie in the right half-plane, we have $\operatorname{sign}(A^{\frac{1}{2}}) = I_n$. Define the iteration

$$\begin{cases} S_0 = A^{\frac{1}{2}}, \\ S_{k+1} = \frac{1}{2}(S_k + S_k^{-1}), \quad k \geq 0, \end{cases} \tag{5.36}$$

which is the Newton sign iteration applied to $A^{\frac{1}{2}}$. By Theorem 5.7, we have

$$\lim_{k\to\infty} S_k = \operatorname{sign}(A^{\frac{1}{2}}) = I_n.$$

Moreover, $X_0 = A = A^{\frac{1}{2}}S_0$, and inductively

$$\begin{aligned} X_{k+1} &= \frac{1}{2}(X_k + X_k^{-1}A) = \frac{1}{2}(A^{\frac{1}{2}}S_k + S_k^{-1}A^{-\frac{1}{2}}A) \\ &= \frac{1}{2}(A^{\frac{1}{2}}S_k + S_k^{-1}A^{\frac{1}{2}}) = \frac{1}{2}A^{\frac{1}{2}}(S_k + S_k^{-1}) = A^{\frac{1}{2}}S_{k+1}. \end{aligned}$$

Note that we have used that $A^{\frac{1}{2}}$ and S_k^{-1} commute, since S_k is a function in A; see Theorem 3.5 (1). We obtain

$$\lim_{k\to\infty} X_k = \lim_{k\to\infty} A^{\frac{1}{2}}S_k = A^{\frac{1}{2}} \lim_{k\to\infty} S_k = A^{\frac{1}{2}}.$$

Moreover,

$$X_{k+1} - A^{\frac{1}{2}} = \frac{1}{2}(X_k + X_k^{-1}A) - A^{\frac{1}{2}} = \frac{1}{2}X_k^{-1}(X_k^2 + A - 2X_kA^{\frac{1}{2}}) = \frac{1}{2}X_k^{-1}(X_k - A^{\frac{1}{2}})^2,$$

where we used that $A^{\frac{1}{2}}$ and X_k commute since both are functions in A. Therefore,

$$\big\|X_{k+1} - A^{\frac{1}{2}}\big\| = \|\frac{1}{2}X_k^{-1}\big(X_k - A^{\frac{1}{2}}\big)^2\| \leq \frac{1}{2}\|X_k^{-1}\|\big\|X_k - A^{\frac{1}{2}}\big\|^2. \qquad \square$$

Similar to the Newton sign iteration, there exist many adaptions, e.g., scaled Newton square root iterations

$$\begin{cases} X_0 = A, \\ X_{k+1} = \frac{1}{2}(\mu_k X_k + \frac{1}{\mu_k}X_k^{-1}A), \quad k \geq 0. \end{cases} \tag{5.37}$$

A common choice for the scaling parameter is $\mu_k = \left|\dfrac{\det(X_k)}{\det(A^{\frac{1}{2}})}\right|^{\frac{1}{n}}$.

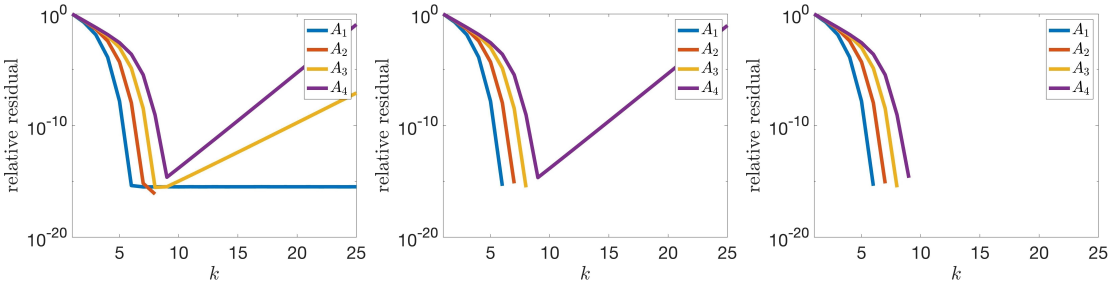


Figure 12: Relative residuals for the Newton square root iteration applied to the matrices $A_1, \ldots, A_4$ in Example 5.12 with tolerances 10^{-16} (left), 10^{-15} (mid), and 10^{-14} (right).

5.3.4 (In-)Stability of the Newton Square Root Iteration

If we implement this method numerically in finite precision arithmetic, we observe that even for well-conditioned Hermitian matrices $A \in \mathbb{C}^{n\times n}$, the Newton square root iteration oftentimes does not converge. This was already stated by Laasonen in 1958 [18, p. 113]:

> "... algorithm (1), if carried out indefinitely, is not stable whenever the ratio of the largest to the smallest eigenvalue of A exceeds the value 9."

We investigate why this occurs, and in particular, how Newton's method fails because of finite precision. Let us start with an example.

Example 5.12. We illustrate the above observation of Laasonen with a numerical example. We consider the matrices

(1) $A_1 \in \mathbb{C}^{201\times 201}$ Hermitian with equispaced eigenvalues in $[1, 4]$,

(2) $A_2 \in \mathbb{C}^{201\times 201}$ Hermitian with equispaced eigenvalues in $[1, 16]$,

(3) $A_3 \in \mathbb{C}^{201\times 201}$ Hermitian with equispaced eigenvalues in $[1, 64]$,

(4) $A_4 \in \mathbb{C}^{201\times 201}$ Hermitian with equispaced eigenvalues in $[1, 256]$,

and apply the Newton square root iteration (5.34). The method is stopped in step k if the relative residual fulfills

$$\frac{\|X_k^2 - A_j\|_F}{\|A_j^2 - A_j\|_F} \leq \varepsilon_i$$

with prescribed tolerances $\varepsilon_1 = 10^{-16}$, $\varepsilon_2 = 10^{-15}$, and $\varepsilon_3 = 10^{-14}$, or if $k_{\max} = 25$ steps were performed. We observe that the method converges quickly within a few iterations. However, depending on the tolerance, the residual again starts to increase at some point. This happens if the condition number $\kappa(A) = \frac{\lambda_{\max}}{\lambda_{\min}}$ is to large, of the desired tolerance is to small.

Note, that above observations are particularly surprising, since Newton's method is locally converging, i.e., if X_k is very close to $A_j^{\frac{1}{2}}$, the convergence is guaranteed by theory. However, this holds not in finite precision as we will elaborate in this section.

Dynamical systems The field of *dynamical systems* studies iterations of the form

$$X_{k+1} = g(X_k), \qquad k \geq 0, \tag{5.38}$$

with some initial point X_0. We then have

$$X_k = g(X_{k-1}) = \ldots = g^k(X_0) \qquad \text{for all } k \geq 0.$$

Definition 5.13. Let $(\mathcal{X}, \|\cdot\|)$ be a finite-dimensional Banach space, $U \subseteq \mathcal{X}$, and $g : U \to \mathcal{X}$ be continuously Fréchet differentiable. A *fixed point* $X_* \in U$ of g, i.e., a point with $g(X_*) = X_*$, is called *attracting*, if

$$\rho(L_g(X_*)) < 1.$$

Moreover, for an attracting fixed point X_*, we define the *basin of attraction* as

$$\mathcal{A}_g(X_*) = \{X \in U \mid \lim_{k\to\infty} g^k(X) = X_*\}. \tag{5.39}$$

Since g is continuously differentiable, the basin of attraction of an attracting fixed point X_* always contains a neighborhood of X_*.

Newton's method applied to a continuously differentiable function $f : U \to \mathbb{C}^{n\times n}$ can also be written in the form of (5.38) using the *Newton map*

$$N_f(X) = X - L_f^{-1}(X, f(X)). \tag{5.40}$$

We then have $X_{k+1} = N_f(X_k) = \ldots = N_f^{k+1}(X_0)$. Moreover, it can be shown that

$$f(X_*) = 0 \qquad \Leftrightarrow \qquad X_* \text{ is an attracting fixed point of } N_f.$$

Since Newton's method is locally convergent, we have

$$\lim_{k\to\infty} X_k = \lim_{k\to\infty} N_f^k(X_0) = X_*,$$

whenever X_0 is sufficient close to X_*.

Example 5.14. Consider the function $f(z) = z^2 - 1$, which has the zeros ± 1. Then

$$N_f(z) = z - \frac{f(z)}{f'(z)} = z - \frac{z^2-1}{2z} = z - \frac{1}{2}z + \frac{1}{2z} = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

and this map has the attracting fixed points ± 1 with basins of attraction

$$\mathcal{A}_{N_f}(1) = \{z \in \mathbb{C} \mid \mathrm{Re}(z) > 0\},$$
$$\mathcal{A}_{N_f}(-1) = \{z \in \mathbb{C} \mid \mathrm{Re}(z) < 0\}.$$

We call the union of the basins of attraction the *Fatou set*. Moreover, we observe a cyclic behavior for $z \in \mathbf{i}\mathbb{R}$ (*Julia set*). Together, the global behavior of Newton's method for f is fully described.

In contrast, for a slightly more complicated function, e.g., $f(z) = z^3 - 1$, the global behavior is much more chaotic and not well understood. In particular, the Julia set has a fractal character.

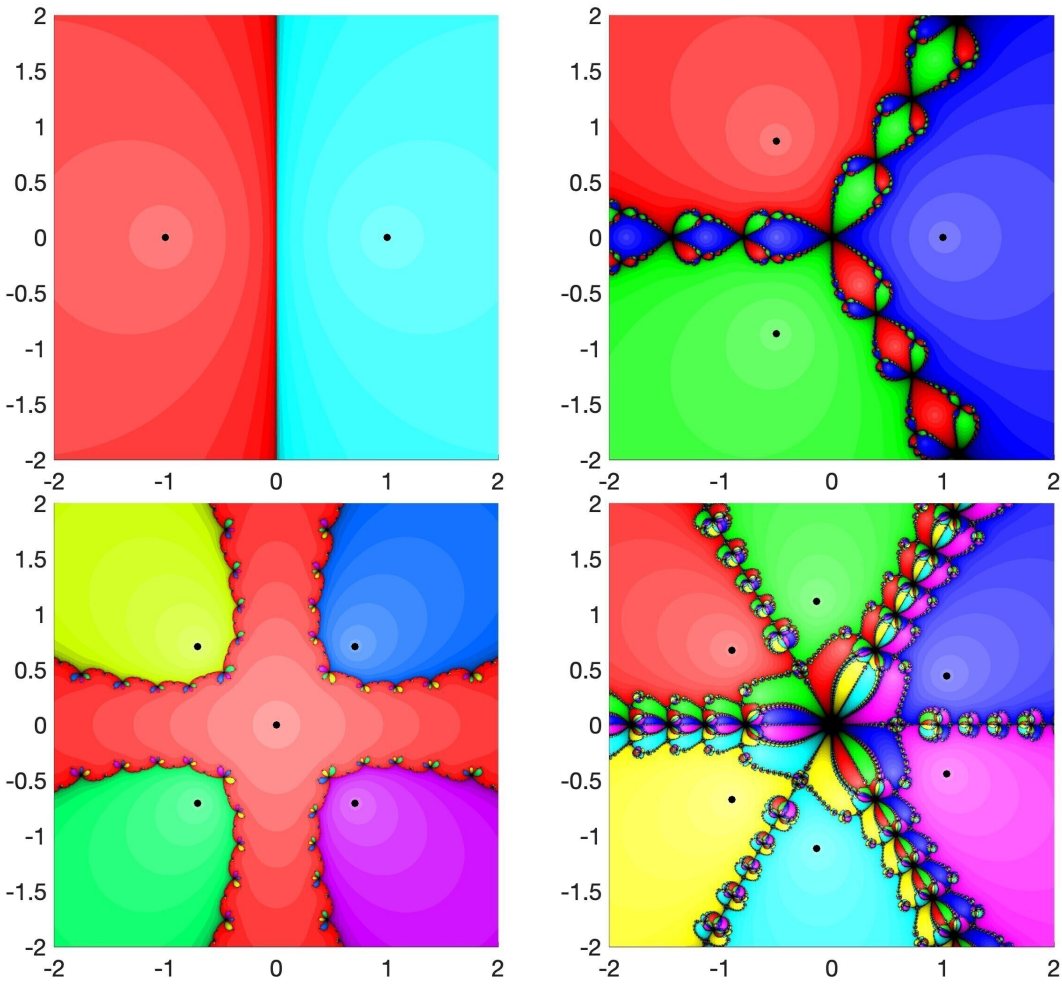


Figure 13: Zeros of f (black dots) and the dynamics of Newton's method for $f(z) = z^2-1$ (top left), $f(z) = z^3 - 1$ (top right), $f(z) = z^5 - z$, and $f(z) = z^6 - z^3 + 2$ (bottom right).

We illustrate the dynamics for four polynomials in Figure 13, where the domain is colored according to which zero of f the Newton method converges with the respective point as initial guess z_0. The black dots indicate the zeros of f. The darker the color, the more iterations were required. Black regions indicate initial points for which Newton's method does not converge (numerically) within prescribed number of iterations. We stop the computation after a maximum of 20 steps or if the residual $|f(z_k)|$ is smaller than 10^{-14}. Note, that any neighborhood around a zero is colored the same, which, once again, illustrates the local convergence of Newton's method.

Instability of the Newton square root iteration

Definition 5.15. Let $(\mathcal{X}, \|\cdot\|)$ be a finite-dimensional Banach space, $U \subseteq \mathcal{X}$, and $g : U \to \mathcal{X}$ be continuously Fréchet differentiable. An iteration

$$X_{k+1} = g(X_k), \qquad k \geq 0,$$

with some $X_0 \in U$ is called *stable* in a neighborhood of a fixed point X_* of g, if there exists a constant $c > 0$ with

$$\left\|L_g^k(X_*)\right\| \leq c \tag{5.41}$$

for all $k \geq 1$.

The inequality (5.41) is fulfilled, if $\rho(L_g(X_*)) < 1$, i.e., if X_* is an attracting fixed point of g. In particular, Newton's method is stable since it is locally convergent. However, the Newton square root iteration (5.34) is not stable in practice as we observed in Example 5.12. Next, we will investigate how this is possible.

The iteration map of the matrix square root iteration is given by

$$g : \mathrm{GL}_n(\mathbb{C}) \to \mathbb{C}^{n\times n}, \quad X \mapsto \frac{1}{2}(X + X^{-1}A).$$

Its Fréchet derivative is given by

$$L_g(X, Y) = \frac{1}{2}(Y - X^{-1}YX^{-1}A).$$

In particular,

$$L_g(A^{\frac{1}{2}}, X) = \frac{1}{2}(X - A^{-\frac{1}{2}}XA^{-\frac{1}{2}}A) = \frac{1}{2}(X - A^{-\frac{1}{2}}XA^{\frac{1}{2}}).$$

Using Lemma 4.13 (2), we receive that the Kronecker form of $L_g(A^{\frac{1}{2}})$

$$\mathcal{K}_g(A^{\frac{1}{2}}) = \frac{1}{2}\big(I_{n^2} - \big(A^{\frac{1}{2}}\big)^T \otimes A^{-\frac{1}{2}}\big),$$

which has the spectrum

$$\Lambda(\mathcal{K}_g(A^{\frac{1}{2}})) = \left\{\frac{1}{2}\Big(1 - \frac{\sqrt{\lambda}}{\sqrt{\mu}}\Big) \mid \lambda, \mu \in \Lambda(A)\right\}$$

by the Theorem of Stephanos (Theorem 4.14). If A is HPD, then $\kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}$ and

$$\rho(\mathcal{K}_g(A^{\frac{1}{2}})) = \frac{1}{2}\big(\kappa_2(A^{\frac{1}{2}}) - 1\big) = \frac{1}{2}\big(\sqrt{\kappa_2(A)} - 1\big)$$

Thus,

$$\rho(\mathcal{K}_g(A^{\frac{1}{2}})) = \rho(L_g(A^{\frac{1}{2}})) < 1 \quad \Leftrightarrow \quad \frac{1}{2}\big(\sqrt{\kappa_2(A)} - 1\big) < 1 \quad \Leftrightarrow \quad \kappa_2(A) < 9,$$

which explains the previously mentioned observation by Laasonen [18].

The crucial issue is that the iteration map g, although it is derived from Newton's method, does not coincide with the Newton map N_f. More precisely, it only coincides with the Newton map N_f on the *centralizer* of A

$$\mathcal{C}_A = \{B \in \mathbb{C}^{n\times n} \mid AB = BA\}. \tag{5.42}$$

Let us illustrate our findings by considering the error propagation. Suppose that A is diagonalizable, i.e., $A = ZDZ^{-1}$ with $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Let

$$X_* \coloneqq A^{\frac{1}{2}} = ZD^{\frac{1}{2}}Z^{-1},$$

and consider the rank-1 perturbation

$$E_0 \coloneqq \varepsilon Z e_j e_i^T Z^{-1}$$

for some $\varepsilon \neq 0$ and distinct $i, j \in \{1, \dots, n\}$. Using the *Sherman-Morrison-Woodburry formula*, we find

$$(X_* + E_0)^{-1} = X_*^{-1} - X_*^{-1}E_0X_*^{-1}, \tag{5.43}$$

and receive

$$\begin{aligned}
E_1 &= g(X_* + E_0) - X_* = \frac{1}{2}(X_* + E_0 + (X_* + E_0)^{-1}A) - X_* \\
&= \frac{1}{2}(X_* + E_0 + (X_*^{-1} - X_*^{-1}E_0X_*^{-1})A) - X_* \\
&= \frac{1}{2}(A^{\frac{1}{2}} + E_0 + A^{\frac{1}{2}} - A^{-\frac{1}{2}}E_0A^{\frac{1}{2}}) - A^{\frac{1}{2}} = \frac{1}{2}(E_0 - A^{-\frac{1}{2}}E_0A^{\frac{1}{2}}) \\
&= \frac{1}{2}(E_0 - \varepsilon A^{-\frac{1}{2}}Ze_je_i^TZ^{-1}A^{\frac{1}{2}}) = \frac{1}{2}(E_0 - \varepsilon ZD^{-\frac{1}{2}}e_je_i^TD^{\frac{1}{2}}Z^{-1}) \\
&= \frac{1}{2}(E_0 - \lambda_i^{\frac{1}{2}}\lambda_j^{-\frac{1}{2}}\varepsilon Ze_je_i^TZ^{-1}) = \frac{1}{2}\Big(1 - \frac{\sqrt{\lambda_i}}{\sqrt{\lambda_j}}\Big)E_0.
\end{aligned}$$

Inductively,

$$E_k \coloneqq g(X_* + E_{k-1}) - g(X_0) = \left(\frac{1}{2}\Big(1 - \frac{\sqrt{\lambda_i}}{\sqrt{\lambda_j}}\Big)\right)^k E_0, \quad k \geq 0.$$

Hence, the Newton square root iteration may diverge, even though we start arbitrarily close to $A^{\frac{1}{2}}$, if

$$\max_{1\leq i,j\leq n}\left|\frac{1}{2}\Big(1 - \frac{\sqrt{\lambda_i}}{\sqrt{\lambda_j}}\Big)\right| \geq 1. \tag{5.44}$$

Stabilizing the Newton square root iteration As the above explained phenomenon is rather unfortunate, we want to improve/stabilize the Newton square root iteration.

One idea is to use a coupled iteration, e.g., the *DB iteration* which is named after Denman and Beavers [9]:

$$\begin{cases} X_{k+1} = \frac{1}{2}(X_k + Y_k^{-1}),\ k \geq 0, & X_0 = A, \\ Y_{k+1} = \frac{1}{2}(Y_k + X_k^{-1}),\ k \geq 0, & Y_0 = I_n. \end{cases} \tag{5.45}$$

Indeed, the DB iteration converges for any $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$ and the iterates fulfill

$$\lim_{k\to\infty} X_k = A^{\frac{1}{2}} \qquad \text{and} \qquad \lim_{k\to\infty} Y_k = A^{-\frac{1}{2}},$$

which can be proven similar to Theorem 5.11.

The DB iteration can also be written as

$$\begin{bmatrix} X_{k+1} \\ Y_{k+1} \end{bmatrix} = g\left(\begin{bmatrix} X_k \\ Y_k \end{bmatrix}\right),\ k \geq 0, \qquad \begin{bmatrix} X_0 \\ Y_0 \end{bmatrix} = \begin{bmatrix} A \\ I_n \end{bmatrix}, \tag{5.46}$$

with the map

$$g : \mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C}) \to \mathbb{C}^{n\times n} \times \mathbb{C}^{n\times n}, \qquad \begin{bmatrix} X \\ Y \end{bmatrix} \mapsto \begin{bmatrix} \frac{1}{2}(X + Y^{-1}) \\ \frac{1}{2}(Y + X^{-1}) \end{bmatrix}. \tag{5.47}$$

Note that for any $B \in \mathrm{GL}_n(\mathbb{C})$, the point $\begin{bmatrix} B \\ B^{-1} \end{bmatrix}$ is a fixed point of g. In particular, $\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}$ is a fixed point of g. Next, we show that $\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}$ is a stable fixed point of g. We have

$$L_g\left(\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}, \begin{bmatrix} X \\ Y \end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix} X - A^{\frac{1}{2}}YA^{\frac{1}{2}} \\ Y - A^{-\frac{1}{2}}XA^{-\frac{1}{2}} \end{bmatrix}. \tag{5.48}$$

Moreover,

$$\begin{aligned} L_g^2\left(\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}, \begin{bmatrix} X \\ Y \end{bmatrix}\right) &= L_g\left(\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}, L_g\left(\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}, \begin{bmatrix} X \\ Y \end{bmatrix}\right)\right) \\ &= L_g\left(\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}, \frac{1}{2}\begin{bmatrix} X - A^{\frac{1}{2}}YA^{\frac{1}{2}} \\ Y - A^{-\frac{1}{2}}XA^{-\frac{1}{2}} \end{bmatrix}\right) \\ &= \frac{1}{4}\begin{bmatrix} X - A^{\frac{1}{2}}YA^{\frac{1}{2}} - A^{\frac{1}{2}}(Y - A^{-\frac{1}{2}}XA^{-\frac{1}{2}})A^{\frac{1}{2}} \\ Y - A^{-\frac{1}{2}}XA^{-\frac{1}{2}} - A^{-\frac{1}{2}}(X - A^{\frac{1}{2}}YA^{\frac{1}{2}})A^{-\frac{1}{2}} \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} X - A^{\frac{1}{2}}YA^{\frac{1}{2}} \\ Y - A^{-\frac{1}{2}}XA^{-\frac{1}{2}} \end{bmatrix} \\ &= L_g\left(\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}, \begin{bmatrix} X \\ Y \end{bmatrix}\right), \end{aligned}$$

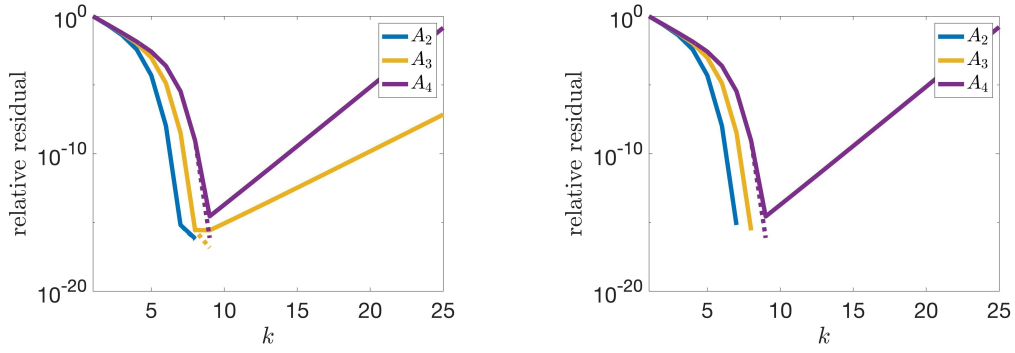


Figure 14: Relative residuals for the Newton square root iteration (solid) and the DB iteration (dotted) applied to the matrices A_2, A_3, and A_4 in Example 5.12 with tolerances 10^{-16} (left), and 10^{-15} (right).

i.e., $L_g\left(\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}\right)$ is idempotent. Therefore, all powers of $L_g\left(\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}\right)$ are bounded, and thus, g is stable in a neighborhood of $\begin{bmatrix} A^{\frac{1}{2}} \\ A^{-\frac{1}{2}} \end{bmatrix}$.

Example 5.16. We compare the Newton square root iteration and the DB iteration for the matrices A_1, A_3, and A_4 from Example 5.12. Figure 14 shows the relative residuals are. We observe that the DB iteration is stable in finite precision in contrast to the Newton square root iteration.

5.4 Scaling and Squaring Algorithm

Finally, we study exemplarily an approximation approach, where we approximate a given function f by a simpler function g and evaluate $g(A)$ instead of $f(A)$. We already faced this idea twice when considering the Hermite interpolation definition (Definition 2.9) and the Taylor series definition (Definition 2.13) of matrix functions. Recall the downsides of these approaches.

- The Hermite interpolation polynomial is hard to compute (e.g., we need to know derivatives of f), unstable, and thus, badly suited for numerical computation. Moreover, the evaluation is expensive.
- In Theorem 5.3, we saw that the m-th Taylor polynomial f_m yields a fairly good approximation, if all eigenvalues of A are in a small disk and if $m \approx n$. Of course, this assumption is rather strong.

Another issue of polynomial approximation, that applies to both approaches above, is that polynomials have a pole at infinity, i.e., they "explode" for $|z| \gg 1$.

We overcome all these issues when computing the matrix exponential $\exp(A)$ and the matrix logarithm $\log(A)$, using the following ideas.

(1) **Padé approximation:** Approximate the function f by a rational function, i.e., we want to find low-degree polynomials p and q with

$$f(z) \approx \frac{p(z)}{q(z)}.$$

Note that this approximation problem is nonlinear, but can easily be linearized via $f(z)q(z) \approx p(z)$ ($\rightarrow$ Section 5.4.1).

(2) **Scaling & Squaring:** Find an appropriate matrix $\widetilde{A}$ (all eigenvalues of $\widetilde{A}$ are in a small disk), such that computing $f(\widetilde{A})$ yields also $f(A)$ ($\rightarrow$ Section 5.4.2).

This idea was already mentioned as "*... one of the most effective we [algorithms] know*" by Moler and Van Loan in their seminal article "Nineteen dubious ways to compute the exponential of a matrix" [22]; see also its update [23].

5.4.1 Padé Approximation

The idea of Taylor approximation is to find a polynomial p with $\deg(p) \leq m$ such that $f(z) - p(z) \in \mathcal{O}(|z|^{m+1})$ for $z \to 0$, i.e., $f^{(j)}(0) = p^{(j)}(0)$ holds for $j = 0, \ldots, m$.

Similarly, for Padé approximation we want to find a rational function

$$r_{km}(z) = \frac{p_{km}(z)}{q_{km}(z)}, \tag{5.49}$$

with $\deg(p_{km}) \leq k$, $\deg(q_{km}) \leq m$, and $q(0) = 1$, such that $f(z) - r_{km}(z) \in \mathcal{O}(|z|^{k+m+1})$, i.e., $f^{(j)}(0) = r_{km}^{(j)}(0)$ holds for $j = 0, \ldots, k+m$. In particular, the $[k/0]$-Padé approximant coincides with the Taylor polynomial of order k.

Definition 5.17. Let f be analytic in a neighborhood of 0. We call a rational function as in equation (5.49) with coprime polynomials p_{km} and q_{km}, $\deg(p_{km}) \leq k$, $\deg(q_{km}) \leq m$ and $q_{km}(0) = 1$ a $[k/m]$-*Padé approximant* of f, if

$$f(z) - r_{km}(z) \in \mathcal{O}(|z|^{k+m+1}) \tag{5.50}$$

for $z \to 0$. If $k = m$, we call $r_m \coloneqq r_{mm}$ a *diagonal Padé approximant*.

Padé approximants have some significant upsides.

(1) Often Padé approximants produce the desired accuracy with less computational costs than Taylor polynomials. We must compute $A^2, \ldots, A^m$ (and one matrix inversion) for the $[m/m]$-Padé approximant instead of $A^2, \ldots, A^{2m}$ for the Taylor polynomial of order $2m$, which has the same asymptotic approximation quality.

(2) Just like for the Taylor approximation, for many "important" functions Padé approximants are known explicitly, e.g., for $\exp(z)$ we have

$$p_{km}(z) = \sum_{j=0}^{k} \frac{(k+m-j)!k!}{(k+m)!(k-j)!} \cdot \frac{z^j}{j!}, \qquad q_{km}(z) = \sum_{j=0}^{m} \frac{(k+m-j)!m!}{(k+m)!(m-j)!} \cdot \frac{(-z)^j}{j!}.$$

Note that $p_{km}(z) = q_{mk}(-z)$ which reflects the identity $\exp(-z) = \frac{1}{\exp(z)}$.

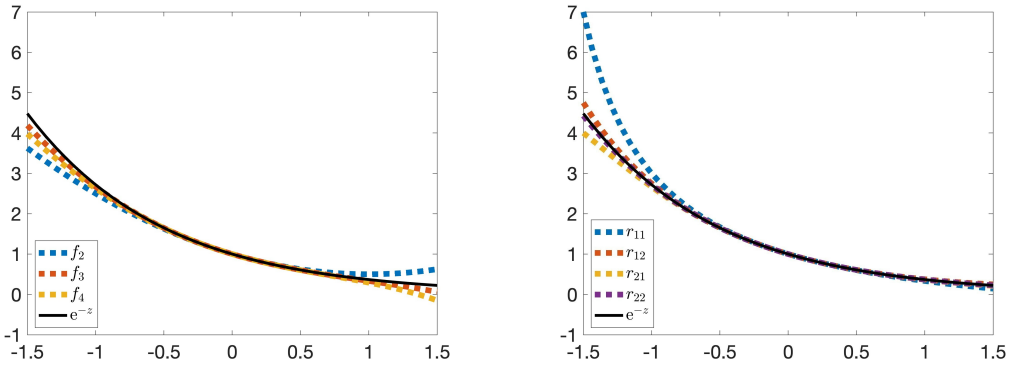


Figure 15: The function $f(z) = \mathrm{e}^{-z}$ and its Taylor polynomials f_m (left) and Padé approximants r_{km} (right).

Let us consider one example, that demonstrates in particular, how we linearize the nonlinear rational approximation problem.

Example 5.18. We want to calculate the [1/1]-Padé approximant of

$$f(z) = \exp(-z) = \sum_{k=0}^{\infty} \frac{(-z)^k}{k!} = 1 - z + \frac{z^2}{2} + \mathcal{O}(|z|^3),$$

which has the form

$$r_1(z) = \frac{a_0 + a_1 z}{1 + b_1 z},$$

if it exists. The linearized approximation problem is

$$\begin{aligned} f(z)q(z) - p(z) &= (1 - z + \frac{z^2}{2} + \mathcal{O}(|z|^3))(1 + b_1 z) - (a_0 + a_1 z) \\ &= 1 - z + \frac{z^2}{2} + b_1 z - b_1 z^2 + \mathcal{O}(|z|^3) - a_0 - a_1 z \overset{!}{\in} \mathcal{O}(|z|^3), \end{aligned}$$

where we used $q(0) = 1$. Comparing coefficients yields the linear system

$$\begin{cases} z^0: & 1 - a_0 = 0 & \Rightarrow a_0 = 1 \\ z^1: & -1 + b_1 - a_1 = 0 & \\ z^2: & \frac{1}{2} - b_1 = 0 & \Rightarrow b_1 = \frac{1}{2} \end{cases}$$

Hence, $a_1 = -\frac{1}{2}$, and therefore,

$$r_1(z) = \frac{1 - \frac{1}{2}z}{1 + \frac{1}{2}z}.$$

Remark 5.19. (1) The existence of a $[k/m]$-Padé approximant is not guaranteed, e.g., $f(z) = z^2$ has no [1/1]-Padé approximant since

$$z^2(1 + b_1 z) - (a_0 + a_1 z) = z^2 + b_1 z^3 - a_0 - a_1 z \notin \mathcal{O}(|z|^3)$$

has no solution for $a_0, a_1, b_1 \in \mathbb{C}$.

(2) Although Padé approximants are preferable in practice, they are not guaranteed to approximate better than Taylor polynomials and vice versa.

- For $f(z) = z^2 + z$, we obviously have $f_2(z) = f(z)$, but $r_{11}(z) = \dfrac{z}{1-z}$ which follows immediately from

$$(z+z^2)(1+b_1z) - (a_0+a_1z) = (1+b_1)z^2 + (1-a_1)z - a_0 + b_1z^3 \overset{!}{\in} \mathcal{O}(|z|^3).$$

- For $f(z) = \dfrac{z}{1-z}$, we have $r_{11}(z) = f(z)$, but $f_2(z) = z + z^2$.

Theorem 5.20. *Let f be analytic in a neighborhood of 0 and $k, m \in \mathbb{N}$. If a $[k/m]$-Padé approximant of f exists, then it is unique.*

Proof. Let $r_{km}(z) = \frac{p_{km}(z)}{q_{km}(z)}$ and $\widetilde{r}_{km}(z) = \frac{\widetilde{p}_{km}(z)}{\widetilde{q}_{km}(z)}$ be two $[k/m]$-Padé approximants of f, i.e.,

$$f(z) - r_{km}(z) \in \mathcal{O}(|z|^{k+m+1}) \qquad \text{and} \qquad f(z) - \widetilde{r}_{km}(z) \in \mathcal{O}(|z|^{k+m+1}).$$

Therefore,

$$r_{km}(z) - \tilde{r}_{km}(z) = \frac{p_{km}(z)}{q_{km}(z)} - \frac{\widetilde{p}_{km}(z)}{\widetilde{q}_{km}(z)} \in \mathcal{O}(|z|^{k+m+1})$$

$$\overset{1=q_{km}(0)=\widetilde{q}_{km}(0)}{\Leftrightarrow} \qquad \frac{p_{km}(z)\widetilde{q}_{km}(z) - \widetilde{p}_{km}(z)q_{km}(z)}{q_{km}(z)\widetilde{q}_{km}(z)} \in \mathcal{O}(|z|^{k+m+1}).$$

Thus, the first $m+k$ derivatives of the numerator $p_{km}(z)\widetilde{q}_{km}(z) - \widetilde{p}_{km}(z)q_{km}(z)$ must vanish at $z = 0$. Since the degree of the numerator is at most $k+m$, we conclude that $p_{km}(z)\widetilde{q}_{km}(z) - \widetilde{p}_{km}(z)q_{km}(z)$ is the zero-polynomial. Hence,

$$p_{km}(z)\widetilde{q}_{km}(z) = \widetilde{p}_{km}(z)q_{km}(z),$$

and therefore $r_{km}(z) = \widetilde{r}_{km}(z)$. □

Remark 5.21. Padé approximants appear also in other methods for computing matrix functions, e.g., in the *family of Padé iterations* for computing the matrix sign function.

Recall from Remark 3.23 that

$$\operatorname{sign}(z) = z(z^2)^{-\frac{1}{2}} = z(1-(1-z^2))^{-\frac{1}{2}} = z(1-\zeta)^{-\frac{1}{2}},$$

where $\zeta = 1 - z^2$. Thus, approximating $\operatorname{sign}(z)$ reduces to the approximation of the function $h(\zeta) = (1-\zeta)^{-\frac{1}{2}}$. Using the $[\ell/m]$-Padé approximant for this purpose, yields the matrix iteration

$$\begin{cases} X_0 = A, \\ X_{k+1} = X_k p_{\ell m}(I_n - X_k^2) q_{\ell m}(I_n - X_k^2)^{-1}, \quad k \geq 0. \end{cases} \tag{5.51}$$

Note that we already faced one instance of the above family, namely the Newton-Schulz iterations which results from the $[1/0]$-Padé approximation. For more information, we refer to [12, Sect. 5.4].

5.4.2 Scaling and Squaring

In this section, we present a method to compute $\exp(A)$ by exploiting the identity

$$\exp(A) = \exp\left(\frac{1}{\sigma}A\right)^{\sigma}, \tag{5.52}$$

for $A \in \mathbb{C}^{n\times n}$ and $\sigma \neq 0$. It can be easily proven using the matrix function induction principle. In particular, we use a special version of (5.52), namely

$$\exp(A) = \exp\left(\frac{1}{2^s}A\right)^{2^s}. \tag{5.53}$$

There are two reasons for using the above identity, although this requires additional costs for squaring the matrix $\exp(\frac{1}{2^s}A)$.

(1) The spectrum of $\frac{1}{2^s}A$ is more clustered than the spectrum of A. Thus, Padé approximants of smaller degrees can be used to approximate $\exp(\frac{1}{2^s}A)$ accurately.

(2) The scaling of the eigenvalues, in particular the ones with negative real part, avoids the "hump effect" when evaluating the exponential at the spectrum.

To illustrate the second upside, let us briefly recall basics of finite precision arithmetic that are known from numerical linear algebra.

Finite precision arithmetic A (normalized) machine (or floating point) number in $F(\beta, t, e_{\min}, e_{\max})$ is of the form

$$\pm(0.d_1 d_2 \dots d_t) \cdot \beta^e, \tag{5.54}$$

where $d_1, \dots, d_t \in \{0, 1, \dots, \beta - 1\}$, $d_1 \neq 0$, and $e_{\min} \leq e \leq e_{\max}$. In particular, the floats in the intervall $[\beta^j, \beta^{j+1}]$ are equispaced with distance $\delta = \beta^{1-t+j}$. Thus, their relative distance is

$$\varepsilon_M = \beta^{1-t}, \tag{5.55}$$

which we call the *machine precision*. Since only finitely man real numbers can be represented as floats, we introduce the *rounding function*

$$\mathrm{fl}(x) \coloneqq \operatorname*{arg\,min}_{z\in F} |x - z|, \tag{5.56}$$

for $x \in \mathbb{R}$. If x is exactly between two floats, there exist sophisticated tiebreak rules. The relative rounding error is bounded by

$$\frac{|x - \mathrm{fl}(x)|}{|x|} < \frac{1}{2}\varepsilon_M =: u, \tag{5.57}$$

which is called *unit roundoff*. For $x, y \in F$, any floating point operation $\circledast \in \{\oplus, \ominus, \odot, \oslash\}$ fulfills

$$x \circledast y = \mathrm{fl}(x * y) = (x * y)(1 + \delta), \tag{5.58}$$

where $\delta \in \mathbb{R}$ with $|\delta| \leq u$.

There are three major problems when working with floats. We will illustrate them with examples in $F \coloneqq F(10, 3, -\infty, \infty)$.

(1) *Not any real numbers is a float.*

Example:

$4321 \neq \mathrm{fl}(4321) = 4320 = +0.432 \cdot 10^4$.

(2) *Floating point operations are not associative.*

Example:

$(5.04 + 5.06) + 5.06 = 10.1 + 5.06 = 15.2$

$5.04 + (5.06 + 5.06) = 5.04 + 10.1 = 15.1$

(3) *Subtraction of nearby floats brings earlier errors into prominence.*

If $x, y \in \mathbb{R}$ have k leading significant digits in common, then $\mathrm{fl}(\mathrm{fl}(x) \ominus \mathrm{fl}(y))$ only has $t - k$ (correct) significant digits. This effect is called *cancellation*.

Example:

Let $x = 101.234$ and $y = 100$ which have two leading digits in common. Then $\mathrm{fl}(x) = 101 = +0.101 \cdot 10^3$ and $\mathrm{fl}(y) = 100 = 0.100 \cdot 10^3$. We have

- $x - y = 1.234$,
- $\mathrm{fl}(\mathrm{fl}(x) \ominus \mathrm{fl}(y)) = \mathrm{fl}(101 \ominus 100) = 1$.

Indeed, only the first $3 - 2 = 1$ significant digits are incorrect.

Note, however, that for floats $x, y \in F$ with $0 \leq y \leq x \leq 2y$, the subtraction is exact by Sterbenz's lemma ([25, p. 138]), i.e., $x \ominus y = x - y$. Thus, the issue is not the subtraction, but earlier errors that now become crucial.

As an example for cancellation, or finite precision issues in a broader sense, we consider the *hump effect* that occurs when computing of the (scalar) exponential function. Similar effects may observed whenever we work with alternating sums.

Example 5.22. For $s \in \{0, 1, 2, 3, 4\}$, we set $x = -2^s$ and compute $\exp(x)$. We compare two methods.

(1) Evaluate $\exp(x) = \sum_{k=0}^{m} \frac{x^k}{k!}$ where m is chosen such that the relative change of the partial sum is smaller than the machine precision, i.e.,

$$\left| \frac{\frac{x^{m+1}}{m!}}{\sum_{k=0}^{m} \frac{x^k}{k!}} \right| \leq \varepsilon_M.$$

(2) Compute $\exp(-1)$ with method (1) and then subsequently square the result to obtain $\exp(-2^s) = \exp(-1)^{2^s}$.

The results are presented in Table 2. We observe the following.

	s	0	1	2	3	4
(1)	relative error	3.0e–16	4.1e–16	1.7e–14	6.6e–12	5.9e–05
	$\max_{1\le k\le m} \frac{\lvert\frac{x^k}{k!}\rvert}{\exp(x)}$	2.7e–00	1.5e+01	5.8e+02	1.2e+06	7.8e+12
	m	19	24	34	50	81
(2)	relative error	3.0e–16	6.1e–16	1.5e–15	2.7e–15	5.5e–15
	$\max_{1\le k\le m} \frac{\lvert\frac{1}{k!}\rvert^{2^s}}{\exp(-1)}$	2.7e–00	2.7e–00	2.7e–00	2.7e–00	2.7e–00
	m	19	19	19	19	19

Table 2: Computations in Example 5.22

- Method (2) is more accurate than method (1).
- For method (1), the larger s the larger m. Hence, we have a significant increase of the computational costs.
- In method (1), we have approximately

$$\text{rel. err.} \le 10^{-16} \cdot \max_{1\le k\le m} \frac{\lvert\frac{x^k}{k!}\rvert}{\exp(x)},$$

  i.e., the absolute value of the largest summand in the exponential series is significant for the number of correct digits. See also Figure 16, where the "humps" are apparent.

The problem with method (1) in the previous example is that the summation of a large summand $\frac{x^k}{k!}$ leads to a loss of accuracy since the last digits of $\frac{x^k}{k!}$, which become relevant later, are affected by earlier roundings. Large summands "absorb" accuracy!

Combining Padé approximants and the identity (5.53) yields the *Scaling and Squaring Algorithm* (Algorithm 4). Note that there are no authors who are solely credited for this method.

Algorithm 4 Scaling and squaring algorithm

Input: $A \in \mathbb{C}^{n\times n}$
Output: $F = \exp(A)$
  Find $s \in \mathbb{N}$ with $\|\frac{1}{2^s}A\|_1 \lesssim 5.4$
  Evaluate [13/13]-Padé approximant $r_{13,13}$ of $\exp(z)$ and set $F = r_{13,13}(\frac{1}{2^s}A)$
  for $k = 1, \dots, s$ **do**
    $F := F^2$
  end for

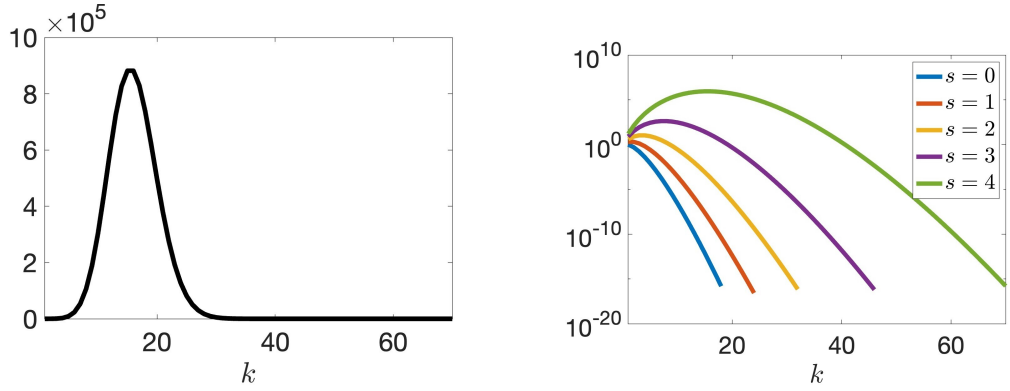


Figure 16: Hump effect. Summands in the exponential series of e^{-2^4} (left) and summands in the exponential series of e^{-2^s} for $s = 0, \ldots, 4$ (right)

Remark 5.23. (1) Obviously, the upper bound of 5.4 for the 1-norm and the degree 13 for the Padé approximation can be changed to different values. The given numbers have proven to be useful in practical computation with double precision arithmetic.

(2) The scaling parameter s can also chosen dynamically which leads to a significant reduction of the computational costs; see the article of Al-Mohy and Higham [2]. Since 2024, their algorithm is used in the built-in routine `expm` in `MATLAB` and replaced the method proposed in Algorithm 4.

5.4.3 Inverse Scaling and Squaring Algorithm

Similar to matrix exponential, the matrix logarithm can be computed using the identity

$$\log(A) = 2^s \log(A^{\frac{1}{2^s}}), \quad s \in \mathbb{N}, \tag{5.59}$$

which leads to the *inverse scaling and squaring algorithm* (Algorithm 5). Note, that here we have to compute the matrix square root of a matrix subsequently. This can be done with the DB iteration for example.

Algorithm 5 Inverse scaling and squaring algorithm

Input: $A \in \mathbb{C}^{n\times n}$ with $\Lambda(A) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$
Output: $F = \log(A)$
  Set $A_0 := A$ and $s := 0$
  while $\|A_s - I_n\|_1 \lesssim 0.246$ **do**
    $s := s + 1$
    $A_s := A_{s-1}^{\frac{1}{2}}$
  end while
  Evaluate the [7/7]-Padé approximant $r_{7,7}$ of $\log(1+z)$ and set $F = r_{7,7}(A_s - I_n)$
  $F := 2^s F$

6 Computation of $f(A)b$

In this chapter, we compute $f(A)b$ for a given large and sparse matrix $A \in \mathbb{C}^{n\times n}$ and a given vector $b \in \mathbb{C}^n$, without explicitly computing $f(A)$. Here, *sparse* means that matrix-vector multiplications can be performed more efficiently.

Example 6.1. Note, that the sparsity of A does not imply the sparsity of $f(A)$. Let $A = J_n(0)$ and $f(z) = \exp(z)$. Then

$$A = \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix} \quad \text{and} \quad f(A) = \begin{bmatrix} 1 & 1 & \cdots & \frac{1}{(n-1)!} \\ & \ddots & \ddots & \vdots \\ & & \ddots & 1 \\ & & & 1 \end{bmatrix}.$$

In A we have $\mathcal{O}(n)$ non-zero entries, while in $f(A)$, we have $\mathcal{O}(n^2)$ non-zero entries.

We consider two approaches for the computation of $f(A)b$.

(1) **Krylov subspace methods:** Similar to the related approaches for solving linear systems or computing eigenpairs, we reduce the problem of computing $f(A)b$ to a smaller subspace, i.e., we evaluate f on a smaller matrix to find an approximation for $f(A)b$.

(2) **Quadrature rules:** We use numerical integration for the Cauchy integral formula for matrix functions

$$f(A)b = \frac{1}{2\pi \mathbf{i}} \int_\Gamma f(z)(zI_n - A)^{-1} b \, \mathrm{d}z.$$

6.1 Krylov Subspace Methods

We briefly recall the basics on Krylov subspace methods from the course Numerical Linear Algebra; see [19]. A comprehensive treatment can be found in [21].

Recall that for $A \in \mathbb{C}^{n\times n}$ and $b \in \mathbb{C}^n$, the k-th *Krylov subspace* is given by

$$\mathcal{K}_k(A, b) := \operatorname{span}\{b, Ab, \ldots, A^{k-1}b\} = \{p(A)b \mid p \in \mathbb{C}[z], \deg(p) \le k-1\}. \tag{6.1}$$

Furthermore, we call $d(A, b) := d$ the *grade* of b with respect to A, if the vectors $b, Ab, \ldots, A^{d-1}b$ are linearly independent and the vectors $b, Ab, \ldots, A^d b$ are linearly dependent. In particular, d is the degree of the minimal polynomial M_A of A and b. The Krylov subspace $\mathcal{K}_k(A, b)$ is a natural choice for searching approximations to $f(A)b$ since $f(A)b = p(A)b \in \mathcal{K}_{d(A,b)}(A, b)$ holds for some polynomial p of degree $d-1$ since d is the grade of b. Thus, we need an orthonormal basis of $\mathcal{K}_k(A, b)$, which can be computed using the Arnoldi algorithm. The Arnoldi algorithm is basically the Gram-Schmidt method applied to $\{b, Ab, \ldots, A^{k-1}b\}$ and its matrix formulation is

$$AV_k = V_{k+1}H_{k+1,k} = V_k H_k + h_{k+1,k} v_{k+1} e_k^T, \tag{6.2}$$

where $V_j = [v_1, \dots, v_j] \in \mathbb{C}^{n\times k}$ is *isometric*, i.e., with orthonormal columns, and $H_{k+1,k} = [h_{ij}] \in \mathbb{C}^{(k+1)\times k}$ is a matrix of the form

$$H_{k+1,k} = \begin{bmatrix} * & \cdots & \cdots & * \\ * & \ddots & & \vdots \\ & \ddots & \ddots & \vdots \\ & & \ddots & * \\ & & & * \end{bmatrix} \in \mathbb{C}^{(k+1)\times k}. \tag{6.3}$$

Since V_k is isometric, we find

$$V_k^H A V_k = H_k, \tag{6.4}$$

where $H_k \in \mathbb{C}^{k\times k}$ is an unreduced upper Hessenberg matrix, i.e., $h_{j+1,j} \neq 0$ for $j = 1, \dots, k$, which consists of the first k rows of $H_{k+1,k}$.

Algorithm 6 Arnoldi algorithm

Input: $A \in \mathbb{C}^{n\times n}$, $b \in \mathbb{C}^n$

Output: Orthonormal basis $\{v_1, \dots, v_d\}$ of $\mathcal{K}_d(A, b)$

$v_1 = \dfrac{b}{\|b\|_2}$

for $j = 1, \dots, k \leq d(A, b) - 1$ **do**

$\tilde{v}_{k+1} = Av_k - \sum_{j=0}^{k} h_{jk} v_j \quad \text{with} \quad h_{jk} = \langle Av_k, v_j \rangle$

$v_{k+1} = \dfrac{\tilde{v}_{k+1}}{h_{k+1,k}}$ with $h_{k+1,k} = \|\tilde{v}_{k+1}\|_2$

end for

Definition 6.2. Let $A \in \mathbb{C}^{n\times n}$, $b \in \mathbb{C}^n$, $k \leq d = d(A, b)$, and let $AV_k = V_{k+1}H_{k+1,k}$ be an Arnoldi decomposition. Moreover, let f be defined on the spectrum of A and on the spectrum of H_k. Then we define the k-th *Arnoldi approximation* for $f(A)b$ by

$$f_k \coloneqq V_k f(H_k) V_k^H b = \|b\|_2 V_k f(H_k) e_1, \tag{6.5}$$

where we used $V_k e_1 = \frac{b}{\|b\|_2}$.

Note, if $n = k = d(A, b)$, the Arnoldi approximation coincides with $f(A)b$, because in that case V_n is unitary and we find

$$f(A)b = f(V_n H_n V_n^H)b = V_n f(H_n) V_n^H b = \|b\|_2 V_n f(H_n) e_1. \tag{6.6}$$

Theorem 6.3. *Let $A \in \mathbb{C}^{n\times n}$, $b \in \mathbb{C}^n$ and $d = d(A, b)$. Moreover, let f be defined on the spectrum of A. Then $f_d = f(A)b$, i.e., the d-th Arnoldi approximation is exact.*

Proof. Since d is the grade of b with respect to A, we have

$$AV_d = V_d H_d$$

with $V_d e_1 = \frac{b}{\|b\|_2}$. Next, let $V \in \mathbb{C}^{n\times(n-d)}$, such that $[V_d, V]$ is unitary. Note, that because of the orthonormality of the columns in $[V_d, V]$, we have $V^H V_d = 0$. Thus,

$$[V_d, V]^H A [V_d, V] = \begin{bmatrix} V_d^H A V_d & V_d^H A V \\ V^H A V_d & V^H A V \end{bmatrix} = \begin{bmatrix} V_d^H V_d H_d & V_d^H A V \\ V^H V_d H_d & V^H A V \end{bmatrix} = \begin{bmatrix} H_d & * \\ 0 & V^H A V \end{bmatrix},$$

and therefore,

$$\begin{aligned} f(A)b &= [V_d, V] f\left(\begin{bmatrix} H_d & * \\ 0 & V^H A V \end{bmatrix} \right) [V_d, V]^H b = [V_d, V] \begin{bmatrix} f(H_d) & * \\ 0 & f(V^H A V) \end{bmatrix} [V_d, V]^H b \\ &= [V_d, V] \begin{bmatrix} f(H_d) & * \\ 0 & f(V^H A V) \end{bmatrix} \|b\|_2 e_1 = \|b\|_2 [V_d, V] \begin{bmatrix} f(H_d) e_1 \\ 0 \end{bmatrix} \\ &= \|b\|_2 V_d f(H_d) e_1. \end{aligned}$$

□

Next, we prove that the k-th Arnoldi approximation is exact for polynomials of degree at most $k-1$.

Theorem 6.4. *Let $A \in \mathbb{C}^{n\times n}$, $b \in \mathbb{C}^n$ and let $AV_k = V_{k+1} H_{k+1,k}$ be an Arnoldi decomposition with $V_k e_1 = \frac{b}{\|b\|_2}$. Then*

$$f(A)b = \|b\|_2 V_k f(H_k) e_1 \tag{6.7}$$

for all polynomials f with $\deg(f) < k$.

Proof. First, we prove the result for $f(z) = z^j$, i.e., we prove

$$A^j b = \|b\|_2 V_k H_k^j e_1$$

by induction. Let $j = 0$, then

$$A^0 b = b = \frac{b}{\|b\|_2} \|b\|_2 = \|b\|_2 V_k e_1 = \|b\|_2 V_k H_k^0 e_1.$$

Next, using the induction hypothesis, we have

$$\begin{aligned} A^j b = A(A^{j-1})b &= \|b\|_2 A V_k H_k^{j-1} e_1 = \|b\|_2 \left(V_k H_k + h_{k+1,k} v_{k+1} e_k^T \right) H_k^{j-1} e_1 \\ &= \|b\|_2 V_k H_k^j e_1 + \|b\|_2 h_{k+1,k} v_{k+1} \big(e_k^T H_k^{j-1} e_1 \big). \end{aligned}$$

Note, that the last term in parentheses is the entry $(k,1)$ entry of H_k^{j-1}. Considering the powers of Hessenberg matrices, it can easily be seen that this entry is zero if $k \geq j+1$. Hence,

$$A^j b = \|b\|_2 V_k H_k^j e_1.$$

Since any polynomial is a linear combination of monomials, the assertion follows for polynomials of degree at most $k-1$. □

To prove the next theorem, we need the following lemma.

Lemma 6.5. *Let $H \in \mathbb{C}^{n\times n}$ be an unreduced upper Hessenberg matrix. Then H is non-derogatory, i.e., the geometric multiplicity of all eigenvalues of H is 1.*

Proof. Let $\lambda \in \Lambda(H)$. Removing the first row and last column of the matrix $H - \lambda I_n$ yields an upper triangular matrix of size $(n-1)\times(n-1)$ with non-zero diagonal entries, and hence, a matrix with rank $n-1$. Therefore, $\operatorname{rank}(H-\lambda I_n) \geq n-1$. Thus,

$$1 \leq g(\lambda, H) = \dim\left(\ker\left(H-\lambda I_n\right)\right) = n - \operatorname{rank}\left(H-\lambda I_n\right) \leq n-(n-1) = 1. \qquad \square$$

By the next theorem, the k-th Arnoldi approximation f_k is the Hermite interpolation polynomial of f on the spectrum of H_k evaluated at A and multiplied with b.

Theorem 6.6. *Let $A \in \mathbb{C}^{n\times n}$, $b \in \mathbb{C}^n$ and let $AV_k = V_{k+1}H_{k+1,k}$ be an Arnoldi decomposition with $V_k e_1 = \frac{b}{\|b\|_2}$. Moreover, let f be defined on the spectrum of A and on the spectrum of H_k. Then there exists a unique polynomial p with $\deg(p) \leq k-1$ that interpolates f on the spectrum of H_k, i.e., with*

$$p^{(j)}(\lambda) = f^{(j)}(\lambda), \qquad j = 0, \ldots, a(\lambda, H_k) - 1$$

for all $\lambda \in \Lambda(H_k)$. Furthermore, the k-th Arnoldi approximation is given by $f_k = p(A)b$.

Proof. Since H_k is an unreduced upper Hessenberg matrix, it is non-derogatory by Lemma 6.5. Thus, there exists no eigenvalue with more than one Jordan block and by Lemma 2.7, p is unique. Moreover,

$$f_k = \|b\|_2 V_k f(H_k)e_1 = \|b\|_2 V_k p(H_k)e_1 \overset{\text{Thm. 6.4}}{=} p(A)b. \qquad \square$$

The next theorem yields an upper bound for the error of the Arnoldi approximation. Here we denote the set of polynomials with a degree less than or equal to m by

$$\mathcal{P}_m \coloneqq \{p \in \mathbb{C}[z] \mid \deg(p) \leq m\}. \tag{6.8}$$

Theorem 6.7 ([3, Prop. 3.1])**.** *Let $A \in \mathbb{C}^{n\times n}$, $b \in \mathbb{C}^n$ and $\Omega \subseteq \mathbb{C}$ be convex and compact with $\Lambda(A) \subseteq \Omega$. Moreover, let f be continuous on Ω and analytic on $\operatorname{int}(\Omega)$. Then*

$$\|f(A)b - f_k\|_2 \leq 23\|b\|_2 \min_{p\in\mathcal{P}_{k-1}} \max_{z\in\Omega} |f(z)-p(z)|. \tag{6.9}$$

Remark 6.8. (1) The bound in (6.9) reminds us of to the similar bounds for the relative error in the A-norm in the CG method, or the bound for the relative 2-norm residual in the GMRES method.

(2) We receive a good approximation to $f(A)b$ if f can be approximated well by a low-degree polynomial on Ω. This is in particular the case if $\Lambda(A)$ is a single cluster, i.e., Ω can be chosen to be small.

(3) Considering a particular polynomial in (6.9) yields special cases of the bound.

Example: Suppose that $\Omega = \{z \in \mathbb{C} \mid |z| \leq \rho\}$ and consider the Taylor polynomial

$$p(z) = \sum_{j=0}^{k-1} \frac{f^{(j)}(0)}{j!} z^j.$$

of order $k-1$. Then

$$\max_{z \in \Omega} |f(z) - p(z)| \lesssim \frac{\rho^k}{k!} \xrightarrow{k \to \infty} 0,$$

which shows $\lim_{k \to \infty} f_k = f(A)b$ in that case.

Remark 6.9. The Arnoldi approach for computing $f(A)b$ has the classical "GMRES issues".

(1) *Computational costs and required storage increase with every iteration.*

A solution to this is to use restarted Arnoldi, i.e., to restart the method after a prescribed number of iterations with a new Krylov subspace and a new initial vector that somehow preserves the previously computed information.

(2) *We have to solve the "reduced" problem $f(H_k)e_1$.*

For this, we can use any method for computing $f(A)$, e.g., the ones from Chapter 5.

6.2 Quadrature Methods

Recall that in numerical integration, we want to approximate an integral

$$\int_a^b f(x) \, \mathrm{d}x \tag{6.10}$$

with only finitely many evaluations of f. For quadrature nodes $a \leq t_0 < \cdots < t_m \leq b$, and weights $\omega_0, \ldots, \omega_m \in \mathbb{C}$, we use the approximation

$$\int_a^b f(x) \, \mathrm{d}x \approx \sum_{j=0}^{n} \omega_j f(t_j), \tag{6.11}$$

e.g., by the *trapezoidal rule*

$$\int_a^b f(x) \, \mathrm{d}x \approx \frac{b-a}{m} \Big(\frac{f(a)}{2} + \sum_{j=1}^{m-1} f(a + j \cdot \tfrac{b-a}{m}) + \frac{f(b)}{2} \Big) =: T_m(f), \tag{6.12}$$

where

$$t_j = a + j \cdot \frac{b-a}{m} \qquad \text{and} \qquad \omega_j = \begin{cases} \frac{1}{2}, & j = 0, \text{ or } j = m, \\ 1, & 1 \leq j \leq m-1. \end{cases}$$

The approximation quality can be estimated by

$$\left| \int_a^b f(x)\,\mathrm{d}x - T_m(f) \right| \leq \frac{b-a}{m^2} \max_{x \in]a,b[} |f''(x)|. \tag{6.13}$$

We use this idea for the Cauchy integral formula for matrix functions

$$f(A)b = \frac{1}{2\pi \mathbf{i}} \int_\Gamma f(z)(zI_n - A)^{-1} b\,\mathrm{d}z. \tag{6.14}$$

Applying a quadrature rule to this integral reduces to solving the shifted linear system

$$(z_j I_n - A)x = b \tag{6.15}$$

for several nodes $z_j \in \Gamma$. Recall that the above linear systems are uniquely solvable since $\Lambda(A) \subseteq \mathrm{int}(\Gamma)$.

For simplicity, we consider a circle with center $c \in \mathbb{C}$ and radius $\rho > 0$ with parametrization

$$\Gamma : [0, 2\pi[\to \mathbb{C}, \qquad t \mapsto c + \rho e^{\mathbf{i}t}.$$

as the Jordan curve in (6.14). Then $\Gamma'(t) = \mathbf{i}\rho e^{\mathbf{i}t} = \mathbf{i}(\Gamma(t) - c)$, and we receive

$$\begin{aligned} f(A)b &= \frac{1}{2\pi\mathbf{i}} \int_\Gamma f(z)(zI_n - A)^{-1} b\,\mathrm{d}z = \frac{1}{2\pi\mathbf{i}} \int_0^{2\pi} \Gamma'(t) f\big(\Gamma(t)\big)\big(\Gamma(t)I_n - A\big)^{-1} b\,\mathrm{d}t \\ &= \frac{1}{2\pi} \int_0^{2\pi} \underbrace{\big(\Gamma(t) - c\big) f\big(\Gamma(t)\big)\big(\Gamma(t)I_n - A\big)^{-1} b}_{=:G(t)}\,\mathrm{d}t = \frac{1}{2\pi} \int_0^{2\pi} G(t)\,\mathrm{d}t. \end{aligned}$$

Applying the trapezoidal rule yields

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} G(t)\,\mathrm{d}t \approx \frac{1}{2\pi} T_m(G) &= \frac{1}{2\pi} \cdot \frac{2\pi}{m} \Big(\frac{G(0)}{2} + \sum_{j=1}^{m-1} G(j\tfrac{2\pi}{m}) + \frac{G(2\pi)}{2} \Big) \\ &= \frac{1}{m} \sum_{j=0}^{m-1} G(j\tfrac{2\pi}{m}), \end{aligned} \tag{6.16}$$

where we used $G(0) = G(2\pi)$.

Theorem 6.10 ([8, p. 138])**.** *Let $k \in \mathbb{N}$ be odd and let $f \in C^k(\mathbb{R})$ be 2π-periodic with $|f^{(k)}(x)| \leq M$ for some $M \geq 0$ and all $x \in \mathbb{R}$. Then*

$$\left| \int_0^{2\pi} f(x)\,\mathrm{d}x - T_m(f) \right| \leq M \cdot \frac{4\pi\zeta(k)}{m^k}. \tag{6.17}$$

Here, ζ denotes the Riemann zeta function for which we have $|\zeta(k)| \leq 2$ for $k \geq 2$, and $\zeta(k) \xrightarrow{k\to\infty} 0$. Thus, we can bound ζ by a constant as well. Taken all together, the term $\frac{4\pi\zeta(k)}{m^k}$ approaches zero quickly with large m and k, but the constant M is potentially very large.

Remark 6.11. To apply Theorem 6.10 to (6.16), we have to bound

$$G^{(k)}(t) = \frac{\mathbf{i}^k}{2\pi} \sum_{j=0}^{k} \underbrace{\big(\Gamma(t) - c\big)^{j+1}}_{\text{(a)}} \sum_{i=0}^{j} \underbrace{c_{ijk}}_{\text{(d)}} \underbrace{f^{(j-i)}\big(\Gamma(t)\big)}_{\text{(c)}} \underbrace{\big(\Gamma(t) I_n - A\big)^{-(i+1)}}_{\text{(b)}} b. \tag{6.18}$$

Several terms may lead to a large error bound in Theorem 6.10:

(a) For a very large radius ρ, this term results in $|G^{(k)}(t)| \lesssim \rho \cdot \left(\frac{\rho}{m}\right)^k$. Thus, widely spread eigenvalues are bad for convergence.

(b) The quantity $\|\big(\Gamma(t) I_n - A\big)^{-(i+1)}\|$ is large, if the eigenvalues of A are close to Γ.

(c) The derivative $f^{(k)}$ may become very large, e.g., for $f(z) = \sqrt{z}$ near $z = 0$.

(d) The constants c_{ijk} grow quickly with k.

In summary, a rapid decay of the bound in Theorem 6.10 for the trapezoidal rule applied to the Cauchy integral formula for matrix functions is not guaranteed. However, this method can be efficient for matrices that have a spectrum in a small disk which is not too close to Γ. If these conditions are not fulfilled, conformal maps can be used to map the spectrum to a more suitable configuration, but this could be the beginning of a new story...

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